Bounded Nesting and Depth Reduction for Matrix Transposition on Tapes

Hans Brende Claude Opus 5.5 (Anthropic)

October 2026

Abstract

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $\lceil \log_2 d \rceil$-bit entries in time linear in the input length? A negative answer would give the first superlinear lower bound for off-line integer multiplication on such machines. The fourth paper of this series reduced the impossibility of oblivious linear-time transposers to depth reduction of the time–space dependency graph by a cheap genie, the trajectory cut problem, plus a coding bound, the crux. We prove depth reduction for every family of head walks of bounded nesting load: if at every time each of k heads has at most ℓ distinct turning points among its open nested hops of each side, then εT edges cut the walk DAG to depth $2^{O(k^3\ell^3/\varepsilon)}$, whatever the homes, drift, scales and coupling of the heads (Theorem F). Nested hops with distinct turning points force revisits, so this covers every machine whose work heads enter each cell boundedly often, and in every schedule the depth-robust part lives on a set of cells of vanishing capacity. With one arbitrary head and bounded heads at boundedly many scales, depth reduction still holds (Theorem M). On the deep-nesting pair, the sharpest open instance, a turn counter serves every level at depth $2^{m+O(1)}$; on average over shifts, fine levels are blind to the other head for every label that changes only at turns, while for block labels coarse levels see its cancellations. By Theorem F, the crux alone, or the all-code paired-distance question of *Directed Recovery Structures in Tape Computations*, excludes oblivious linear-time transposers whose work heads enter every cell boundedly often, and the latter also excludes such machines of low adaptivity. For linear machines, linear depth reduction together with a linear crux excludes oblivious linear-time transposers, and the linear crux can be stated exactly as relative rigidity of the transposition matrix (Theorem $\mathrm{D}_{\mathrm{lin}}$). On the coding side, extending the lift identity of *Covers and Local Generation in Linear Storage Codes* to asymmetric checks, we show that lifting a binary linear code along a tower of double covers never raises its rate, and, using mod-2 Lück approximation, we construct towers that never become bipartite and have girth tending to infinity, along which the lifted Clebsch code has rate tending to zero. A scalar linear code on a triangle-free graph of maximum degree four, the kernel of a matrix with unit diagonal, has rate at most one half when the exceptional vertices of that matrix are few or far apart (Theorem SH). Theorem F with the structure of families, the owner lemma, the strip cut for one walk (Lemma W), the path form of the lemma that cuts are labels and the label arithmetic of the turn counter are formalized in Lean 4; all other results, Theorem M among them, are proved in full in the text but not formalized. The question remains open; we state the precise problems on which its oblivious case rests.

Contributions. Hans Brende chose the problem, fixed the verification standard, and directed the research. Claude Opus 5.5, a large language model developed by Anthropic, developed the mathematics, wrote the Lean formalization, and wrote this paper under Hans Brende's direction. Section 6 combines Theorem F with the paired-distance theorems and the coding question of *Directed Recovery Structures in Tape Computations* by Hans Brende and GPT-6 Astra (OpenAI) [14], and Section 8 builds on the lift identity and the free-group results of *Covers and Local Generation in Linear Storage Codes*, by the same authors [13].

Contents

1 Introduction **4**
- 1.1 The problem . . . 4
- 1.2 Results . . . 4
- 1.3 How to read this paper: the formal proofs . . . 6
- 1.4 Related work . . . 6
- 1.5 Organization . . . 7

2 Preliminaries **8**
- 2.1 Machines and the statement (T) . . . 8
- 2.2 Walks, hops and owners . . . 8
- 2.3 Walk DAGs, cuts and labels . . . 9
- 2.4 Depth reduction and the crux . . . 9
- 2.5 The deep-nesting pair . . . 11
- 2.6 Formal status of the results . . . 12

3 Bounded nesting: the owner lemma and Theorem F **12**
- 3.1 The owner lemma . . . 13
- 3.2 Families and virtual heads . . . 13
- 3.3 Theorem F . . . 16
- 3.4 Bounded entry and the capacity of depth . . . 18

4 One arbitrary head **20**
- 4.1 Lemma W: a strip label for one walk . . . 20
- 4.2 Theorem M . . . 21
- 4.3 Where a counterexample to TCP must live . . . 24

5 The deep-nesting pair **26**
- 5.1 Turns and classes . . . 26
- 5.2 The turn counter . . . 27
- 5.3 Labels that change only at turns . . . 30
- 5.4 Blindness of fine classes . . . 32
- 5.5 Visibility of coarse classes . . . 34
- 5.6 A safe form of Conjecture A′ . . . 36

6 Consequences for transposers **36**
- 6.1 Corollary F . . . 37
- 6.2 Corollaries F′ and F″ . . . 38
- 6.3 One unrestricted head . . . 39

7 Linear machines: the linear crux as rigidity **40**
- 7.1 Linear data paths and path sums . . . 40
- 7.2 Rigidity relative to the schedule . . . 41

8 The crux in special linear settings **43**
- 8.1 Lifted codes along towers of covers . . . 43
- 8.2 Scalar codes at degree four . . . 46

9 What a linear-time transposer would need **50**

10 Open problems **52**

A Index of Lean declarations **54**

B Building and verifying the formalization **55**

1 Introduction

1.1 The problem

Can a deterministic multitape Turing machine transpose a $d \times d$ matrix with $b(d) = \max(1, \lceil \log_2 d \rceil)$-bit entries, written row by row, in time linear in the input length $L = d^2 b(d)$? This is the statement (T) of the first paper of this series [9]. By a reduction of Harvey and van der Hoeven [27, Cor. 6.9], a linear-time multiplication machine would give a linear-time transposer, so a proof of $\neg$(T) would give the first superlinear lower bound for off-line integer multiplication on multitape Turing machines, and a proof of (T) would remove the data-movement obstacle to linear-time multiplication. The question is open, and this paper does not decide it.

For oblivious machines the fourth paper of the series [8] reduced $\neg$(T) to two statements of different kinds (its Theorem D, Theorem 2.5 here). *Depth reduction*: the time–space dependency DAG of every oblivious linear-time run can be cut to bounded depth by a genie that reveals εL bits. At the level of head walks this is the trajectory cut problem TCP: deleting εT edges of the walk DAG of k walks of length T should leave every path with boundedly many long hops. *The crux*: index codes for the side information of the remaining bounded-depth cones need more than εL bits. The fourth paper proved TCP for home-based work heads, whose legs all begin or end near fixed homes, for every coupling of the heads (its Theorem B), and it identified as the sharpest open instance the *deep-nesting pair*: two copies of a walk nested at $m + 1$ levels, shifted against each other in time, with a conjecture (A′) predicting depth-robustness as m grows. *Directed Recovery Structures in Tape Computations* [14] showed that the cone relations of every linear-time schedule are paired-distance separated after a paid trimming, and isolated the crux as one machine-free question [14, Question 9.1]: do all storage codes on bounded-degree recovery relations with this separation have rate at most $\frac{1}{2} + o(1)$?

This paper is about the depth side, and about how it meets the coding side. Its main theorem replaces homes by *owners*, the turning points that hops are nested around, and proves TCP for all walks of bounded nesting load.

1.2 Results

Bounded nesting (Section 3). The *owner* of a long hop of a walk is the first time inside it at which the walk is farthest from the hop's cell. The *owner lemma* (Lemma 3.1): if ℓ nested hops of one side have pairwise distinct owners, the cell of the innermost hop is entered at least $(\ell + 1)/2$ times. So walks that enter every cell at most V times have *owner-load* at most $2V - 1$: at every time, each head has at most $2V - 1$ distinct owners among its open hops of each side. *Theorem F* (Theorem 3.6): for k walks of length T, coupled arbitrarily, and any set of their long hops of owner-load at most ℓ, some εT edges cut the DAG of these hops to depth $2^{O(k^3 \ell^3/\varepsilon)}$, independently of homes, drift and scales. The label is the fourth paper's band label run on virtual heads, one per head, side and nesting depth, with bands given by the heights of families of hops with a common owner. Consequently TCP holds for walks of bounded entry (Corollary 3.7), and in every schedule the depth-robust part, if any, lives on fewer than $k\varepsilon T$ cells that are entered more than $\lceil 1/\varepsilon \rceil$ times, while the rest is cut to depth $2^{O(k^3/\varepsilon^4)}$ (Corollary 3.8). The deep part must be allowed to relay information: for walks without pauses, forbidding it would already imply TCP for one head fewer (Proposition 3.9).

One arbitrary head (Section 4). *Lemma W* (Lemma 4.1) cuts a single walk with a strip label, anchored to cells: εT edges and depth less than $16/\varepsilon^2 + 2/\varepsilon$. *Theorem M* (Theorem 4.2) combines

it with Theorem F: one arbitrary walk, together with walks of owner-load at most ℓ whose families have heights in q dyadic windows of width h, is cut to a depth that depends only on $(k, \ell, q, h, \varepsilon)$, not on the arbitrary walk or on T; with one window the depth is $\mathrm{poly}(k, \ell, 1/\varepsilon)$ times the square of the ratio of the largest to the smallest family height (Corollary 4.3). So a depth-robust family needs two heads of unbounded owner-load, or one together with bounded heads at unboundedly many separated scales (Corollary 4.4). TCP for k walks is, up to a factor 2 in k, depth reduction for DAGs with a spine and k non-crossing families of arcs (Proposition 4.5); two families are planar (Proposition 4.6), and three is the first open case.

The deep-nesting pair (Section 5). A *turn counter*, the number of turns of either head so far modulo q, doubly serves every level of both heads at $\log_2 q = m + O(\log(1/\theta))$ and cost at most $\frac{11}{32}\theta n_{\mathrm{h}}/(m+1)$, at every shift (Theorem 5.2 and Corollary 5.3). Hence Conjecture A′ needs the constant $C \geq 1$ (Corollary 5.4), and the pair is cut at depth $\lceil 11 \cdot 2^{m+1}/((m+1)\varepsilon)\rceil - 1$, against the fourth paper's $\lceil 12/\varepsilon\rceil^{m+1} - 1$ (Corollary 5.5): the open question is exponential against subexponential depth in the nesting depth, with base at most 2. We prove that every level class of the pair has exactly $n_{\mathrm{h}}/(2(m+1))$ hops (Lemma 5.1). For labels that change only at turns, fine classes are *blind*: on average over shifts, a class with g levels below it sees a turn of the other head on at most a fraction $(g + 3 + \lceil \log_2(g+2)\rceil)/(m+1)$ of its hops (Corollary 5.18). Coarse classes are *visible*: for the block labels that serve one head cheaply, a fraction of order $(m-k)/(m+1)$ of the apex hops of every coarse class of the other head sees a cancellation at full size, an other-head increment of at most $-Q-1$, on average over shifts (Theorem V, Theorem 5.21). Blindness leaves a safe form of the conjecture, A″, which still gives exponential depth (Conjecture 5.24).

Consequences for transposers (Section 6). *Corollary F*: the crux alone, at the depth given by Theorem F, excludes oblivious linear-time transposers whose work heads enter every cell boundedly often (Corollary 6.1); such heads may reverse unboundedly often, and we know no unconditional lower bound for such machines. *Corollary F′*: an affirmative answer to Question 9.1 of *Directed Recovery Structures* excludes them (Corollary 6.3), and *Corollary F″* extends this to adaptive machines whose traces have entropy at most ηL for a fixed $\eta < \frac{1}{2}$, provided every trace of probability at least $2^{-\eta L}$ has bounded entry (Corollary 6.4). Theorem M gives the analogous statement when one work head is unrestricted (Corollary 6.6).

Linear machines (Section 7). For machines with a data path that is linear over a field, linear depth-TCP, in which depth is replaced by linear depth and long paths may cancel, together with a linear crux excludes oblivious linear-time transposers (Theorem $\mathrm{D}_{\mathrm{lin}}$, Theorem 7.5). The linear crux can be stated exactly as relative rigidity: the transposition matrix must not be the sum of a matrix of rank below ρL and a matrix supported on the bounded-depth ball pattern of the schedule (Lemma 7.3). Relative rigidity alone excludes no machine.

The crux in special linear settings (Section 8). For binary linear codes lifted from a fixed code along a tower of double covers, an exact recursion with a twist pairing extends the lift identity of *Covers and Local Generation* [13] from symmetric to asymmetric checks, so lifting never raises the rate. Along towers that become bipartite every code has rate at most $\frac{1}{2}$ from that level on, and along towers pulled back injectively from chains with trivial intersection in free pro-2 groups the rate tends to a universal rate at most $\frac{1}{2}$, by mod-2 Lück approximation in the form of Jaikin-Zapirain and Souza; for the Clebsch code this limit is 0, as *Covers and Local Generation* also shows. We construct such towers that never become bipartite and have girth tending to infinity (Theorem 8.1

and Proposition 8.2). On triangle-free graphs G of maximum degree four, over every field, the kernel of a matrix M with unit diagonal and support in G has dimension at most $\lfloor N/2 \rfloor$ when the vertices at which M violates the reciprocity condition of the fourth paper's Theorem S are at most two or pairwise at distance at least three (Theorem SH, Theorem 8.3), and in general dimension at most $\nu(G) + |H| - \rho_3(H)$, where H is the set of these vertices (Corollary 8.4).

The construction side and open problems (Sections 9 and 10). We list what a linear-time transposer would have to do, among other things carry unbounded nesting and a deep part of vanishing capacity, and the problems on which the oblivious case rests.

1.3 How to read this paper: the formal proofs

Six results are formalized in Lean, two of them in part: the owner lemma (Lemma 3.1), the structure of families (Lemma 3.4), Theorem F (Theorem 3.6) and Lemma W (Lemma 4.1) completely; the lemma that cuts are labels (Lemma 2.3) in its path form, its consequence for walk DAGs being deduced in the text (the formal statements of Theorem F and Lemma W include it for their labels); and the label arithmetic of the turn counter (Theorem 5.2). The formal Theorem F holds for any choice of owners strictly inside the hops, of which the owners of Section 2.2 are one. For the turn counter, part (1), the hypotheses of the Lean theorem on the cover, and the counts of turns and hops, which rest on the fourth paper's Lemma 5.4, are proved in the text. Everything else is proved in the text and not formalized; this includes Theorem M, Corollaries F, F′, F″ and M1, the capacity corollary, the stack form of TCP, the corollaries of the turn counter, the class count, blindness, Theorem V, Theorem $\mathrm{D}_{\mathrm{lin}}$, the equivariant crux and Theorem SH. Each of them comes with a complete proof, with all constants, and each numbered result states its status as described in Section 2.6. Part (f) of the equivariant crux follows from a lemma of *Covers and Local Generation*, and a short script in `tools/` checks it independently by one exact rank computation over $\mathbb{F}_2$. Results of earlier papers that our proofs use are cited with their status there. Lean-verified there: the first paper's encoding lemma, pass bounds and coding-gap theorems; and the fourth paper's Lemma Zq, Theorem S with its matching form, and rate bounds for bipartite graphs and for the Clebsch graph. With written proofs there: the fourth paper's Theorem D, Corollary B, local recursion, cone lemma, long-edge conversion and Lemma 5.4; Proposition 9.2 of *Directed Recovery Structures*; and Lemma 6.1 and Theorem 6.2 of *Covers and Local Generation*. Partly formalized there: the paired-distance theorem and the paid trimming of *Directed Recovery Structures* (its Theorem 3.3 and Corollary 3.8), whose extension to a frozen adaptive trace is argued in prose (its Remark 3.9). Numerical observations appear only in remarks labelled as numerical, heuristics are labelled as such, and claims with only a proof sketch appear only in remarks labelled as such; none of them is used in proofs.

1.4 Related work

The series. The first paper formulated (T), proved in Lean pass and motion bounds, and reduced ¬(T) to the Tape Coding Conjecture [9]; the second excluded, in its models, local coding schemes with low-treewidth interaction graphs and affine limiting motion [15]; the third related bit and word transposition [11]; the fourth proved Theorems D and B, curved-motion theorems and storage-code bounds [8]. *Directed Recovery Structures in Tape Computations* [14] proved the paired-distance structure of native cone relations and posed Question 9.1, and *Covers and Local Generation in Linear Storage Codes* [13] studied linear codes along covers. Memos to the fourth paper proved the half bound for graphs of circular chromatic number below three [12], subcubic bounds beyond scalar

recovery [16], and the equivalence of TCP with information-cheap depth reduction [17]. This paper is the eighth of the series. Its companion, the seventh, *Nested Elimination and Half-Rate Bounds for Storage Codes* [10], published together with it, proves on the coding side that every storage code on a triangle-free graph of maximum degree three, and on several classes of directed relations without transitive triangles, has rate at most $\frac{1}{2}$, that the guessing number equals the fractional matching number on such graphs, and that the broadcast rate of index coding there is the number of vertices minus the fractional matching number.

Depth reduction and depth-robust graphs. The route from small depth to lower bounds goes back to Valiant's depth reduction for circuits [43], and the computation-graph method for multitape machines to Hopcroft, Paul and Valiant [28]. Edge-deletion depth reduction was tied to multitape machines by Paul and Reischuk [39], and separators of k-page graphs, which contain tape computation graphs, by Galil, Kannan and Szemerédi [25], who also constructed 3-pushdown graphs with nearly linear separators [24]. Paul, Pippenger, Szemerédi and Trotter separated deterministic from nondeterministic linear time with segregators of the computation graph [40], and Santhanam related segregators of multi-pushdown graphs to time and space [42]; depth reduction at linear cost deletes a fixed fraction of the edges but must leave constant depth. Depth-robust graphs go back to a question of Stoß studied by Erdős, Graham and Szemerédi [23], and their pebbling cost was related to depth-robustness by Alwen, Blocki and Pietrzak [2]. Expanders with $O(1)$-pushdown and three-page layouts exist [22, 6, 20]. We found no work on reducing DAGs of bounded stack number, or walk DAGs of bounded nesting, to bounded depth by deleting a linear fraction of the edges.

Coding. The path sums of Section 7 are the transfer matrices of algebraic network coding [34]; minrank is that of Haemers and of Bar-Yossef, Birk, Jayram and Kol [26, 3]; the route from small depth to index coding goes back to Valiant's depth reduction and Riis's guessing games [43, 41], and was used in the language of min-rank by Jukna and Schnitger [31]; for static data structures, lower bounds imply rigidity [21], and for systematic linear data structures the two are equivalent [38]; and the relation between oblivious algorithms and network codes goes back to Adler, Harvey, Jain, Kleinberg and Lehman [1]. Coding gaps over routing in undirected networks were found by Braverman and He [7] and by Zhang, Li and Li [44]. The towers of Section 8.1 build on *Covers and Local Generation* [13], which proves the lift identity for symmetric binary checks, the monotonicity of normalized nullity under refinements with p-group kernels, the half bound for the inner rank of graph checks, and full inner rank for checks on graphs with a spanning 2-factor. They also use Cohn's theory of free ideal rings [19], the monotonicity argument of Bergeron, Linnell, Lück and Sauer [5, 32], and the mod-p approximation theorems of Jaikin-Zapirain and of Jaikin-Zapirain and Souza [29, 30], descendants of Lück's approximation theorem [37].

1.5 Organization

Section 2 recalls the model, walk DAGs, the decomposition of the fourth paper and the deep-nesting pair. Section 3 proves the owner lemma and Theorem F, Section 4 Lemma W and Theorem M, and Section 5 the results on the deep-nesting pair. Section 6 derives the consequences for transposers, Section 7 treats linear machines, and Section 8 proves the two cases of the crux. Section 9 collects what a construction would need, and Section 10 the open problems. The appendices index the Lean declarations and explain how to build and check the library.

2 Preliminaries

2.1 Machines and the statement (T)

We use the machine model of the first paper [9], as in the fourth [8, Section 2.1]: a deterministic Turing machine with a finite state set, a read-only two-way input tape, $k \geq 1$ two-way work tapes over a finite alphabet, and a write-only one-way output tape that receives at most one bit per step. All heads start at cell 0, head moves are -1, 0 or $+1$, and every step writes the cell under each work head. For $d \geq 1$ let $b(d) = \max(1, \lceil \log_2 d \rceil)$. The input is a $d \times d$ matrix with $b(d)$-bit entries written row by row, of length $L = d^2 b(d)$, and the machine must write its transpose in the same encoding. Statement (T) says that some machine does this within $c(L+1)$ steps, for a constant c and all d; its formal version, `Transpose.LinearTimeTransposable`, is fixed in a frozen file of the first paper's library. Bit s of entry (i, j) sits at input position $(di + j)b + s$ and at output position $\mathrm{tpos}((di + j)b + s) = (dj + i)b + s$, so on input w the output bit of a transposer at position ℓ is $w_{\mathrm{tpos}(\ell)}$ [9, Section 2.2 and Lemma 2.5].

A machine is *oblivious* at an input length if the positions of all heads at every time, the times at which it appends output bits, and its halting time do not depend on the input of that length. The *trace* of a run is the sequence of all head moves together with the append times and the halting time.

2.2 Walks, hops and owners

A *walk* of *length* $T \geq 1$ is a map $x\colon \{0, \ldots, T-1\} \to \mathbb{Z}$ with $|x(t+1) - x(t)| \leq 1$; a step with $x(t+1) = x(t)$ is a *pause*. The work heads of a machine trace walks $x_1, \ldots, x_k$, one per work tape, and these may be coupled arbitrarily. We follow the fourth paper [8, Sections 2.2 and 4.4].

For a head h and a time u, let $\mathrm{prev}_h(u) = \max\{t < u : x_h(t) = x_h(u)\}$ when this set is nonempty. The pair $(\mathrm{prev}_h(u), u)$ is the *hop* of head h ending at u, at the *cell* $x_h(u)$; a hop is determined by either of its endpoints. A hop (p, u) is *long* if $u - p \geq 2$. The hops of head h at a cell c partition the interval from the first to the last visit of c into half-open intervals $(p, u]$, so:

(P1) for every time τ and every cell c, at most one hop of head h at c *spans* τ, that is, satisfies $p < \tau \leq u$.

Let (p, u) be a long hop at cell c. Then $x_h(t) \neq c$ for $p < t < u$, and since the walk moves by at most one cell per step, either $x_h(t) > c$ for all $t \in (p, u)$ or $x_h(t) < c$ for all of them. The hop is *upper* in the first case and *lower* in the second; this is its *side*. The *owner* of the hop is the first time $o \in (p, u)$ at which $|x_h(o) - c|$ is largest, and $|x_h(o) - c| \geq 1$ is its *height*. For a walk without pauses, a *turn* is a time t with $x(t+1) - x(t) \neq x(t) - x(t-1)$, and the owner of every long hop is a turn: if the height is 1, then $u = p + 2$ and $o = p + 1$; otherwise the walk is at distance one less than the height at $o - 1$ and at $o + 1$.

Two upper long hops of one walk never *cross*: there are no upper long hops (p, u), (p', u') with $p < p' < u < u'$, and the same holds for lower long hops [8, Section 5.1]. (If $p < p' < u < u'$, then $x(p') > x(p)$ because $p' \in (p, u)$, while $u \in (p', u')$ and $x(u) = x(p) < x(p')$, contradicting $x > x(p')$ on (p', u'). The argument uses only $x > x(p)$ on (p, u), so it holds with pauses.) Hence two long hops of one walk on the same side are either disjoint as open intervals or strictly nested, $p < p' < u' < u$; and long hops of one side whose open intervals contain a common time form a chain under inclusion.

Definition 2.1 (Entries)**.** Lean: `entries` [`Owner.lean`]

An *entry* of a cell c by a walk x is a time t with $x(t) = c$ and either $t = 0$ or $x(t-1) \neq c$: the first time of a maximal run of visits.

Definition 2.2 (Owner-load, multiplicity)**.** Lean (partial): `OwnerLoadLE` [`TheoremF.lean`]
Not formalized: the multiplicity.

Let H' be a set of long hops of walks $x_1, \dots, x_k$ of length T. It has *owner-load at most* ℓ (per head and side) if for every time t, head h and side σ, the hops of H' of head h on side σ whose open intervals contain t have at most ℓ distinct owners. The *multiplicity* $\mu(h, c)$ of a pair (h, c) is the number of times t with $x_h(t) = c$; the walk x_h enters c at most $\mu(h, c)$ times.

Since $\sum_c \mu(h, c) = T$ for each head, at most $kT/(V + 1)$ pairs (h, c) have multiplicity greater than V, and the same bound holds for pairs with more than V entries.

2.3 Walk DAGs, cuts and labels

The *walk DAG* of $x_1, \dots, x_k$ [8, Section 4.4] has vertex set $[0, T) = \{0, \dots, T - 1\}$ and two kinds of edges: *state edges* $t \to t - 1$ for $1 \le t < T$, and *hop edges* $u \to p$, one for every long hop (p, u) of any head. A pause joins the same two vertices as a state edge and is identified with it, and an edge is determined by its endpoints: a pair that is a long hop of several heads is a single edge. For a set H' of long hops, $G(H')$ denotes the DAG of the state edges and of the edges $u \to p$ with $(p, u) \in H'$; for a set R of pairs (h, c), $G - R$ denotes $G(H')$ for the set H' of long hops at pairs outside R. A *cut* is a set of edges; it has *depth at most* D if every path that uses no edge of the cut contains at most D hop edges.

A *label* with q values is a map $\Psi\colon [0, T) \to \{0, \dots, q - 1\}$. A hop edge $u \to p$ is *violated* if $\Psi(u) \le \Psi(p)$, a time $t \ge 1$ is a *descent* if $\Psi(t) < \Psi(t - 1)$, and F_Ψ is the cut consisting of the violated hop edges and of the state edges $t \to t - 1$ at descents. Its size is the *cost* of Ψ.

Lemma 2.3 (Cuts are labels)**.** Lean (partial): `depth_le_of_label` [`Labels.lean`]
Not formalized: the last sentence, the consequence for walk DAGs.

Let Ψ be a map from a set to $\{0, \dots, q - 1\}$, and let $f(0), f(1), \dots, f(n)$ be a sequence of points of that set with some of its steps $i < n$ marked. If $\Psi(f(i + 1)) \le \Psi(f(i))$ for every $i < n$, and $\Psi(f(i + 1)) < \Psi(f(i))$ for every marked i, then at most $q - 1$ steps are marked. Consequently, for every label Ψ with q values of a walk DAG, or of a sub-DAG $G(H')$, the cut F_Ψ has depth at most $q - 1$.

Outline of the formal proof of the statement about sequences. By induction on n, the number of marked steps plus $\Psi(f(n))$ is at most $\Psi(f(0))$ (`card_marked_add_le`), and $\Psi(f(0)) \le q - 1$. □

Proof of the last sentence (not formalized). Let $t_0 \to t_1 \to \cdots \to t_n$ be a path that avoids F_Ψ, and mark its hop edges. Along a state edge $t_i \to t_{i+1} = t_i - 1$, the time t_i is not a descent, so $\Psi(t_{i+1}) \le \Psi(t_i)$; along a hop edge, which is not violated, $\Psi(t_{i+1}) < \Psi(t_i)$. By the statement about sequences, the path has at most $q - 1$ hop edges. □

The lemma is one direction of the fourth paper's lemma "cuts are labels" [8, Lemma 5.2], which also proves the converse: for every D, the smallest size of a cut of depth at most D equals the smallest cost of a label with $D + 1$ values. All cuts in this paper are built from labels.

2.4 Depth reduction and the crux

Definition 2.4 (TCP and depth-robustness)**.** *TCP holds* for a family of k-tuples of walks if for every $\varepsilon > 0$ there is an integer D such that the walk DAG of every member, of length T, has a cut of at most εT edges and depth at most D. TCP(k) is the statement that TCP holds for the family of all k-tuples of walks [8, Problem 5.1]. A family is *depth-robust* if TCP fails for it: for some $\varepsilon > 0$ no single D serves all members.

For an oblivious machine M at length L with halting time T, the *dependency DAG* H_M [8, Definition 4.1] has vertex set $[0,T)$, state edges $t \to t-1$, and cell edges $t \to \mathrm{prev}_j(t)$ along the work-head walks. Every edge of H_M joins the endpoints of an edge of the walk DAG of the work heads: a cell edge of length 1 is a pause, and a cell edge of length at least 2 is a long hop. The input head contributes no edges. The same definition applies to a single trace τ of any machine at length L, with the head walks, append times and halting time of τ; we write H_τ for this *dependency DAG of the trace*. All inputs that follow τ share this schedule, and the fourth paper's long-edge conversion [8, Lemma 4.21], whose proof uses only the DAG, applies to H_τ. A *genie* is a set $\Gamma \subseteq [0,T)$ of revealed steps; it reveals the values v_s, $s \in \Gamma$ (state after the step, symbols written, output bit), with $\beta \geq \beta(M)$ bits per value, at *cost* $\beta|\Gamma|$. A path $t_0 \to \cdots \to t_m$ of H_M is Γ-free if $t_1, \ldots, t_m \notin \Gamma$, and Γ has *full depth at most* D if every Γ-free path has at most D edges. The cone lemma [8, Lemma 4.4 and Corollary 4.5], with written proofs there, says that output bit ℓ, appended at step t_ℓ, is a function of the revealed word $G_\Gamma(w) = (v_s(w))_{s\in\Gamma}$ and of the input bits at the positions $I_\Gamma(\ell)$ read at the ends of Γ-free paths from t_ℓ, and that $|I_\Gamma(\ell)| \leq (k+1)^{D+1}$ if Γ has full depth at most D.

The fourth paper calls *depth-TCP* the statement that genies of cost εL and bounded full depth exist for all oblivious machines with k work tapes running in time $C(L+1)$ [8, Definition 4.7]. A *side-information system* at size d is a family $I = (I(\ell))_{\ell\in[0,L)}$ of subsets of $[0,L)$, and an *index code* for I with m bits is a map $G\colon \{0,1\}^L \to \{0,1\}^m$ such that $G(w)$ and $w|_{I(\ell)}$ determine $w_{\mathrm{tpos}(\ell)}$. For a class $\mathcal{S}$ of systems and $\rho \geq 0$, $\mathrm{Crux}(\mathcal{S},\rho)$ says that for every $\eta > 0$ there is d_0 such that every index code for every system of $\mathcal{S}$ at a size $d \geq d_0$ has at least $(\rho-\eta)L$ bits; $\mathcal{S}_{k,C,\beta,D}$ is the class of the systems I_Γ of oblivious machines with k work tapes, value width at most β and halting time at most $C(L+1)$ that append exactly L bits, for genies of full depth at most D [8, Definition 4.8]. Every fibre of an index code for I is a binary storage code for the relation $N_I(p) = I(\mathrm{tpos}(p))$, in which every symbol is a function of the symbols in its recovery set [8, Remark 4.9(b)].

Theorem 2.5 (Theorem D of the fourth paper)**.** Cited: the fourth paper [8, Theorem 4.10], with a complete written proof there; not formalized.

If depth-TCP$(k,C,\beta,\varepsilon,D)$ *and* $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D},\rho)$ *hold with* $0<\varepsilon<\rho$*, then for every machine with* k *work tapes and value width at most* β *there is* d_0 *such that for no* $d \geq d_0$ *is it oblivious at length* $L = d^2 b(d)$ *with halting time at most* $C(L+1)$ *and a transposer of all* $d \times d$ *matrices.*

Why it matters. Theorem D splits a proof that no oblivious machine transposes in linear time into a structural half, depth-TCP, and a coding half, the crux. Sections 3 to 5 work on the structural half through cuts of walk DAGs, Sections 7 and 8 on the coding half, and Section 6 combines the two. An oblivious linear-time transposer must violate, for its own parameters, one of the two hypotheses.

The structural hypothesis is a statement about walk DAGs. Every application below converts a cut of the walk DAG into a genie, as in the proof of the fourth paper's Corollary B.

Lemma 2.6 (From cuts to genies)**.** Not formalized; a complete proof is given below.

Let M *have* k *work tapes and value width at most* β*, let* $C \geq 1$ *and* $0 < \varepsilon \leq 1$*, and put* $P = \lceil 4\beta C/\varepsilon \rceil$*. Suppose that at length* $L \geq 1$ *the run of* M *along a trace of halting time* $T \leq C(L+1)$ *is given, and that the walk DAG of its work-head walks on* $[0,T)$ *has a cut of at most* $\varepsilon T/(4\beta C)$ *edges and depth less than* D_1*. Then the dependency DAG of that trace has a genie of cost at most* εL *and full depth at most* $D_1 P - 1$*.*

Proof (not formalized). Let F be the cut and Γ_0 the set of steps s at which some edge $t \to s$ of F ends, so $|\Gamma_0| \leq |F|$. Every edge of a Γ_0-free path of the dependency DAG ends outside Γ_0, so its endpoints

are not those of an edge of F; read in the walk DAG, the path avoids F, and its edges of length at least 2 are long hops. Hence every Γ_0-free path has at most $D_1 - 1$ edges of length greater than 1. By the long-edge conversion of the fourth paper [8, Lemma 4.21] (a written proof there), with $\mu = 1 < P$ and $D' = D_1 - 1$, the genie $\Gamma = \Gamma_0 \cup \{t < T : t \equiv P - 1 \pmod P\}$ has full depth at most $D_1 P - 1$ and $|\Gamma| \le |F| + \lfloor T/P \rfloor$. Its cost is at most $\varepsilon T/(4C) + \beta T/P \le \varepsilon T/(2C) \le \varepsilon(L+1)/2 \le \varepsilon L$. □

The lemma uses the trace only through its dependency DAG. When the machine is oblivious and transposes, the revealed word of the genie is an index code with at most εL bits for the system $I_\Gamma \in \mathcal{S}_{k,C,\beta,D_1 P}$, by the cone lemma and correctness, exactly as in the proof of Theorem D.

2.5 The deep-nesting pair

We recall the instance that the fourth paper regards as the sharpest open case of TCP(2), with the parameter $g = 1$ throughout. Let $s_0 > m \ge 0$ be integers, $\Lambda = 2^{s_0}$ and $R_i = 2^{s_0 - i}$ for $0 \le i \le m$, so $R_0 = \Lambda$ and $R_m \ge 2$. A *level-m stroke* of direction $\sigma \in \{-1, 1\}$ is a run of R_m steps equal to σ. For $i < m$, a *level-i stroke* of direction σ is a *segment* of R_i steps equal to σ followed by a V of level $i+1$: a level-$(i+1)$ stroke of direction $-\sigma$ (the *back stroke*) followed by a level-$(i+1)$ stroke of direction σ (the *forward stroke*). The walk $U = U(s_0, m, 1)$ satisfies $U(0) = 0$, is periodic with period

$$P = 4(m+1)\Lambda,$$

and its steps over one period are those of four level-0 strokes of directions $1, -1, -1, 1$ [8, Definition 5.3]. A level-i stroke has $n_i R_i$ steps, where $n_i = m - i + 1$, and moves the head by σR_i; U has no pauses. The first time of a back stroke is the *apex* time of its V, the first time of its forward stroke the *valley* time; the V is *attached* at its *base*, the cell where it starts and ends, and its *valley cell* is at distance R_i from the base, for a V of level i. The turns of U are exactly the apex and valley times of all V's, each of the level of its V, and the first times of the second and fourth level-0 strokes of every period, of level 0, at the *tips* $\pm\Lambda$ [8, Lemma 5.4], with a written proof there. A period contains 2^{i+1} V's of level i for $1 \le i \le m$, as counted in the proof of the fourth paper's Proposition 5.13, hence 2^{i+2} turns of level i and two tips; all turn times and turn positions are multiples of R_m [8, Remark 5.23].

For an integer shift ξ the pair is $x_1 = U$ and $x_2(t) = x_1(t+\xi)$; the *deep-nesting pair* $\mathrm{DNP}(s_0, m, 1)$ of the fourth paper is the pair for a particular odd shift, the golden shift [8, Definition 5.9], and we call the pair for any odd shift a deep-nesting pair. A pair (p, u) is a hop of head 2 if and only if $(p + \xi, u + \xi)$ is a hop of head 1. For $r \ge 1$ the *r-fold cover* is the pair on the cyclic time axis $\mathbb{Z}_T$, $T = rP$; its hops are the residues of the hops on $\mathbb{Z}$, every time is the earlier and the later endpoint of exactly one hop of each head, and there are $n_\mathrm{h} = 2T$ hops in all. Each head visits every cell of $[-\Lambda, \Lambda]$ in every period, so every time lies in the half-open interval $(p, u]$ of exactly $2(2\Lambda + 1)$ hops of the two heads, one per head and cell. A *label* on the cover is a map $G \colon \mathbb{Z}_T \to \{0, \dots, q-1\}$; a hop (p, u) is violated if $G(u) \le G(p)$, a time t is a descent if $G(t) < G(t-1)$, and $\mathrm{cost}(G)$ is the number of violated hops of both heads (counted per head) plus the number $\mathrm{desc}(G)$ of descents. The *error* of a hop is $e(p, u) = (G(u) - G(p) - 1) \bmod q \in \{0, \dots, q-1\}$.

Lemma 2.7 (Lemma Zq of the fourth paper)**.** Cited: the fourth paper [8, Lemma 5.15]; Lean-verified there (`MotionDepthCodes.lemma_Zq`, `MotionDepthCodes.card_descents_eq`).

On the r-fold cover, every label satisfies $q \cdot \mathrm{cost}(G) = n_\mathrm{h} + \sum_{hops} e + q \cdot \mathrm{desc}(G)$ *and* $q \cdot \mathrm{desc}(G) = \sum_{t \in \mathbb{Z}_T} \chi_t$*, where* $\chi_t = (G(t) - G(t-1)) \bmod q$.

The *level* of a hop is the least level of a turn of its own head strictly inside it (inside any lift to $\mathbb{Z}$); the owner is such a turn, so levels are defined. The hops of head h of level i form the *class*

(h, i). A label *θ-serves* level i for head h if the class (h, i) has total error at most $\theta q n_{\mathrm{h}}/(2(m+1))$, and *doubly θ-serves* it if it θ-serves it for both heads [8, Section 5.7]. By Lemma 5.1 below, every class has exactly $n_{\mathrm{h}}/(2(m+1))$ hops, so this says that the mean error of the class is at most θq.

Conjecture 2.8 (Conjecture A′ of the fourth paper)**.** *There are $\theta_0 > 0$ and constants C, C', and for all s_0 and m an odd shift ξ, such that for all r and q every label G with $\mathrm{cost}(G) \le n_{\mathrm{h}}/8$ on the r-fold cover of the pair $x_1 = U(s_0, m, 1)$, $x_2 = x_1(\cdot + \xi)$, has at most $C \log_2 q + C'$ doubly θ_0-served levels [8, Conjecture 5.25].*

By the per-class split of Lemma Zq [8, Corollary 5.16], $\mathrm{cost}(G)$ is at least the total error of all hops divided by q, so a level that is not doubly θ_0-served contributes more than $\theta_0 n_{\mathrm{h}}/(2(m+1))$ to $\mathrm{cost}(G)$; hence a label of cost at most $\varepsilon n_{\mathrm{h}}$ with $\varepsilon \le 1/8$ and $\varepsilon < \theta_0/2$ doubly serves more than $(m+1)(1 - 2\varepsilon/\theta_0)$ levels. Under Conjecture A′ this forces $q \ge \exp(c(1 - 2\varepsilon/\theta_0)(m+1) - c')$ for constants $c, c' > 0$, and the same statement on the walk DAG of many periods would make the pairs depth-robust as $m \to \infty$, so that TCP(2) fails [8, Section 5.7]. The fourth paper's clock labels cut every pair at depth $\lceil 12/\varepsilon \rceil^{m+1} - 1$ with εT edges [8, Proposition 5.13].

2.6 Formal status of the results

Every numbered result names its formal status in one of four ways. A line

`Lean: declaration, ...` [`file`]

gives the Lean declarations of a fully formalized result. The declarations live in the namespace `BoundedNesting` and the files in the directory `lean/BoundedNesting/`. The text after such a statement, labelled *Outline of the formal proof*, is a guide to the formal proof; it is a convenience, not a proof. A line beginning Lean (partial) marks a result whose formal counterpart covers all but an explicitly named part, which the text proves; on a definition, it names the notions that have no Lean counterpart. The words Not formalized mark a result with no formal counterpart; it comes with a complete proof, labelled *Proof (not formalized)*. A line beginning Cited marks a result of an earlier paper of the series that is used without proof, and says where it is proved and whether it is Lean-verified there. Where the English statement simplifies, the Lean statement is authoritative. Results from the literature are cited. Numerical observations appear only in remarks labelled as numerical, heuristics are labelled as such, and claims that have only a proof sketch are stated in remarks beginning *Proof sketch only*; none of them is used in proofs. Numbered definitions carry a status line only when they have a Lean counterpart.

The library depends on Lean 4 and Mathlib (both v4.34.1) and on the Lean library of the fourth paper, which imports those of the first two papers and the repository's shared library `lean-machine`, all through relative paths. It contains no `sorry`, no `native_decide` and no unsafe evaluation, and every declaration cited here depends only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B). Appendix A lists every declaration of this library cited in the paper.

3 Bounded nesting: the owner lemma and Theorem F

The fourth paper's Theorem B cuts the walk DAG of any coupling of home-based walks with a lexicographic label: one digit per occupied group of scales, each counting legs measured from a fixed home [8, Theorem 4.20]. Its hypothesis, linear leg area about fixed homes, fails when heads turn often, far from every home, on legs much shorter than their distance from home. This section replaces homes by owners. Lemma 3.1 shows that nested structures with distinct turning points

force revisits, and Theorem F (Theorem 3.6) proves depth reduction for every family of walks of bounded *owner-load* (Definition 2.2), whatever their homes, drift and scales. Its label is Theorem B's band label run on *virtual heads*, one per head, side and nesting depth, with the band of a virtual head given by the height of its current family of hops instead of a distance from home. Section 3.4 derives the consequences at walk level: TCP for walks of bounded entry, and the confinement of depth-robustness to a set of cells of vanishing capacity.

3.1 The owner lemma

Lemma 3.1 (Nesting with distinct owners costs visits)**.** Lean: `owner_entries`, `card_crossings_le` [`Owner.lean`]

Let x be a walk, $\ell \geq 1$, and let (p_i, u_i), $0 \leq i < \ell$, be upper long hops of x that are strictly nested, $p_i < p_{i+1}$ and $u_{i+1} < u_i$, with pairwise distinct owners. Then for every time $T' \geq u_0$, the cell of the innermost hop $(p_{\ell-1}, u_{\ell-1})$ has at least $(\ell+1)/2$ entries at times at most T'. By the reflection $x \mapsto -x$, the same holds for lower long hops.

Outline of the formal proof. The Lean statement concerns upper hops; the lower case follows by the reflection $x \mapsto -x$, which maps lower long hops to upper ones with the same owners and does not change entries. In Lean a walk is a map $x\colon \mathbb{N} \to \mathbb{Z}$ with $|x(t+1) - x(t)| \leq 1$ (`IsWalk`); a walk of length T is extended by staying at $x(T-1)$, which adds no entries before time T. Upper long hops and owners are the structures `IsUpperHop` and `IsOwner`, and entries and crossings are `entries` and `crossings`. Let c_i be the cell of (p_i, u_i) and $c = c_{\ell-1}$. Since $p_{i+1} \in (p_i, u_i)$, the cells increase inwards, $c_0 < \cdots < c_{\ell-1} = c$. Every owner o_i is a maximum of x over $(p_i, u_i) \supseteq (p_{\ell-1}, u_{\ell-1})$, where $x \geq c+1$, so $x(o_i) \geq c+1$. For $i < \ell - 1$, o_i lies outside $[p_{i+1}, u_{i+1}]$: at the endpoints $x = c_{i+1} \leq c$, and if o_i lay strictly inside (p_{i+1}, u_{i+1}) it would also be the first maximum over that interval, that is o_{i+1} (`IsOwner.unique`), contrary to distinctness. So the walk crosses the edge between c and $c+1$ at some step in (p_i, p_{i+1}), between o_i and p_{i+1}, or in $[u_{i+1}, u_i)$, between u_{i+1} and o_i. The innermost hop crosses it at the steps $p_{\ell-1}$ and $u_{\ell-1} - 1$. These $\ell + 1$ crossings occur at distinct steps, because the intervals $(p_i, p_{i+1}) \cup [u_{i+1}, u_i)$ are pairwise disjoint and disjoint from $[p_{\ell-1}, u_{\ell-1})$. Finally, every crossing of the edge either enters c or leaves it, each maximal run of visits to c is entered once and left at most once, and so the number of crossings is at most twice the number of entries (`card_crossings_le`). □

Corollary 3.2 (Bounded entry bounds the owner-load)**.** Not formalized; a complete proof is given below.

Let $x_1, \ldots, x_k$ be walks of length T and H' a set of their long hops such that, for every hop of H' of head h at a cell c, the walk x_h enters c at most V times. Then H' has owner-load at most $2V - 1$. In particular, if every walk enters every cell at most V times, the set of all long hops has owner-load at most $2V - 1$.

Proof (not formalized). Fix a time t, a head h and a side, say upper. The hops of H' of head h on this side whose open intervals contain t form a chain (Section 2.2). Choosing one hop for each of their ℓ distinct owners gives a strictly nested chain of upper long hops with distinct owners, and by Lemma 3.1 with $T' = T - 1$ its innermost cell has at least $(\ell+1)/2$ entries. This cell is the cell of a hop of H' of head h, so $(\ell+1)/2 \leq V$. Lower hops are symmetric. □

3.2 Families and virtual heads

Definition 3.3 (Families)**.** Lean: `side`, `sw`, `fam`, `famP`, `famU`, `famd`, `famDepth`, `vcode`, `famData` [`Families.lean`]

Let H' be a set of long hops of walks $x_1, \dots, x_k$ of length T. For a head h, a side σ and a time o, the *family* of (h, σ, o) is the set of hops of H' of head h on side σ with owner o, when it is nonempty; every hop of H' lies in exactly one family. All hops of a family F contain its owner o_F, so they form a chain. The *span* of F is its outermost hop (P_F, U_F), at the cell c_F; its *height* is $d_F = |x_h(o_F) - c_F| \geq 1$; its *band* is $\lambda(d_F)$, where $\lambda(d) = \lfloor \log_2 d \rfloor$, so $2^{\lambda(d)} \leq d < 2^{\lambda(d)+1}$; and its *depth* is the number of families of (h, σ) whose span contains the span of F as an interval, F included. A *virtual head* is a triple $v = (h, \sigma, i)$ such that some family of (h, σ) has depth i; its families are those of (h, σ) of depth i. We write N for the number of virtual heads.

Lemma 3.4 (Structure of families)**.** Lean: `fam_cell_mem`, `fam_cell_injOn`, `card_fam_le`, `fam_long`, `famP_ne`, `fam_no_cross`, `famDepth_le`, `vcode_lt`, `vh_eq`, `famData_disj`, `famData_valid` [`Families.lean`]

Lean: `Fam.Valid.unique_span`, `Fam.Valid.sum_d_le`, `Fam.Valid.overlap` [`Clock.lean`]

Let H' have owner-load at most ℓ.

(a) The cells of the hops of a family F are distinct and lie between c_F, included, and $x_h(o_F)$, excluded, so $|F| \leq d_F$; moreover $U_F - P_F \geq 2d_F$.

(b) Distinct families of (h, σ) have distinct spans, and their spans do not cross.

(c) Every family has depth at most ℓ. Hence $N \leq 2k\ell$.

(d) Two distinct families F, F' of one virtual head satisfy $U_F \leq P_{F'}$ or $U_{F'} \leq P_F$.

(e) For every virtual head v and time τ, at most one family F of v has $\tau \in (P_F, U_F]$, and $\sum_{F \text{ of } v} d_F \leq (T-1)/2$.

(f) (Overlap.) *For every time τ and integer $\theta \geq 0$, at most $N(2^\theta - 1)$ hops $(p, u) \in H'$ with $p < \tau \leq u$ lie in families of band less than θ.*

Outline of the formal proof. In Lean the walks are maps $x_h \colon \mathbb{N} \to \mathbb{Z}$ (`IsWalk`), extended beyond T as in Lemma 3.1; the hops of H' are triples (h, p, u) such that (p, u) is a long hop of x_h before T (`IsHop`); and the owners may be any assignment of times strictly inside the hops, the owners of Section 2.2 being one (`hopOwner_spec`). A hop is upper if the walk steps up after its start (`side`); seen from its side, that is, after the reflection $x \mapsto -x$ for lower hops (`sw`), it lies strictly above its cell (`sw_above`), so all hops of a family are upper hops of one walk. Families, spans, heights and depths are `fam`, `famP`, `famU`, `famd` and `famDepth`; the span is the hop of the family that starts first, and it also ends last (`fam_nested`, `fam_span`). With the heads numbered $0, \dots, k-1$ and $\sigma = 1$ for the upper side and 0 for the lower one, a virtual head (h, σ, i) is coded by the number $(i-1) + (2h + \sigma)\ell < 2k\ell$ (`vcode`, `vcode_lt`, `vh_eq`). The formal family clock runs on all $2k\ell$ codes, the unused ones being virtual heads without families, of band 0 at all times; so the formal N is its bound $2k\ell$. The data of all families form an instance `famData` of abstract family data (`Fam`), and `famData_valid` proves the hypotheses `Fam.Valid` of the family clock: every hop lies in the span of its family with the owner strictly inside, a hop that ends at the end of its span is the span, (d), (a), and spans end before T. Parts (e) and (f) are proved from these hypotheses alone (`Fam.Valid.unique_span`, `Fam.Valid.sum_d_le`, `Fam.Valid.overlap`).

(a) (`fam_cell_mem`, `fam_cell_injOn`, `card_fam_le`, `fam_long`.) Let F be upper; the lower case is the same seen from its side. If $(p, u) \supsetneq (p', u')$ are hops of F, then $p' \in (p, u)$, so $x_h(p') > x_h(p)$: the cells increase strictly inwards. The innermost hop contains o_F, where the walk is above its cell, so all cells lie in $[c_F, x_h(o_F) - 1]$, and $|F| \leq d_F$. The walk moves by at most one cell per step and goes from c_F at P_F to $x_h(o_F)$ at o_F and back to c_F at U_F, so $o_F - P_F \geq d_F$ and $U_F - o_F \geq d_F$.

(b) (`famP_ne`, `fam_no_cross`.) The span of F is a hop of F, with owner o_F; equal spans give equal owners and hence equal families. Since a hop is determined by its start, distinct spans even start at distinct times. Spans are long hops of one head on one side, which do not cross.

(c) (`famDepth_le`.) Let $t \in (P_F, U_F)$, a nonempty interval by (a). Every family F' whose span contains the span of F has t in the open interval of its span, a hop of H' of head h on side σ with owner $o_{F'}$, and distinct such families have distinct owners. So there are at most ℓ of them. A virtual head is then a triple (h, σ, i) with $i \le \ell$.

(d) (`famData_disj`, `disj_aux`.) By (b) we may assume $P_F < P_{F'}$. Suppose that $P_{F'} < U_F$. Since the spans do not cross, $U_{F'} \le U_F$, and $U_{F'} \ne U_F$, since a hop is also determined by its end; so $(P_{F'}, U_{F'}) \subsetneq (P_F, U_F)$. Every family whose span contains the span of F also contains the span of F', and F' contains its own span but not that of F; so F' would have larger depth than F (`famDepth_lt`). Hence $U_F \le P_{F'}$.

(e) (`Fam.Valid.unique_span`, `Fam.Valid.sum_d_le`.) The first claim is (d). The half-open spans $(P_F, U_F]$ of the families of v are therefore disjoint subsets of $\{1, \dots, T-1\}$, and by (a) each has at least $2d_F$ elements.

(f) (`Fam.Valid.overlap`.) A hop (p, u) of a family F of a virtual head v with $p < \tau \le u$ satisfies $P_F \le p < \tau \le u \le U_F$. By (e) at most one family of v qualifies, and if its band is less than θ, then by (a) it has at most $d_F < 2^{\lambda(d_F)+1} \le 2^\theta$ hops. Summing over the N virtual heads gives the bound. □

Definition 3.5 (The family clock)**.** Lean: `lam`, `grp`, `flr`, `Fam.band`, `Fam.group`, `Fam.occ`, `Fam.rk`, `Fam.cnt`, `Fam.digit`, `Fam.clock` [`Clock.lean`]

Let H' have owner-load at most ℓ and $N \ge 1$ virtual heads, numbered $1, \dots, N$.

- The *band* of a virtual head v at time t is $b_v(t) = \lambda(d_F)$ for the family F of v with the largest $P_F \le t$, and $b_v(t) = 0$ if no family of v starts at or before t. It changes only at the start times $P_F \ge 1$ of the families of v, the *jumps* of v.
- The *events* are the owners o_F of all families of all virtual heads, one per family, with *scale* $\lambda(d_F)$.
- Fix integers $a \ge 1$ and $K \ge 2$, an *offset* $r \in \{0, \dots, a-1\}$ and $\omega \in \{0, \dots, K-1\}^N$. The *group* of v at t is $G_v(t) = \lfloor (b_v(t) + r)/a \rfloor$, and the *floor* of a group $G \ge 0$ is $\theta(G) = \max(aG - r, 0)$; a band b lies in group G exactly when $\theta(G) \le b < \theta(G+1)$, and the floors at least 1 are the integers congruent to $-r$ modulo a.
- The *occupied groups* at t are the distinct values among $G_1(t), \dots, G_N(t)$, listed from the top as $G_1(t) > \cdots > G_{m(t)}(t)$, with floors $\theta_i(t) = \theta(G_i(t))$; here $m(t) \le N$.
- The *counts* are $n_i(t) = \#\{\text{events at times at most } t \text{ of scale at least } \theta_i(t)\}$.
- The *family clock* is the label

$$\Psi(t) = \sum_{i=1}^{m(t)} \left((n_i(t) + \omega_i) \bmod K \right) K^{N-i} \ \in \ [0, K^N).$$

So $\Psi(t)$ is the integer whose base-K digits, most significant first, are $\psi_i(t) = (n_i(t) + \omega_i) \bmod K$ for $i \le m(t)$ and 0 for $m(t) < i \le N$, and labels compare as these digit strings compare lexicographically. Two bands $b < b'$ lie in different groups exactly when some floor $\theta \ge 1$ satisfies $b < \theta \le b'$; for a uniformly random offset each integer $\theta \ge 1$ is a floor with probability $1/a$. This is the band label of the fourth paper's Theorem B [8, Definition 4.19], with virtual heads in place of heads, family bands in place of distances from home, and owners in place of leg starts.

3.3 Theorem F

Theorem 3.6 (Theorem F: depth reduction at bounded owner-load)**.** Lean: `theoremF_owner`, `theoremF`, `DF` [`TheoremF.lean`]

Let $k, \ell \geq 1$, $\varepsilon \in (0,1]$ and $T \geq 1$, let $x_1, \dots, x_k$ be walks of length T, coupled arbitrarily, and let H' be a set of their long hops of owner-load at most ℓ. Then the DAG $G(H')$ has a cut of at most εT edges and depth less than

$$D_{\mathrm{F}}(k,\ell,\varepsilon) = 2^{\,2k\ell\left(\lceil 64k^2\ell^2/\varepsilon\rceil + 5 + \lceil \log_2(k^2\ell^2/\varepsilon)\rceil\right)} = 2^{O(k^3\ell^3/\varepsilon)}.$$

The bound does not depend on T, on the coupling, on homes or on the scales of the walks, and pauses are allowed.

Outline of the formal proof. The theorem for the owners of Section 2.2 (`hopOwner`) is `theoremF_owner`. It is an instance of `theoremF`, which holds for every assignment of owners strictly inside the hops and for every $\varepsilon > 0$. Owner-load is `OwnerLoadLE`, and the hop edges of $G(H')$ are the pairs of `hopPairs`. The cut consists of the state edges $t \to t-1$ at the times of a subset of $[1,T)$ and of a set of hop pairs, and the depth bound concerns every path $f(0) \to \cdots \to f(n)$ that avoids the cut, as in `strip_cut_depth`. The family clock is defined for abstract family data (`Fam.clock`) and applied to `famData`, whose hypotheses `Fam.Valid` hold by Lemma 3.4, with $N = 2k\ell$ as explained there; digit positions and the coordinates of ω are numbered from 0, so that position $i-1$ carries the digit ψ_i. The expectation is formalized as a count: summed over the a offsets and the K^N choices of ω, the violated hops and the descents number at most $K^N \cdot 4N^2T + aK^{N-1}2^{a+1}N^2T$ (`Fam.Valid.cost_sum_le`), which is aK^N times the right side of (1), and some (r,ω) does at most as well as the average (`Fam.Valid.exists_cheap`). The bounds below are lemmas of that count.

Let r and ω be uniformly random and independent, and let F_Ψ be the cut of the family clock in $G(H')$ (violated hop edges and state edges at descents). By Lemma 2.3 it has depth at most $K^N - 1$; in Lean, for a label with N digits in $[0,K)$ compared lexicographically (`Fam.depth_le_of_lex`, `Fam.digit_lt`). We show

$$\mathbb{E}|F_\Psi| \;\leq\; \frac{4N^2T}{a} + \frac{2^{a+1}N^2T}{K}. \tag{1}$$

We bound the number of violated hops of H', counted per hop, which is at least the number of violated hop edges, and the number of descents. Throughout, $\eta = (p,u)$ is a hop of H' of the virtual head w, in the family F, and $G_\eta = \lfloor(\lambda(d_F) + r)/a\rfloor$ is the group of its band.

Consistency (`Fam.Valid.band_of_mem`, `Fam.Valid.band_at_end`). We have $G_w(p) = G_\eta$, and $G_w(u) = G_\eta$ unless η is the span of F and the next family F' of w starts at $P_{F'} = U_F$. Indeed, $P_F \leq p < u \leq U_F$, and by Lemma 3.4(d) no family of w starts in (P_F, U_F), so $b_w = \lambda(d_F)$ on $[P_F, U_F)$; the hops of F other than its span have $u < U_F$, by strict nesting. Call η *inconsistent* if $G_w(u) \neq G_\eta$.

The jumps that change a group (`Fam.gjumps`, `Fam.Valid.sum_gjumps`). A jump of v from band b to band b' changes the group of v with probability at most $|b-b'|/a \leq (b+b')/a$ (`card_grp_ne_le`). At the jump that starts a family F'', the new band is $\lambda(d_{F''})$ and the old band is that of the previous family of v, or the initial band 0; each family occurs at most once as a new band and at most once as an old band (`Fam.Valid.sum_old_le`). So, since $\lambda(d) < d$, $\sum_{\text{jumps of } v}(b+b') \leq 2\sum_F \lambda(d_F) \leq T$ (`Fam.Valid.sum_jumps_lam`), and the jumps that change a group number at most NT/a in expectation.

(α) Inconsistent hops (`Fam.incons`, `Fam.Valid.card_incons_le`). We count them all as violated. By consistency, an inconsistent hop is the span of its family F, and the next family F' of w

starts at U_F with a jump that changes the group of w; distinct inconsistent hops give distinct such jumps. So their expected number is at most NT/a.

(β) Consistent hops whose higher groups differ (`Fam.higher`, `Fam.Valid.exists_charge`, `Fam.Valid.card_higher_le`, `Fam.Valid.sum_higher_le`). Let $\mathcal{U}(t)$ be the set of occupied groups above G_η at time t, and suppose η is consistent and $\mathcal{U}(p) \neq \mathcal{U}(u)$; we count it as violated. Some group $G' > G_\eta$ is occupied at exactly one of the times p, u, by a virtual head v that is not in G' at the other time; $v \neq w$, since w is in G_η at both times. So $G_v(p) \neq G_v(u)$ and $\max(G_v(p), G_v(u)) > G_\eta$. The group of v changes only at its jumps, and so some jump $\tau \in (p, u]$ of v moves it between two groups whose larger one, G°, exceeds G_η (`exists_change`: going back from u, the first step at which the group of v changes between two groups the larger of which exceeds G_η). Let the jump move the band of v from b to b'. The floor $\theta^\circ = \theta(G^\circ)$ satisfies $\min(b, b') < \theta^\circ \leq \max(b, b')$, and since $G^\circ \geq G_\eta + 1$, $\lambda(d_F) < \theta(G_\eta + 1) \leq \theta^\circ$. So η spans τ and its family has band less than θ°. Charge η to the jump (v, τ). By the overlap bound of Lemma 3.4, at most $N(2^{\theta^\circ} - 1)$ hops are charged to one jump. Given r, θ° is the largest floor in $(\min(b, b'), \max(b, b')]$, and a jump that changes no group receives no charge. Over the a offsets at which the jump changes a group, these floors are distinct integers in $(\min(b, b'), \max(b, b')]$, since the floors at least 1 for the offset r are the integers congruent to $-r$ modulo a (`sum_floor_pow_le`); so the expected charge of a jump is at most $(N/a)2^{M+1}$, where $M = \max(b, b')$ and $2^M \leq 2^b + 2^{b'}$. As above, each family occurs at most once as a new band and at most once as an old band, so $\sum_{\text{jumps of } v} 2^M \leq 1 + 2\sum_F 2^{\lambda(d_F)} \leq 1 + 2\sum_F d_F \leq T$ (`Fam.Valid.sum_jumps_pow`). The expected number of such hops is therefore at most $(2N/a)\sum_v T = 2N^2T/a$.

(γ) The remaining hops (`Fam.calm`, `Fam.Valid.card_viol_calm_le`, `Fam.Valid.sum_calm_growth_le`, `Fam.Valid.sum_viol_calm_le`). Now η is consistent and $\mathcal{U}(p) = \mathcal{U}(u)$. Then G_η has the same position $j = |\mathcal{U}(p)| + 1$ in the lists of occupied groups at p and at u, and the groups above it, hence their floors θ_i, $i \leq j$, agree. Put $\Delta_i = n_i(u) - n_i(p)$ for $i \leq j$; then $\Delta_i \geq 0$, because the floor is the same at both times. Moreover $\Delta_j \geq 1$: the owner $o_F \in (p, u)$ is an event of scale $\lambda(d_F) \geq \theta(G_\eta) = \theta_j$. Say that digit i *wraps* if $\lfloor (n_i(p) + \omega_i)/K \rfloor \neq \lfloor (n_i(u) + \omega_i)/K \rfloor$. If no digit $i \leq j$ wraps, then $\psi_i(u) = \psi_i(p) + \Delta_i$ for $i \leq j$, the first position $i \leq j$ with $\Delta_i > 0$ carries a larger digit at u, all earlier digits agree, and $\Psi(u) > \Psi(p)$ (`Fam.clock_lt_of_no_wrap`). Given r, digit i wraps with probability at most Δ_i/K (`Fam.card_wrap_le`, `Fam.card_wrap_union`), so η is violated with probability at most $\sum_{i \leq j} \Delta_i/K$.

We bound $\sum_\eta \sum_{i \leq j(\eta)} \Delta_i(\eta)$ over these hops, for every r, by exchanging the order of summation. $\Delta_i(\eta)$ is the number of events $o' \in (p, u]$ of scale at least θ_i. Fix an event o' of scale s. A pair (η, i) counts o' only if η spans o' and $\theta(G_\eta) \leq \theta_i \leq s$; the floors θ_i, $i \leq j(\eta)$, are distinct, since θ is strictly increasing on groups. If $\theta(G_\eta) \leq \varphi$, then $\lambda(d_{F(\eta)}) < \theta(G_\eta + 1) \leq \theta(G_\eta) + a \leq \varphi + a$. So, by the overlap bound with $\theta = \varphi + a$, the pairs counting o' number at most

$$\sum_\varphi N(2^{\varphi + a} - 1) \;<\; N2^{a+1}2^s,$$

the sum over the floors $\varphi \leq s$, which are distinct integers in $[0, s]$. Each family is the family of one event, and $\sum_{o'} 2^{s(o')} = \sum_v \sum_F 2^{\lambda(d_F)} \leq \sum_v \sum_F d_F \leq NT/2$. Hence these hops contribute at most $N2^{a+1} \cdot NT/(2K) = 2^a N^2 T/K$ in expectation.

Descents (`Fam.still`, `Fam.card_desc_still_le`, `Fam.sum_desc_still_le`, `Fam.gchange`, `Fam.Valid.card_gchange_le`). Consider the state edge $t \to t - 1$. If no virtual head changes group at step $t - 1 \to t$, the occupied groups and their floors are the same at $t - 1$ and t, and $n_i(t) \geq n_i(t - 1)$ for every i; as in (γ), a descent needs a wrap (`Fam.clock_le_of_no_wrap`), which given r has probability at most $\sum_i (n_i(t) - n_i(t - 1))/K$. Summed over t this is at most N times the number of events divided by K, and there are at most $\sum_v \sum_F d_F \leq NT/2$ events; so these descents number

at most $N^2T/(2K)$ in expectation. Otherwise some virtual head has a jump at t that changes its group, and we count the edge as cut; these edges number at most NT/a in expectation, by the bound on the jumps that change a group.

Total (`Fam.violated_subset`, `Fam.descents_subset`, `Fam.Valid.cost_sum_le`). Adding the four bounds, $\mathbb{E}|F_\Psi| \le (2NT + 2N^2T)/a + (2^a + 1/2)N^2T/K$, which gives (1). (The formal count bounds the two terms with K together: by $N2^{a+1}\sum_F d_F/K$ and by N/K per family, at most $2^{a+1}N^2T/K$ in all.)

Parameters (`theoremF`, `cost_le_eps`, `ceil_le_two_pow_clog`). By Lemma 3.4(c), $N \le 2k\ell$, so the right side of (1) is at most $16k^2\ell^2T/a + 2^{a+3}k^2\ell^2T/K$. Take $a = \lceil 64k^2\ell^2/\varepsilon \rceil$ and $K = 2^{a+4}\lceil k^2\ell^2/\varepsilon \rceil$. Then the first term is at most $\varepsilon T/4$ and the second at most $\varepsilon T/2$, and some choice of (r, ω) gives a cut of at most εT edges. Its depth is at most $K^N - 1 < K^{2k\ell}$, and since $\lceil x \rceil \le 2^{\lceil \log_2 x \rceil}$ for $x > 0$,

$$K^{2k\ell} \le 2^{2k\ell\left(\lceil 64k^2\ell^2/\varepsilon \rceil + 4 + \lceil \log_2(k^2\ell^2/\varepsilon) \rceil\right)} < D_{\mathrm{F}}(k, \ell, \varepsilon).$$

In `DF` the ceiling of the logarithm is taken in $\mathbb{N}$, which agrees with the integer ceiling for $\varepsilon \le 1$, since $k^2\ell^2/\varepsilon \ge 1$. Pauses are state edges, and the argument uses only that hops are long and that owners lie strictly inside their hops. □

Why it matters. Theorem F proves the walk form of the structural hypothesis of Theorem D for every family of bounded owner-load, for every coupling, and with no reference to positions: neither homes nor leg area enter. Together with the owner lemma it shows that a depth-robust family of walks, if one exists, must contain heads with unboundedly many simultaneously open nested structures with distinct turning points, hence, by Lemma 3.1, cells entered unboundedly often. This is the structure of the deep-nesting pair (Section 2.5), which Theorem F does not cut at a depth independent of its nesting depth. Theorem F is consistent with the fourth paper's nesting barrier for band rules [8, Remarks 5.6 and 5.7], which concerns rules whose load does not grow with the nesting depth; the load of the family clock is $N \le 2k\ell$.

3.4 Bounded entry and the capacity of depth

Corollary 3.7 (Bounded entry implies TCP)**.** Not formalized; a complete proof is given below.

Let $V \ge 1$. TCP holds for every family of k-tuples of walks in which every walk enters every cell at most V times, in particular for walks of multiplicity at most V, with $D(\varepsilon) = D_{\mathrm{F}}(k, 2V - 1, \varepsilon)$ for $\varepsilon \in (0, 1]$.

Proof (not formalized). By Corollary 3.2 the set of all long hops has owner-load at most $2V - 1$, and $G(H')$ is the walk DAG; apply Theorem 3.6. □

The corollary is incomparable with Theorem B: home-based sweeps $0 \to n \to 0$, repeated, enter cells near home unboundedly often, while walks made of V's of varying sizes that drift away from every home enter every cell at most three times but have superlinear leg area about every cell.

Corollary 3.8 (The capacity of depth)**.** Not formalized; a complete proof is given below.

Let $x_1, \dots, x_k$ be walks of length T, let $\varepsilon \in (0, 1]$ and $V = \lceil 1/\varepsilon \rceil$, and let R be the set of pairs (h, c) such that x_h enters c more than V times. Then $|R| \le kT/(V + 1) < k\varepsilon T$; at every time τ at most $|R|$ hops at pairs of R span τ; and $G - R$ has a cut of at most εT edges and depth less than $D_{\mathrm{F}}(k, 2\lceil 1/\varepsilon \rceil - 1, \varepsilon) = 2^{O(k^3/\varepsilon^4)}$.

Proof (not formalized). Each walk has at most T entries in all, which bounds $|R|$. By (P1), at most one hop at each pair spans a given time. The DAG $G - R$ is $G(H')$ for the set H' of long hops at pairs entered at most V times, which by Corollary 3.2 has owner-load at most $2V - 1 \le 3/\varepsilon$; apply Theorem 3.6. □

Why it matters. Every schedule of linear length splits, with no property of transposition used, into a *shallow* part, the hops at boundedly entered cells, which a cut of εT edges reduces to bounded depth, and a *scratch* R of fewer than $k\varepsilon T$ cells, which carries fewer than $k\varepsilon T$ symbols across every time cut. Paths that avoid the cut may still alternate between at most D_{F} shallow hops and hops through the scratch. So depth-robustness of a linear-time schedule can only live in a set of cells of vanishing capacity that its heads visit over and over. On the deep-nesting family the trade-off between $|R|/T$ and the depth is of order $1/\log D$, and no better under the walk-DAG form of Conjecture A′ (Proposition 5.8). The next proposition shows that the alternation cannot be forbidden at no cost: for walks without pauses, the statement $\mathrm{RF}(k+1)$ that forbids it implies $\mathrm{TCP}(k)$ in genie form, while $\mathrm{TCP}(k+1)$ in genie form implies $\mathrm{RF}(k+1)$ with $R = \emptyset$.

Proposition 3.9 (Relays must be allowed)**.** Not formalized; a complete proof is given below.

Restrict to walks without pauses, and let $\mathrm{RF}(k)$ *be the statement: for every* $\varepsilon > 0$ *there are* $D(\varepsilon)$ *and* $\delta(\varepsilon)$*, with* $\delta(\varepsilon) \to 0$ *as* $\varepsilon \to 0$*, such that for all* k *walks of length* T *there are a set* R *of pairs with* $|R| \le \delta(\varepsilon)T$ *and a set* $\Gamma \subseteq [0, T)$ *with* $|\Gamma| \le \varepsilon T$ *such that, for every path* $t_0 \to \cdots \to t_n$ *of the walk DAG with* $t_1, \dots, t_n \notin \Gamma$ *and every* i *with* $D(\varepsilon) < i \le n$*, if* $t_{i-1} \to t_i$ *is a hop edge then every hop that it carries lies at a pair of* R*. Then* $\mathrm{RF}(k+1)$ *implies* $\mathrm{TCP}(k)$ *in genie form: for every* $\varepsilon' > 0$ *there is* D' *such that every* k *walks of length* T *have a set* $\Gamma' \subseteq [0, T)$ *with* $|\Gamma'| \le \varepsilon' T$ *such that every path whose vertices after the first avoid* Γ' *has at most* D' *edges.*

Proof (not formalized). For $\varepsilon' \ge 1$ take $\Gamma' = [0, T)$. Let $0 < \varepsilon' < 1$, let M be the least odd integer with $M \ge 4/\varepsilon'$, and choose $\varepsilon \le \varepsilon'/8$ with $(M-1)\delta(\varepsilon) \le \varepsilon'/8$. Let $x_1, \dots, x_k$ be walks of length $T \ge 16/\varepsilon'$. Define an extra walk z on $[0, T)$ blockwise: for $t \in [jM, (j+1)M)$, $z(t) = j + ((t - jM) \bmod 2)$. Inside block j it alternates between j and $j+1$, ending at j because M is odd, and the next block starts at $j+1$; so z has no pauses. Cell j is visited at most M times, z has at most $T/M + 2$ first visits, and every time that is not a first visit of z is the later endpoint of a long hop of z.

Apply $\mathrm{RF}(k+1)$ to $(x_1, \dots, x_k, z)$ at ε, obtaining R and Γ. Let Δ be the set of times s at which some path of length greater than $D(\varepsilon)$, with all vertices after the first outside Γ, ends. For such a path ending at s, if s were not a first visit of z, the pair $(z, z(s))$ were not in R and $\mathrm{prev}_z(s)$ were not in Γ, then appending the hop edge $s \to \mathrm{prev}_z(s)$ would give a path that violates RF at its last edge. So every $s \in \Delta$ is a first visit of z (at most $T/M + 2$ times), or the later endpoint of a hop of z at a pair of R (at most $(M-1)|R| \le (M-1)\delta(\varepsilon)T$ times, since each pair of z carries at most $M - 1$ hops), or has $\mathrm{prev}_z(s) \in \Gamma$ (at most $|\Gamma| \le \varepsilon T$ times, since prev_z is injective).

Put $\Gamma' = \Gamma \cup \Delta$. A path of length greater than $D(\varepsilon)$ whose vertices after the first avoid Γ' would avoid Γ, so its last vertex would lie in $\Delta \subseteq \Gamma'$, a contradiction. Hence such paths have at most $D(\varepsilon)$ edges in the walk DAG of $(x_1, \dots, x_k, z)$, and therefore in that of $(x_1, \dots, x_k)$, which has the same vertices and fewer edges. Finally $|\Gamma'| \le 2\varepsilon T + T/M + 2 + (M-1)\delta(\varepsilon)T \le (\varepsilon'/4 + \varepsilon'/4 + \varepsilon'/8)T + 2 \le \varepsilon' T$, using $T \ge 16/\varepsilon'$. For $T < 16/\varepsilon'$ every path has fewer than $16/\varepsilon'$ edges, so $D' = \max(D(\varepsilon), \lceil 16/\varepsilon' \rceil)$ works. □

The conversion is the reason the scratch of Corollary 3.8 must be allowed to relay: a busy extra head of bounded multiplicity and space about T/M turns any relay-free certificate into a genuine genie for the other heads.

Remark 3.10 (Depth-robust schedules carry capacity)**.** Proof sketch only. Alwen, Blocki and Pietrzak showed that a DAG that keeps a path with d vertices after the deletion of any e vertices has cumulative pebbling cost greater than about ed [2]; graphs of this kind go back to Erdős, Graham and Szemerédi [23]. Pebbling the walk DAG of k walks in time order keeps, at step t, pebbles on the current and the previous step and on the earlier endpoints of the hops that span t, at most $\mathrm{Live}(t)+2$ pebbles, where $\mathrm{Live}(t)$ is the number of hops spanning t. Hence, if a family is depth-robust at (ε, D) in genie form, then $\sum_t \mathrm{Live}(t)$ is at least about $\varepsilon TD - 2T$: the mean time-cut capacity of a depth-robust schedule grows linearly with the depth that it forces. Corollary 3.8 bounds the capacity of the deep part from above; on the deep-nesting pairs both bounds are compatible, with capacity about $T/(m+1)$ and, under Conjecture A′, depth exponential in m.

4 One arbitrary head

Theorem F needs every head to have bounded owner-load. One head of arbitrary owner-load is harmless on its own: the walk DAG of a single walk is planar, and TCP(1) holds by separators [8, Section 5.1]. Lemma 4.1 gives TCP(1) in the form of a label that is anchored to cells rather than to times, and Theorem M (Theorem 4.2) combines it with the family clock: one arbitrary walk plus walks of bounded owner-load whose families use boundedly many scales are depth-reducible. Section 4.3 collects what Theorems F and M, and a stack formulation of TCP, say about where a depth-robust family must live.

4.1 Lemma W: a strip label for one walk

Fix integers $w \geq 2$ and $r \geq 1$, a *line residue* $\rho \in \{0, \ldots, w-1\}$ and an *offset* $\omega \in \{0, \ldots, r-1\}$. The *lines* are the cells congruent to ρ modulo w, and the *strips* are the sets of $w-1$ consecutive cells strictly between consecutive lines. For a walk x of length T, the *line times* are the times t at which $x(t)$ is a line. For a time t that is not a line time, let $S(t)$ be the strip of $x(t)$ and $c(t)$ the number of times $t' \leq t$ with $x(t') \in S(t)$; the *strip label* is $\Psi(t) = (c(t) + \omega) \bmod r$. The *strip cut* consists of the edges of the walk DAG with an endpoint at a line time, the hop edges $u \to p$ with $\Psi(u) \leq \Psi(p)$, and the state edges $t \to t-1$ with $\Psi(t) < \Psi(t-1)$. (The value of the label at line times does not matter, since every edge at a line time is cut.)

Lemma 4.1 (Lemma W: the strip cut for one walk)**.** `Lean: strip_cut, strip_cut_depth` [`OneWalk.lean`]

Let x be a walk of length T and let $w \geq 2$ and $r \geq 1$ be integers. For every line residue and offset, every path of the walk DAG of x that avoids the strip cut contains at most $r-1$ hop edges. For some line residue and offset, the strip cut has at most $4T/w + wT/r$ edges.

Outline of the formal proof. In Lean a walk is a map $\mathbb{N} \to \mathbb{Z}$ (`IsWalk`); a walk of length T is extended by staying at $x(T-1)$, and since hops, line times, strip counts and cut edges are taken at times below T, the extension changes none of them. Hops are the pairs of consecutive visits of a cell at distance at least two in time (`IsHop`, `hops`); lines, strips, the strip count and the strip label are `IsLine`, `strip`, `stripCount` and `stripLabel`, and the cut consists of `cutStates` and `cutHops`. *Depth.* Along a path that avoids the cut, the label does not increase along state edges and decreases strictly along hop edges, so Lemma 2.3 bounds the number of hop edges by $r-1$. *Size.* Some residue has at most T/w line times, by averaging over the w residues (`exists_line_residue`). A *wrap* is a time τ, not a line time, with $c(\tau) + \omega \equiv 0 \pmod r$; for some offset there are at most T/r wraps, again by averaging (`exists_offset`). A hop at a cell that is not a line has both endpoints in one strip S, and c rises by at least one from p to u, at the visit u; so if the hop is cut, the

count of S passes a multiple of r at a visit to S in $(p, u]$, a wrap of S. At each wrap at most $w-1$ hops of the strip span it, one per cell, by (P1) (`hop_unique`). Two adjacent cells that are not lines lie in one strip (`strip_eq_of_adj`), so a state edge between non-line times is cut only at a wrap. Hence at most $2L + W$ state edges and at most $L + (w-1)W$ hop edges are cut, where L and W are the numbers of line times and of wraps (`card_cutStates_le`, `card_cutHops_le`), and $3L + wW \le 3T/w + wT/r$. □

For $0 < \varepsilon \le 4$, with $w = \lceil 8/\varepsilon \rceil \ge 2$ and $r = \lceil 2w/\varepsilon \rceil$, the strip cut has at most εT edges and depth at most $r - 1 < 2w/\varepsilon \le 16/\varepsilon^2 + 2/\varepsilon$. The label is local to the strip of the current cell: the two endpoints of a hop read the counter of the same strip. So the label is not a band rule in the sense of the fourth paper's nesting barrier [8, Definition 5.5], and that barrier does not apply to it.

4.2 Theorem M

Let x_0 be any walk of length T; its long hops are called *chords*. Let H' be a set of long hops of further walks $x_1, \dots, x_k$ of length T, of owner-load at most ℓ, with families, bands and events as in Section 3.2, and put $N = 2k\ell$, an upper bound for the number of virtual heads (Lemma 3.4). Let G be the DAG on $[0, T)$ with the state edges, all chords and the hop edges of H'; an edge is determined by its endpoints.

Theorem 4.2 (Theorem M: one arbitrary head)**.** Not formalized; a complete proof is given below.
Let $q \ge 1$ and $h \ge 0$ be integers and $0 \le \beta_1 < \beta_2 < \cdots < \beta_q$ integers, and suppose that every family F of H' is assigned a window $i(F) \in \{1, \dots, q\}$ with $\beta_{i(F)} \le \lambda(d_F) \le \beta_{i(F)} + h$. Let $\varepsilon \in (0, 1]$ and put

$$w = 2^{\lceil \log_2(8(4+2k)/\varepsilon) \rceil}, \qquad K_0 = \lceil 8(w+1)/\varepsilon \rceil, \qquad K = \Big\lceil \frac{8}{\varepsilon}\Big(q + qN + \frac{64q^2N^2 2^h}{\varepsilon} + qwN + N^2 2^{h+1}\Big)\Big\rceil.$$

Then G has a cut of at most εT edges and depth less than $D_{\mathrm{M}}(k, \ell, q, h, \varepsilon) = K_0 K^{2q}$. The bound does not depend on T, on x_0 or on the β_i, and pauses are allowed.

For $q = 0$, that is, $H' = \emptyset$, Lemma 4.1 applies. Windows may touch or overlap; any cover of a finite set of bands by q windows of width h can moreover be replaced by the greedy cover, in which each window starts at the smallest band not yet covered, with at most q windows of the same width.

Proof (not formalized). Widths and lines. Put $W_0 = w$ and, for $1 \le i \le q$, let W_i be the least power of 2 that is at least $\max(\mathrm{lo}_i, w)$, where $\mathrm{lo}_i = 64qN2^{\beta_i + h}/\varepsilon$. The lo_i increase with i, so $w = W_0 \mid W_1 \mid \cdots \mid W_q$, and $W_i \le 2\max(\mathrm{lo}_i, w) \le 2(\mathrm{lo}_i + w)$. Draw ρ uniformly from $\{0, \dots, W_q - 1\}$. The *level-i lines* are the cells congruent to ρ modulo W_i; every level-i line is a level-i' line for $i' \le i$. A *level-i strip* is the set of cells strictly between two consecutive level-i lines, so it has $W_i - 1$ cells and every level-i strip lies inside a level-$(i+1)$ strip. Let $Z = \{t : x_0(t) \text{ is a level-0 line}\}$, and for $t \notin Z$ let $\sigma_i(t)$ be the level-i strip of $x_0(t)$.

Counters and label. The level-0 counter of a level-0 strip S is $c^0_S(t) = \#\{t' \le t : x_0(t') \in S\}$, and for $i \ge 1$ the level-i counter of a level-i strip S counts *entries* into S, $c^i_S(t) = \#\{t' \le t : x_0(t') \in S \text{ and } x_0(t'-1) \text{ is a level-}i \text{ line}\}$. For the walks of H', let $n_i(t)$ be the number of events at times at most t of scale at least β_i. Choose independent uniform offsets $\omega_0 \in [0, K_0)$ and $\omega_1, \dots, \omega_q, \omega'_1, \dots, \omega'_q \in [0, K)$, and put $d_0(t) = (c^0_{\sigma_0(t)}(t) + \omega_0) \bmod K_0$, $d_i(t) = (c^i_{\sigma_i(t)}(t) + \omega_i) \bmod K$ and $e_i(t) = (n_i(t) + \omega'_i) \bmod K$ for $1 \le i \le q$. The label $\Psi(t)$, for $t \notin Z$, is the mixed-radix number with the *static* digit order, most significant first,

$$d_q,\ e_q,\ d_{q-1},\ e_{q-1},\ \dots,\ d_1,\ e_1,\ d_0,$$

so it has K_0K^{2q} values; put $\Psi(t) = 0$ for $t \in Z$. The cut F consists of all edges with an endpoint in Z, the hop edges $u \to p$ (chords or hops of H') with $\Psi(u) \le \Psi(p)$, and the state edges at descents. Every edge at Z is cut, so the value of Ψ on Z is never compared, and by Lemma 2.3 F has depth less than K_0K^{2q}. A digit *wraps* at a step if its count passes a value congruent to minus its offset modulo its radix; a digit whose count rises by Δ at a step wraps there with probability at most Δ divided by its radix.

The deciding level of a chord. Let $\gamma = (p, u)$ be a chord at a cell c that is not a level-0 line (chords at lines have both endpoints in Z). Its *excursion* $E(\gamma) = \{x_0(t) : p < t < u\}$ is a nonempty interval of cells next to c, on one side of c. Let $\mathrm{lev}(\gamma)$ be the largest $i \in \{0, \dots, q\}$ such that $E(\gamma)$ contains a level-i line, and 0 if it contains none. If $\mathrm{lev}(\gamma) = i < q$, then $E(\gamma)$ contains no level-$(i+1)$ line, so $E(\gamma) \cup \{c\} \subseteq \sigma_{i+1}(c)$ (*confinement*).

(M-a) State edges. Let $t, t-1 \notin Z$. Then $x_0(t-1)$ and $x_0(t)$ are equal or adjacent cells that are not lines, so they lie in the same strip at every level. No counter of level $i \ge 1$ changes at t, since an entry at t needs $x_0(t-1)$ on a line; the level-0 counter rises by one, and n_i by the number of events at t of scale at least β_i. So $\Psi(t) < \Psi(t-1)$ only if some digit wraps at t.

(M-b) Chords. Let $\gamma = (p, u)$ be a chord at a cell c that is not a line, with $i = \mathrm{lev}(\gamma)$. (i) At p and at u every head-0 digit reads the counter of the same strip $\sigma_{i'}(c)$, and counters do not decrease. (ii) The level-i counter of $\sigma_i(c)$ rises strictly in $(p, u]$: for $i = 0$ by the visit at u; for $i \ge 1$, $E(\gamma)$ contains the nearest level-i line y_* to c on the side of the excursion, and if t_1 is the last time in (p, u) with $x_0(t_1) = y_*$, then $x_0(t_1 + 1)$ is the neighbour of y_* on the side of c, since the walk reaches c at u without returning to y_*; this neighbour is a cell of $\sigma_i(c)$, so $t_1 + 1 \in (p, u]$ is an entry into $\sigma_i(c)$. (iii) If $i < q$, then by confinement x_0 visits no level-$(i+1)$ line at the times $p, \dots, u-1$, so no counter of a level greater than i changes in $(p, u]$. (iv) The event counts do not decrease. Hence, if no digit at a position at or above d_i wraps in $(p, u]$, the most significant digit whose count changed has risen without wrapping, and $\Psi(u) > \Psi(p)$. By (iii), γ is violated only if (c1) the level-i counter of $\sigma_i(c)$ wraps at a time in $(p, u]$, or (c2) some $e_{i'}$ with $i' > i$ wraps in $(p, u]$.

(M-c) Hops of H'. Let $\eta = (p, u) \in H'$ lie in a family F with window $i = i(F)$, and $p, u \notin Z$. If x_0 visits no level-i line during (p, u), it stays in one level-i strip on $[p, u]$, hence in one level-i' strip for every $i' \ge i$, and enters none of them in $(p, u]$; so the digits $d_{i'}$, $i' \ge i$, do not change. The counts $n_{i'}$ do not decrease, and n_i rises, because the owner $o_F \in (p, u)$ is an event of scale $\lambda(d_F) \ge \beta_i$. So η is violated only if (d1) x_0 visits a level-i line during (p, u), or (d2) some $e_{i'}$ with $i' \ge i$ wraps in $(p, u]$.

Cost. All expectations are over ρ and the offsets; each violated edge is charged to one of the causes above, and the classes may overlap, which only over-counts.

(a) *Edges at Z.* A vertex has at most 2 state edges, 2 chords and $2k$ hops of H'. Since $\rho \bmod w$ is uniform, $\mathbb{E}|Z| = T/w$, and these edges cost at most $(4 + 2k)T/w \le \varepsilon T/8$.

(b) *Descents.* By (M-a), they cost at most $T/K_0 + \sum_i \#\{\text{events of scale} \ge \beta_i\}/K \le T/K_0 + qNT/(2K)$, because there are at most $\sum_v \sum_F d_F \le NT/2$ events (Lemma 3.4(e)).

(c) *(c1).* A wrap at time τ of the level-i counter of a strip S is charged only by chords at cells of S that span τ, at most $W_i - 1$ of them by (P1). At level 0 there are at most T/K_0 wraps in expectation, costing at most $(w-1)T/K_0$. At level $i \ge 1$ every entry follows a visit to a level-i line, there are T/W_i such visits in expectation, and each entry is a wrap with probability at most $1/K$; so each level $i \ge 1$ costs at most T/K.

(d) *(c2).* Let $e_{i'}$ wrap at τ. The chords charged to it have level $i < i'$ and span τ, so by confinement $x_0(\tau) \in E(\gamma) \cup \{c\} \subseteq \sigma_{i+1}(c) \subseteq \sigma_{i'}(c)$: all of them lie at cells of the level-i' strip containing

$x_0(\tau)$, and by (P1) there are at most $W_{i'} - 1$ of them. Events of scale at least $\beta_{i'}$ are owners of families of height at least $2^{\beta_{i'}}$, so by Lemma 3.4(e) there are at most $NT/2^{\beta_{i'}+1}$ of them. The cost is at most $\sum_{i'} W_{i'} NT/(2^{\beta_{i'}+1}K) \le \sum_{i'} (\mathrm{lo}_{i'} + w)NT/(2^{\beta_{i'}}K) \le q(64qN^2 2^h/\varepsilon + wN)T/K$.

(e) *(d1).* Charge η to its first visit $\tau \in (p, u)$ to a level-i line. The hops of window i that span τ lie in families of band at most $\beta_i + h$, so by the overlap bound of Lemma 3.4 there are fewer than $N2^{\beta_i+h+1}$ of them. There are $T/W_i \le T/\mathrm{lo}_i$ visits to level-i lines in expectation, so the cost is at most $\sum_i N2^{\beta_i+h+1}T/\mathrm{lo}_i = \varepsilon T/32$.

(f) *(d2).* An event o' of scale s raises the counts $n_{i'}$ with $\beta_{i'} \le s$, and a wrap of $e_{i'}$ at o' exposes the hops of windows at most i' that span o', whose bands are at most $\beta_{i'} + h$: fewer than $N2^{\beta_{i'}+h+1}$ of them. Since the $\beta_{i'}$ are distinct integers, $\sum_{\beta_{i'} \le s} 2^{\beta_{i'}} < 2^{s+1}$, so the cost is at most $\sum_{o'} N2^{h+2}2^{s(o')}/K \le N^2 2^{h+1}T/K$, using $\sum_{o'} 2^{s(o')} \le \sum_v \sum_F d_F \le NT/2$.

Total. The edges at Z cost at most $\varepsilon T/8$; the level-0 terms, $T/K_0 + (w-1)T/K_0 = wT/K_0$, at most $\varepsilon T/8$; all terms with denominator K together at most $(qN + q + 64q^2N^2 2^h/\varepsilon + qwN + N^2 2^{h+1})T/K \le \varepsilon T/8$; and (d1) at most $\varepsilon T/32$. The sum is less than $\varepsilon T/2$, and some choice of ρ and the offsets attains at most the expectation. □

The chords never see the digits of the bounded walks except through wraps, because the digit positions are static and the event counts are global. A chord of a strip that x_0 has left is decided at the coarsest level whose line it crosses, by the re-entry into its own strip there; above that level it is confined, which bounds its exposure in (c2) by a strip width rather than by the number of live chords. The hops of the bounded walks compare only head-0 digits at or above their window, whose lines are far apart compared with their heights.

Corollary 4.3 (Corollary M1: scale-ratio form)**.** Not formalized; a complete proof is given below. *If all family heights of H' lie in $[d_{\min}, d_{\max}]$, then G has a cut of at most εT edges and depth*

$$O(k\,(N^2 d_{\max}/d_{\min} + kN)^2/\varepsilon^6) = \mathrm{poly}(k, \ell, 1/\varepsilon) \cdot (d_{\max}/d_{\min})^2.$$

In particular: (i) bounded walks whose families have height at most $d_{\max}$, together with any x_0, are cut at depth $\mathrm{poly}(k, \ell, 1/\varepsilon)\, d_{\max}^2$; (ii) if all families except at most two per virtual head have their band in one window of width h, the depth is bounded in terms of k, ℓ, h and ε alone; this covers zigzags of any amplitude, even growing with T.

Proof (not formalized). Take $q = 1$, $\beta_1 = \lambda(d_{\min})$ and $h = \lambda(d_{\max}) - \lambda(d_{\min})$, so that $2^h \le 2d_{\max}/d_{\min}$; then $K_0 = O(k/\varepsilon^2)$ and $K = O((N^2 2^h + kN)/\varepsilon^2)$, and Theorem 4.2 gives depth less than $K_0 K^2$. For (i), $d_{\min} \ge 1$. For (ii), the exceptional families are those truncated at the ends of $[0, T)$; with them the bands are covered by at most $1 + 2N$ windows of width h, and Theorem 4.2 applies with $q \le 1 + 2N$. A zigzag between two cells at distance A has owner-load one per side, and all its families have height A except the truncated ones, so $h = 0$. □

Why it matters. Theorem M allows one head to be anything: of unbounded owner-load, nested at every scale, drifting, deep everywhere. So a depth-robust family must contain either two heads of unbounded owner-load, as the deep-nesting pair does, or one such head together with bounded heads whose families occupy unboundedly many separated scales (Corollary 4.4). For machines this becomes Corollary 6.6.

4.3 Where a counterexample to TCP must live

Corollary 4.4 (Where depth-robustness lives)**.** Not formalized; a complete proof is given below.

Let $\mathcal{W}$ be a depth-robust family of k-tuples of walks. Then the owner-load of the set of all long hops is unbounded on $\mathcal{W}$. If moreover all walks but one have owner-load at most a fixed ℓ on $\mathcal{W}$, then for every fixed h the least number of windows of width h covering the family bands of the bounded walks is unbounded on $\mathcal{W}$; in particular the ratio of their largest to their smallest family height is unbounded.

Proof (not formalized). If the owner-load were bounded, Theorem 3.6 would give a single depth bound for each ε. If, for some h, the bands of the bounded walks of every member were covered by at most q windows of width h, Theorem 4.2, applied with the unbounded walk as x_0, would do the same. A bounded ratio $d_{\max}/d_{\min}$ gives such a cover with $q = 1$. □

The structure of walk DAGs that these theorems exploit is a stack structure. Call an *s-stack DAG* a DAG on $[0, N)$, with edges determined by their endpoints, consisting of the spine edges $t \to t-1$ and of *arcs* $u \to p$ with $u - p \geq 2$, the arcs being partitioned into s families; in each family no two arcs cross (there are no arcs (p, u), (p', u') with $p < p' < u < u'$), and every vertex is the left endpoint of at most one arc of each family and the right endpoint of at most one arc of each family. A cut is a set of edges, and its depth is the largest number of arcs on a path that avoids it. Let $\mathrm{SR}(s)$ be the statement that for every $\varepsilon > 0$ there is D such that every s-stack DAG on $[0, N)$ has a cut of at most εN edges and depth at most D.

Proposition 4.5 (TCP is depth reduction for stack DAGs)**.** Not formalized; a complete proof is given below.

(a) The walk DAG of k walks, pauses allowed, is a $2k$-stack DAG.

(b) Every s-stack DAG H on $[0, N)$ is realized by s walks of length $2N$ without pauses: every arc (p, u) of family i is the upper long hop $(2p, 2u)$ of walk i. Consequently a cut of size c and depth D of their walk DAG yields a cut of H of size at most c and depth at most D.

Hence $\mathrm{TCP}(k)$ *for walks without pauses implies* $\mathrm{SR}(k)$*, with the depth of* $\mathrm{TCP}(k)$ *at* $\varepsilon/2$*, and* $\mathrm{SR}(2k)$ *implies* $\mathrm{TCP}(k)$ *for walks with pauses; in particular the forms of TCP with and without pauses are equivalent up to a factor* 2 *in the number of walks.*

Proof (not formalized). (a) The upper long hops of one walk do not cross, nor do its lower long hops (Section 2.2). Each time is the left endpoint of at most one long hop of each walk and the right endpoint of at most one. A pair of times that is a long hop of several walks is a single edge, kept in one of its families, and pauses are spine edges.

(b) For $0 \leq t \leq N$ let $d_i(t)$ be the number of arcs of family i that contain t strictly inside, so $d_i(0) = d_i(N) = 0$. By the degree condition, $|d_i(t+1) - d_i(t)| \leq 1$: $d_i(t+1) - d_i(t)$ is 1 if t is the left endpoint of an arc and 0 otherwise, minus 1 if $t+1$ is the right endpoint of an arc and 0 otherwise. The two endpoints of an arc (p, u) have the same value of d_i: an arc (p', u') of the family that contains p strictly inside has $u' > u$, since $u' < u$ would cross and $u' = u$ would violate the degree condition, so it also contains u, and conversely. Every time strictly inside the arc has a larger value. Put $x_i(2t) = 2d_i(t)$ and $x_i(2t+1) = \min(x_i(2t), x_i(2t+2)) + 1$ for $t < N$. Consecutive even values differ by 0 or ± 2, so x_i is a walk without pauses. At $2p$ and $2u$ it is at $2d_i(p)$, and strictly between them it is at least $2d_i(p) + 1$, so $(2p, 2u)$ is an upper long hop of x_i. A spine edge $t \to t-1$ maps to the state edges $2t \to 2t-1 \to 2t-2$ and an arc $u \to p$ to the hop edge $2u \to 2p$; distinct

edges of H have disjoint images. Given a cut F of the walk DAG, cut an edge of H if some edge of its image lies in F. This cut has at most $|F|$ edges, and a path of H that avoids it maps to a path that avoids F with as many long hops as the path has arcs.

For the consequences, a k-stack DAG on $[0, N)$, padded with empty families, is realized by k walks of length $2N$, and a cut of $\varepsilon N = (\varepsilon/2)(2N)$ edges of their walk DAG pulls back; conversely (a) applies to k walks. □

In this language, Theorem F says that s-stack DAGs are depth-reducible when the arcs open at any time in any family come from boundedly many distinct owners, the owner of an arc being that of its hop in (b). Forgetting the orientation of the edges, s-stack DAGs are essentially the s-pushdown graphs of Galil, Kannan and Szemerédi [25]. Part (a) of Proposition 4.5 is the classical fact that a tape is two pushdown stores, so that the computation graphs of multitape machines, the objects of the computation-graph method of Hopcroft, Paul and Valiant [28], are $O(1)$-pushdown graphs; part (b) is the converse, a pushdown store being a tape whose head sits at the height of the stack. The stack view of tape computations was also used by Paul, Pippenger, Szemerédi and Trotter [40] and by Santhanam [42]. SR(s) is the depth analogue of the separator question of Galil, Kannan and Szemerédi, who showed that sublinear separators for 3-pushdown graphs, sublinear separators for k-page graphs, and subquadratic simulations of nondeterministic two-tape machines by nondeterministic one-tape machines stand or fall together [25]. We found no work on reducing DAGs of bounded stack number, or walk DAGs of bounded nesting, to bounded depth by deleting a linear fraction of the edges.

Proposition 4.6 (Two stacks are planar)**.** Not formalized; a complete proof is given below.
SR(2) *holds, with* $D(\varepsilon) = O(\varepsilon^{-2})$.

Proof (not formalized). Draw the vertices of a 2-stack DAG in order on a line, the spine edges on the line, the arcs of one family above it and those of the other below it. Arcs of one family do not cross, and arcs that share an endpoint can be drawn without crossing, so the underlying graph is planar; its maximum degree is at most $2 + 2 \cdot 2 = 6$. By the planar separator theorem of Lipton and Tarjan [35], applied recursively as in [36], some set of $O(N/\sqrt{M})$ vertices leaves components of at most M vertices. Cutting the at most six edges at each of these vertices cuts $O(N/\sqrt{M})$ edges, and a path that avoids the cut uses no vertex of the separator, except possibly as a path with no edge, and so stays inside one component and has fewer than M arcs. Taking M of order ε^{-2} gives the claim. □

Why they matter. Proposition 4.6 recovers TCP(1), and Proposition 4.5 places the open part of TCP at three or four stacks: SR(3) is the first open case, and if TCP(2) fails, as Conjecture A′ with its walk-DAG analogue predicts for the deep-nesting pairs, then SR(4) fails. From three stacks on, separators no longer help. Galil, Kannan and Szemerédi constructed 3-pushdown graphs whose separators have $\Omega(n/\log^{(s)} n)$ vertices, with $\log^{(s)}$ the s-fold iterated logarithm, for every fixed s [24]. Dvir and Wigderson obtained $O(1)$-pushdown expanders from monotone expanders, answering a question of Galil, Kannan and Szemerédi [22], and Dujmović, Sidiropoulos and Wood built bounded-degree expanders with three-page book embeddings from the constant-degree monotone expanders of Bourgain and Yehudayoff [20, 6]; each page of that embedding is a non-crossing matching, so with a spine added these expanders are 3-stack DAGs. Depth-robustness, however, needs arcs that span many scales, and within boundedly many non-crossing families such arcs must nest; Theorem F says that the nesting must then come from unboundedly many distinct owners.

Remark 4.7 (A candidate counterexample)**.** Heuristic. Let $x_0 = U(s_0, m, 1)$, a nested hierarchy, and let x_1 be a sequential one: a walk made of V's, each a segment of 2^{j+1} steps followed by a back stroke and a forward stroke of 2^j steps, with the exponents j running through $s_m, \dots, s_0$ with frequencies proportional to 2^{-j}, so that every scale gets the same time, as every level of U does. Then x_1 has owner-load one per side and enters every cell at most three times. Theorem F cuts x_1 alone at depth $2^{O(1/\varepsilon)}$ and Lemma 4.1 cuts x_0 alone at depth $O(\varepsilon^{-2})$, while Theorem M cuts the pair at depth $O(4^m)\,\mathrm{poly}(1/\varepsilon)$, with one window of width $h = m$. If the pair needs depth $2^{\Omega(m)}$ at some fixed ε, then TCP(2) fails already for one deep head and one head of owner-load one. The fourth paper's forced-shape heuristics for the deep-nesting pair apply with the small V's of x_1 in the role of the second hierarchy; no lower-bound technique is available, and packing certificates cannot certify depth-robustness [8, Proposition 5.14].

5 The deep-nesting pair

The deep-nesting pairs of Section 2.5 have two heads of unbounded owner-load as m grows, so they escape Theorems F and M, and they remain the sharpest open instance of TCP(2). This section proves three things about them. A single turn counter serves every level of both heads at $\log_2 q = m + O(\log(1/\theta))$ and cuts the pair at depth $2^{m+O(1)}/\varepsilon$ (Section 5.2); so Conjecture A′ can hold only with $C \geq 1$, and the depth question is exponential against subexponential in m. Fine levels are *blind*: on average over shifts, a class with g levels below it sees a turn of the other head on at most a fraction $(g + 3 + \lceil \log_2(g+2) \rceil)/(m+1)$ of its hops, for every label that changes only at turns (Section 5.4). Coarse levels are *visible*: for the block labels that serve one head cheaply, a fixed fraction of the apex hops of every coarse class of the other head sees a cancellation at full size, an other-head increment of at most $-Q-1$, on average over shifts (Section 5.5). Blindness leaves a weaker form of Conjecture A′, which still gives exponential depth (Conjecture 5.24).

Throughout, $x_1 = U(s_0, m, 1)$ and $x_2 = x_1(\cdot + \xi)$ for an integer shift ξ, with $s_0 > m$, $\Lambda = 2^{s_0}$, $R_i = 2^{s_0 - i}$, $n_i = m - i + 1$ and $P = 4(m+1)\Lambda$, as in Section 2.5.

5.1 Turns and classes

Let B be a V of level $k \geq 1$ of a head, attached at its base b to a stroke of direction σ, with apex α, valley ν and valley cell $v = b - \sigma R_k$. Its back stroke is a segment from b to v followed, if $k < m$, by the V C_{k+1} of level $k+1$ attached at v; for $k+1 < l \leq m$ let C_l be the V of level l attached at the end of the segment of the forward stroke of C_{l-1}. By [8, Lemma 5.4(a)], which has a written proof there, C_l is attached at v, has valley cell $v + \sigma R_l$, stays between v and $v + \sigma R_l$, and ends at time ν, since C_{k+1} ends the back stroke of B and C_{l+1} ends the forward stroke of C_l; its valley time is $\nu - n_l R_l$. We call $C_{k+1}, \dots, C_m$ the *chain* of B, and put $R_{m+1} = 0$.

Lemma 5.1 (Turns and classes)**.** Not formalized; a complete proof is given below.

(a) Each head has $2^{m+3} - 6$ turns per period, and the pair has $N_P = 2^{m+4} - 12$.

(b) For $1 \leq k \leq m$, the hops of level k of a head are exactly the following $2R_k$ hops for each V B of level k of that head, in the notation above: the apex hops *$(\alpha - d, \alpha + d)$, $1 \leq d \leq R_k$, and the* valley hops *at the cells $v + \sigma x$, $1 \leq x \leq R_k$, whose later endpoint is $\nu + x$. For $d < R_k$ the only turn of the head in the half-open interval $(\alpha - d, \alpha + d]$ of an apex hop is α. For $k \leq l \leq m$ and $R_{l+1} < x < R_l$, the valley hop at $v + \sigma x$ is*

$$(\nu - (m-l)R_l - x,\ \nu + x),$$

and the turns of the head in its half-open interval are ν and the turns of C_{l+1} (only ν if $l = m$).

(c) On the r-fold cover each head has exactly $4r\Lambda$ hops of each level $i \in \{0, \dots, m\}$; so every class has $n_{\mathrm{h}}/(2(m+1))$ hops, and the hops of (b) other than those listed with $d < R_k$ or $R_{l+1} < x < R_l$, the boundary hops*, are a fraction $(m-k+2)/(2R_k)$ of the class.*

Proof (not formalized). (a) The turns are the apex and valley times of the V's, two per V, and two tips per period; a period contains 2^{i+1} V's of level i (Section 2.5). So each head has $2 + \sum_{i=1}^{m} 2^{i+2} = 2^{m+3} - 6$ turns per period.

(b) The V B is attached at the end of the segment of a stroke of level $k-1$, which has $R_{k-1} = 2R_k$ steps in direction σ and ends at b at time α; the back stroke starts with a segment of R_k steps in direction $-\sigma$. So for $1 \le d \le R_k$ the cell $b - \sigma d$ is visited at $\alpha - d$ and at $\alpha + d$ and not in between, and for $d < R_k$ the only turn in $(\alpha - d, \alpha + d]$ is α. For the valley hops, the forward stroke of B starts with a segment from v to b, so $v + \sigma x$ is visited at $\nu + x$ for $1 \le x \le R_k$, and before that time, after ν, it is not visited. Let $R_{l+1} < x < R_l$. If $l = k$, the chain stays between v and $v + \sigma R_{k+1}$, so the last visit before ν is by the back segment, at time $\alpha + R_k - x = \nu - (m-k)R_k - x$, using $\nu = \alpha + n_k R_k$. If $l > k$, the forward segment of C_l runs from $v + \sigma R_l$ to v starting at time $\nu - n_l R_l$, and afterwards C_{l+1} stays between v and $v + \sigma R_{l+1}$; so the last visit is at $\nu - n_l R_l + R_l - x = \nu - (m-l)R_l - x$, after the valley time of C_l. In both cases the turns in the half-open interval are those of C_{l+1} and ν, because $\nu + x$ lies before the apex of the V of the forward stroke, at $\nu + R_k$.

The listed hops have level exactly k. The only turn strictly inside an apex hop is α, since the times before α in the hop belong to the parent's segment and those after it to the back segment. A valley hop at $v + \sigma x$ has its earlier endpoint at or after α, because the back segment visits its cell at time $\alpha + R_k - x$, and its later endpoint is $\nu + x \le \nu + R_k$; so its interior lies in the span of B, whose turns have level at least k, and it contains the turn ν of level k. Each listed hop contains exactly one turn of level k, the apex or the valley of B, so it determines B and its type; within B, the apex hops are distinguished by d and the valley hops by their cells. So the listed hops are distinct, and every head has at least $2^{k+1} \cdot 2R_k = 4\Lambda$ hops of level k per period, for $1 \le k \le m$.

(c) A hop has level 0 exactly when it contains a tip strictly inside. At a tip τ the head is at $\pm\Lambda$, and for each of the other 2Λ cells of $[-\Lambda, \Lambda]$ exactly one hop of the head at that cell has τ in its half-open interval, hence in its open interval. No hop contains two tips: between consecutive tips the head sweeps all of $[-\Lambda, \Lambda]$, the second and third level-0 strokes of a period sweeping from Λ to $-\Lambda$ and the fourth and first from $-\Lambda$ to Λ, and a hop at $\pm\Lambda$ would contain the tip at its own cell. So each head has exactly 4Λ hops of level 0 per period. With (b), every level has at least 4Λ hops per period and head, and there are exactly $P = 4(m+1)\Lambda$ hops per period and head, one ending at every time. Hence every level has exactly 4Λ, the list of (b) is complete, and on the r-fold cover every class has $4r\Lambda = n_{\mathrm{h}}/(2(m+1))$ hops. The boundary hops are $m - k + 2$ of the $2R_k$ hops of each V: $d = R_k$ and $x = R_l$ for $k \le l \le m$. □

The fourth paper defined service with the normalization $n_{\mathrm{h}}/(2(m+1))$ per class on the strength of numerical counts [8, Section 5.7]; Lemma 5.1(c) proves that every class has exactly this size. All turn times of head 1 are multiples of R_m, and those of head 2 are congruent to $-\xi$ modulo R_m; so for odd ξ no time is a turn of both heads.

5.2 The turn counter

Theorem 5.2 (Turn counter)**.** Lean (partial): `turnCounter_cost` [`TurnCounter.lean`]
Not formalized: part (1), the hypotheses of the Lean theorem on the cover, and the counts of Lemma 5.1(a) and Section 2.5, which rest on the fourth paper's Lemma 5.4 and its count of V's.

Let ξ be any shift, $q \geq 3$, and r a multiple of $q/\gcd(q, N_P)$. On the r-fold cover let $\mathcal{N}(t)$, for $t \in [0, T)$, be the number of turns of either head at times in $[0, t]$, and $G(t) = \mathcal{N}(t) \bmod q$. Then:

(1) every hop (p, u) has error $e = (\kappa - 1) \bmod q \leq \kappa - 1$, where $\kappa \geq 1$ is the number of turns of either head in the lifted interval $(p, u]$;

(2) $\mathrm{desc}(G) = rN_P/q$ and $q \cdot \mathrm{cost}(G) \leq rN_P(4\Lambda + 3)$.

Outline of the formal proof of the label arithmetic. The Lean theorem `turnCounter_cost` concerns a finite set of times with a cyclic predecessor bijection and a finite set of hops arranged in closed chains, as in Lemma Zq (`MotionDepthCodes.lemma_Zq`). It assumes a label G with values in $[0, q)$ that advances, modulo q, by $\chi_t \in [0, q)$ at every time t and by $\kappa_h \geq 1$ along every hop h, and it concludes $q \cdot \mathrm{cost}(G) \leq \sum_h \kappa_h + \sum_t \chi_t$ and $q \cdot \mathrm{desc}(G) = \sum_t \chi_t$. Its proof: Lemma Zq writes $q \cdot \mathrm{cost}(G)$ as the number of hops plus the sum of the reduced errors plus $\sum_t \chi_t$, and each reduced error $(G(u) - G(p) - 1) \bmod q = (\kappa_h - 1) \bmod q$ is at most $\kappa_h - 1$. □

Proof of (1), of the hypotheses and of the counts (not formalized). The times form the cycle $\mathbb{Z}_T$ under $t \mapsto t - 1$, and the hops of each head at each cell form closed chains, each hop followed by the hop that starts at its later endpoint, since every time is the earlier and the later endpoint of exactly one hop of each head (Section 2.5). Extend $\mathcal{N}$ to $\mathbb{Z}$ by $\mathcal{N}(t + T) = \mathcal{N}(t) + rN_P$. Since r is a multiple of $q/\gcd(q, N_P)$, q divides rN_P, so $G \equiv \mathcal{N} \pmod q$ on all of $\mathbb{Z}$. Hence G advances at each time t by the number χ_t of turns at t, at most $2 < q$, and along each hop by the number κ of turns in its lifted half-open interval, which is at least 1 because the owner of the hop is a turn strictly inside it (Section 2.2). This gives (1), since $e \equiv \kappa - 1 \pmod q$ and $\kappa - 1 \geq 0$. Each turn is counted once in $\sum_t \chi_t$, so $\sum_t \chi_t = rN_P$ by Lemma 5.1(a). Each turn lies in the half-open intervals of exactly $2(2\Lambda + 1)$ hops (Section 2.5), so $\sum_h \kappa_h = rN_P \cdot 2(2\Lambda + 1)$. The Lean theorem then gives $\mathrm{desc}(G) = rN_P/q$ and $q \cdot \mathrm{cost}(G) \leq rN_P(4\Lambda + 2) + rN_P$. □

Corollary 5.3 (The counter serves every level)**.** Not formalized; a complete proof is given below.

In the setting of Theorem 5.2, let $0 < \theta < 1$. If $q \geq N_P(1 + 1/(2\Lambda))/\theta$, for instance if $q \geq 20 \cdot 2^m/\theta$, then every level is doubly θ-served. If moreover $q \geq 32 \cdot 2^m/\theta$ and $\theta \leq 1/3$, then $\mathrm{cost}(G) \leq \frac{11}{32}\,\theta\, n_{\mathrm{h}}/(m + 1) \leq n_{\mathrm{h}}/8$.

Proof (not formalized). By Theorem 5.2(1), the total error of all hops is at most $\sum_h (\kappa_h - 1) < rN_P(4\Lambda + 2)$, so every class has total error below this. It is at most $\theta q n_{\mathrm{h}}/(2(m + 1)) = 4\theta q r \Lambda$ as soon as $q \geq N_P(4\Lambda + 2)/(4\theta\Lambda) = N_P(1 + 1/(2\Lambda))/\theta$; and $N_P < 2^{m+4}$, $\Lambda \geq 2$. By Theorem 5.2(2), $\mathrm{cost}(G)/n_{\mathrm{h}} \leq N_P(4\Lambda + 3)/(8(m+1)\Lambda q)$, which for $q \geq 32 \cdot 2^m/\theta$ is at most $\theta(4\Lambda + 3)/(16(m+1)\Lambda) \leq \frac{11}{32}\theta/(m + 1)$, since $(4\Lambda + 3)/\Lambda \leq 11/2$. □

Corollary 5.4 (Conjecture A′ needs $C \geq 1$)**.** Not formalized; a complete proof is given below.

If Conjecture A′ holds with constants (θ_0, C, C'), then $C \geq 1$.

Proof (not formalized). Put $\theta = \min(\theta_0, 1/3)$ and $q = 2^{m+5+\lceil \log_2(1/\theta) \rceil} \geq 32 \cdot 2^m/\theta$. For every m, every $s_0 > m$ and every shift, in particular the one provided by the conjecture, Corollary 5.3 gives a label on a cover of cost at most $n_{\mathrm{h}}/8$ that doubly θ_0-serves all $m + 1$ levels. Hence $m + 1 \leq C(m + 5 + \lceil \log_2(1/\theta) \rceil) + C'$ for all m, and letting $m \to \infty$ gives $C \geq 1$. □

Corollary 5.5 (Cutting the pair with a counter)**.** Not formalized; a complete proof is given below.

For every shift, every $\varepsilon > 0$ and every multiple T of P, the walk DAG of the pair on $[0, T)$ has a cut of at most εT edges and depth at most $\lceil 11 \cdot 2^{m+1}/((m + 1)\varepsilon) \rceil - 1$. For $m \geq 1$ and $\varepsilon \leq 1$ this is smaller than the depth $\lceil 12/\varepsilon \rceil^{m+1} - 1$ of the fourth paper's clock labels.

Proof (not formalized). Let $\mathcal{N}(t)$ be the number of turns of either head at times in $[0,t]$, $\Psi(t) = \mathcal{N}(t) \bmod q$, and call t a *wrap time* if $\lfloor \mathcal{N}(t)/q \rfloor > \lfloor \mathcal{N}(t-1)/q \rfloor$. A hop (p,u) with no wrap time in $(p,u]$ is not violated, since then $\Psi(u) - \Psi(p) = \mathcal{N}(u) - \mathcal{N}(p) \geq 1$ by its owner. Descents occur only at wrap times. There are at most $\mathcal{N}(T-1)/q \leq (T/P)N_P/q$ wrap times, and each lies in the half-open intervals of at most $2(2\Lambda+1)$ hops. So $\mathrm{cost}(\Psi) \leq (T/P)N_P(4\Lambda+3)/q$, which is at most εT for $q \geq N_P(4\Lambda+3)/(4(m+1)\Lambda\varepsilon)$; this holds for $q = \lceil 11 \cdot 2^{m+1}/((m+1)\varepsilon) \rceil$, because $N_P < 2^{m+4}$ and $(4\Lambda+3)/(4\Lambda) \leq 11/8$. Lemma 2.3 gives depth at most $q-1$. For the comparison, $11 \cdot 2^{m+1}/(m+1) \leq 12^{m+1}/6$ for $m \geq 1$ (at $m=1$ it reads $22 \leq 24$, and from m to $m+1$ the left side less than doubles while the right side grows twelvefold), so with $x = 12^{m+1}/\varepsilon \geq 144$ we get $\lceil 11 \cdot 2^{m+1}/((m+1)\varepsilon) \rceil \leq \lceil x/6 \rceil < x \leq (12/\varepsilon)^{m+1} \leq \lceil 12/\varepsilon \rceil^{m+1}$ for $\varepsilon \leq 1$. □

Why they matter. The counter shows what Conjecture A′ must look like: all $m+1$ levels are doubly served at $\log_2 q = m + O(\log(1/\theta))$, within the cost budget, at every shift. So the conjecture, if true, is sharp up to the constant C. The least depth $D(\varepsilon)$ of a cut of εT edges of the walk DAG satisfies $\log_2 D(\varepsilon) \leq m + \log_2(1/\varepsilon) + O(1)$ unconditionally, by Corollary 5.5, while Conjecture A′ with its walk-DAG analogue would give $\log_2 D(\varepsilon) \geq c(m+1) - c'$ for small ε, with $c \leq 1$ by the counter. So the open question is exponential against subexponential depth in the nesting depth m, and the base of an exponential is at most 2, not the c/ε of the clock labels.

Remark 5.6 (Numerical: the evidence for Conjecture A′)**.** The fourth paper quotes, as evidence for Conjecture A′, the numbers 1, 1, 2 and 2 or 3 of doubly 0.1-served levels of optimized labels at $m=3$, $s_0=6$ and $q = 8, 16, 32, 64$ [8, Remark 5.26]. On the walk DAG of eight periods at the golden shift, the turn counter doubly serves 1, 1, 2 and 3 levels at the same q, and at $m=6$, $s_0=10$ it serves $\log_2 q - 3$ levels for $\log_2 q = 7, 8, 9$. The quoted evidence is therefore essentially what the counter does; it is consistent with Conjecture A′ with $C \approx 1$, but it cannot discriminate. (Not all of these labels meet the cost budget; the comparison is of served counts against q.)

The counter is a special case of a general criterion.

Proposition 5.7 (The counter criterion)**.** Not formalized; a complete proof is given below.

Let $x_1, \ldots, x_k$ be walks of length T without pauses, with $\mathcal{N}$ turns in all and at most H hops spanning any time. Then for every $\varepsilon > 0$ the walk DAG has a cut of at most εT edges and depth at most $\max(0, \lceil \mathcal{N}(H+1)/(\varepsilon T) \rceil - 1)$. *Hence TCP holds for every family with $(\mathcal{N}/T) \cdot H$ bounded.*

Proof (not formalized). Let $\mathcal{N}(t)$ count the turns of all walks at times in $[0,t]$, each walk separately, and $\Psi(t) = \mathcal{N}(t) \bmod q$ with $q = \max(1, \lceil \mathcal{N}(H+1)/(\varepsilon T) \rceil)$. As in Corollary 5.5, a hop with no wrap time in its half-open interval contains its owner, a turn, and is not violated; there are at most $\lfloor \mathcal{N}/q \rfloor$ wrap times, each a descent at most once and inside at most H hops. So the cost is at most $\lfloor \mathcal{N}/q \rfloor (H+1) \leq \varepsilon T$, and the depth is at most $q-1$. □

The criterion is incomparable with the fourth paper's Theorem B: two zigzags of fixed width whose centres drift monotonically satisfy it but are not home-based. It fails on the deep-nesting pairs as $m \to \infty$, where $(\mathcal{N}/T) \cdot H \approx 2^{m+4}(4\Lambda+2)/P \approx 2^{m+4}/(m+1)$.

The counter also pins down the capacity trade-off of Corollary 3.8 on the deep-nesting family. For two walks write $\mathrm{Cap}(D, \delta, \varepsilon)$ for the statement: for all pairs of walks of length T there are a set R of pairs with $|R| \leq \delta T$ and a cut of $G - R$ with at most εT edges and depth at most D.

Proposition 5.8 (Capacity on the deep-nesting family)**.** Not formalized; a complete proof is given below.

(a) Let $0 < \varepsilon < 1/2$, $\delta(\varepsilon) = 2/\log_2(1/\varepsilon)$ and $D(\varepsilon) = \lceil 11/\varepsilon^2 \rceil - 1$. For every m, every $s_0 > m$, every shift and every multiple T of P, the pair on $[0,T)$ has a set R of pairs with $|R| \le \delta(\varepsilon)T$ and a cut of $G - R$ with at most εT edges and depth at most $D(\varepsilon)$.

(b) Suppose there are $\varepsilon_0 > 0$, an integer $N_0 \ge 1$ and a function D_* such that for every $m \ge 0$ there are $s_0 > m$ and an odd shift for which the walk DAG of the pair on $[0, N_0P)$ has no cut of at most $\varepsilon_0 N_0 P$ edges and depth less than $D_*(m)$ (a walk-DAG form of Conjecture A′). If $\mathrm{Cap}(D(\varepsilon), \delta(\varepsilon), \varepsilon)$ holds for all ε, then $D(\varepsilon) \ge D_*(m)$ whenever $\varepsilon \le \varepsilon_0/2$ and $4(m+1)N_0\delta(\varepsilon) \le \varepsilon_0$. In particular, if $D_*(m) \ge 2^{c(m+1)-c'}$, then $\delta(\varepsilon) \ge c\varepsilon_0/(4N_0(\log_2 D(\varepsilon) + c + c'))$ whenever $\varepsilon \le \varepsilon_0/2$ and $\delta(\varepsilon) \le \varepsilon_0/(4N_0)$.

Proof (not formalized). (a) If $m+1 \ge 2/\delta(\varepsilon)$, let R be the set of all pairs visited; then $|R| = 2(2\Lambda+1)$ and $|R|/T \le (4\Lambda+2)/(4(m+1)\Lambda) \le (5/4)/(m+1) \le \frac{5}{8}\delta(\varepsilon)$, every hop is at a pair of R, and the empty cut of $G - R$ has depth 0. If $m + 1 < 2/\delta(\varepsilon) = \log_2(1/\varepsilon)$, let $R = \emptyset$; Corollary 5.5 gives depth at most $\lceil 11 \cdot 2^{m+1}/((m+1)\varepsilon) \rceil - 1 \le \lceil 11/\varepsilon^2 \rceil - 1$, since $2^{m+1} < 1/\varepsilon$.

(b) Take the instance for m and $T = N_0P$, and let R and a cut F of $G - R$ be given by the capacity statement. Each pair is visited at most $2(m+1)N_0$ times in $[0,T)$, so it carries fewer than $2(m+1)N_0$ hops. Let F_0 be the set of hop edges of G all of whose hops lie at pairs of R; then $|F_0| \le 2(m+1)N_0|R| \le \varepsilon_0 T/2$, and every edge of G outside F_0 is an edge of $G - R$. So $F \cup F_0$ is a cut of G with at most $\varepsilon T + \varepsilon_0 T/2 \le \varepsilon_0 T$ edges and depth at most $D(\varepsilon)$, and the hypothesis gives $D(\varepsilon) \ge D_*(m)$. For the second statement put $y = \varepsilon_0/(4N_0\delta(\varepsilon)) \ge 1$ and $m + 1 = \lfloor y \rfloor$; then $\log_2 D(\varepsilon) \ge c\lfloor y \rfloor - c' \ge c(y-1) - c'$, that is, $y \le (\log_2 D(\varepsilon) + c + c')/c$. □

So on the deep-nesting family, under the walk-DAG form of Conjecture A′ assumed in (b), the scratch must have size about $T/\log D$ when the rest is cut at depth D, and (a) achieves this: there $\log_2 D(\varepsilon) \approx 2\log_2(1/\varepsilon) + 3.5$ and $\delta(\varepsilon) \approx 4/\log_2 D(\varepsilon)$.

5.3 Labels that change only at turns

Let ξ be odd. A label G on the cover, or on the walk DAG of a window, is *turn-supported* if there are integers $J_h(\tau)$ at the turn times τ of each head h such that $G(t) - G(t-1) \equiv J_h(t) \pmod q$ when t is a turn of head h, and $G(t) \equiv G(t-1)$ when t is not a turn. This class contains the counter ($J \equiv 1$), the fourth paper's clock labels, and every function of the vector of turn counts. The *potential* of head h is $\Phi_h(t) = \sum J_h(\tau)$ over the turns $\tau \le t$ of head h (on a window, or on $\mathbb{Z}$ for periodic jumps, normalized by $\Phi_h(0) = 0$), so $G \equiv \Phi_1 + \Phi_2 + \mathrm{const} \pmod q$. For a hop of head h', the *other-head increment* is $F = \Phi_h(u) - \Phi_h(p)$, where $h \ne h'$.

For a V of head h let $M(V)$ be the sum of the jumps of head h at the turns in the span of V, with $M(\emptyset) = 0$; let V^{b} and V^{f} be the V's attached at the ends of the segments of its back and forward strokes (none if V has level m), and put

$$A_V = J_h(\alpha_V) - 1, \qquad E_V = J_h(\nu_V) + M(V^{\mathrm{b}}) - 1, \qquad c_{\mathrm{b}}(V) = 1 + A_V, \qquad c_{\mathrm{f}}(V) = 2 + A_V + E_V.$$

Lemma 5.9 (Path sums)**.** Not formalized; a complete proof is given below.
Let t be a time in a level-0 stroke S_0 of head h, and let $V_1 \supset V_2 \supset \cdots \supset V_j$ be the V's whose spans contain t, with V_i of level i and V_1 the V of S_0. Then

$$\Phi_h(t) - \Phi_h(\alpha_{V_1} - 1) = \sum_{i=1}^{j} c_{\delta_i}(V_i),$$

where $\delta_i = \mathrm{b}$ if t lies in the back stroke of V_i and $\delta_i = \mathrm{f}$ otherwise; for t in the segment of S_0, $j = 0$. Moreover $M(V) = c_{\mathrm{f}}(V) + M(V^{\mathrm{f}})$.

Proof (not formalized). By induction on the depth inside one V. The apex α_V is the first time of V. For t in the segment of the back stroke, the only turn in $[\alpha_V, t]$ is α_V, so the increment over $(\alpha_V - 1, t]$ is $J_h(\alpha_V) = c_{\mathrm{b}}(V)$; for t in V^{b} it is $c_{\mathrm{b}}(V)$ plus the increment from $\alpha_{V^{\mathrm{b}}} - 1$ to t. For t in the segment of the forward stroke the increment is $J_h(\alpha_V) + M(V^{\mathrm{b}}) + J_h(\nu_V) = c_{\mathrm{f}}(V)$, by the definition of E_V, and for t in V^{f} it is $c_{\mathrm{f}}(V)$ plus the increment from $\alpha_{V^{\mathrm{f}}} - 1$ to t. At level m there are no inner V's and the same computation applies with $M(\emptyset) = 0$. The mass identity is the case where t is the last time of V. The segment of S_0 contains no turn after its first time. □

Lemma 5.10 (Band decomposition)**.** Not formalized; a complete proof is given below.
Let G be turn-supported, and let B be a V of head h' of level $k \geq 1$, with chain $C_{k+1}, \ldots, C_m$.

(a) For $1 \leq d < R_k$ the apex hop $(\alpha - d, \alpha + d)$ has error $(A_B + F) \bmod q$.

(b) For $k \leq l \leq m$ and $R_{l+1} < x < R_l$, the valley hop at $v + \sigma x$ has error

$$\Big(E_B - \sum_{i=k+1}^{l} (A_{C_i} + E_{C_i} + 2) + F\Big) \bmod q.$$

Here F is the other-head increment of the hop.

Proof (not formalized). The error is $(G(u) - G(p) - 1) \bmod q$, and $G(u) - G(p) \equiv \Phi_{h'}(u) - \Phi_{h'}(p) + F$. (a) By Lemma 5.1(b) the only turn of head h' in $(\alpha - d, \alpha + d]$ is α, with jump $1 + A_B$. (b) By Lemma 5.1(b) the increment of $\Phi_{h'}$ is $M(C_{l+1}) + J_{h'}(\nu)$, and $J_{h'}(\nu) = E_B + 1 - M(C_{k+1})$. Since $C_{i+1} = C_i^{\mathrm{f}}$, the mass identity of Lemma 5.9 gives $M(C_{k+1}) - M(C_{l+1}) = \sum_{i=k+1}^{l} (A_{C_i} + E_{C_i} + 2)$. □

When $F = 0$ on the hops concerned (the *single-head regime*), A_B is the apex error of B and $c_{\mathrm{b}} - 1$, $c_{\mathrm{f}} - c_{\mathrm{b}} - 1$ are errors of B's own hops. So a V whose own errors lie in $[0, E]$ contributes at least 1 and 2 to the path sums of Lemma 5.9: the potential of a head that serves many of its levels rises along every path, and it can come back down only through V's with an apex or band error near q. Serving many levels cheaply needs cancellation, and the next labels cancel in blocks.

Definition 5.11 (Block labels)**.** Let $a \geq 1$ and $K \geq 0$ be integers. A level $l \in [1, m]$ is a *canceller* if $m - l \equiv a \pmod{a + 1}$ and *served* otherwise. Define integers A_l, E_l from level m upwards: for a served level l, $A_l = K$ and $E_l = K + \sum_{i=l+1}^{c-1} (A_i + E_i + 2)$, where c is the first canceller below l, or $m + 1$ if there is none; for a canceller c, $A_c = 0$ and $E_c = -Q_c - 2$ with $Q_c = \sum_{i=c+1}^{c'-1} (A_i + E_i + 2)$, where c' is the next canceller below c, or $m + 1$. The *block label* $B(a, K)$ on head h has the jumps $J_h(\alpha_V) = 1 + A_{\mathrm{lev}V}$, $J_h(\nu_V) = E_{\mathrm{lev}V} + 1 - M(V^{\mathrm{b}})$, computed from level m upwards, and $J_h = 1$ at the tips; so $A_V = A_{\mathrm{lev}V}$ and $E_V = E_{\mathrm{lev}V}$.

Lemma 5.12 (Block masses)**.** Not formalized; a complete proof is given below.
In $B(a, K)$, a served level with j served levels below it in its block has $A + E + 2 = 2^j(2K + 2)$. Every canceller has a served levels directly below it, and $Q_c = Q := (2^a - 1)(2K + 2)$. Hence $c_{\mathrm{b}}(V) = 1 + K$ and $c_{\mathrm{f}}(V) = 2^j(2K + 2)$ at a served V, and $c_{\mathrm{b}}(V) = 1$ and $c_{\mathrm{f}}(V) = -Q$ at a canceller V: the path sums depend only on the level. In the single-head regime, and if $K + Q < q$, every apex hop with $d < R_k$ of a served V has error K and every valley hop of Lemma 5.10(b) of a served V has error in $[K, K + Q]$.

Proof (not formalized). Write $T(j)$ for $A + E + 2$ at a served level with j served levels below it in its block. Then $T(0) = 2K + 2$ and $T(j) = 2K + 2 + \sum_{j' < j} T(j')$, so $T(j) = 2^j(2K + 2)$. A canceller c satisfies $m - c \equiv a \pmod{a + 1}$ and $m - c \geq 0$, so $m - c \geq a$ and the levels

$c+1,\dots,c+a$ are served and below c in one block; hence $Q_c = \sum_{j<a} T(j) = Q$. The values of c_b and c_f follow. For the errors, write $S(i_1,i_2) = \sum_{i=i_1}^{i_2}(A_i+E_i+2)$. A served level k has $E_k = K + S(k+1,c-1)$ with c the first canceller below k. For $k \le l < c$ the band error of Lemma 5.10(b) is $K + S(l+1,c-1) \in [K, K+Q)$. For $c \le l < c'$, with c' the next canceller, it is $K - S(c,l) = K + Q - S(c+1,l) = K + S(l+1,c'-1) \in [K, K+Q]$, because $A_c + E_c + 2 = -Q$ and $S(c+1,c'-1) = Q$; and the pattern repeats below c'. All values lie in $[0,q)$, so no reduction occurs. □

So, in the single-head regime, one head carrying $B(a,K)$ serves the $a/(a+1)$ fraction of its levels that are served, with errors in $[K, K+Q]$, and its violated own hops are essentially the valley hops of the canceller classes. Heuristically, its cost per own hop is then about $1/(2(a+1))$, which for $a=4$ is $1/10$, inside the budget $n_\mathrm{h}/8$ for every m at fixed q (see also Remark 5.13). So no argument that uses only one head's own structure can be expected to prove Conjecture A′. At the aligned shift the second head inherits the first head's errors.

Remark 5.13 (Numerical: the single-head cost of block labels)**.** Exact evaluations on covers gave costs per own hop of 0.05 to 0.09 for block labels with $a=4$, $q=256$ and $4 \le m \le 10$.

Proposition 5.14 (The aligned shift)**.** Not formalized; a complete proof is given below.

Let $\xi = 1$, put the same jumps on both heads, $J_2(t) = J_1(t+1)$, and for a hop (a',b') of head 1 write $I(a',b') = \Phi_1(b') - \Phi_1(a')$. The hops of head 2 are the pairs $(a'-1, b'-1)$ for the hops (a',b') of head 1, and

$$G(b'-1) - G(a'-1) \equiv 2I(a',b') - J_1(b')[b' \text{ a turn}] + J_1(a')[a' \text{ a turn}],$$
$$G(b') - G(a') \equiv 2I(a',b') + J_1(b'+1)[b'+1 \text{ a turn}] - J_1(a'+1)[a'+1 \text{ a turn}].$$

So every hop whose shifted endpoints are not turns has error $(1+2e) \bmod q$, where $e = (I-1) \bmod q$ is the single-head error of the corresponding hop of head 1; each turn affects at most four hops.

Proof (not formalized). Head 2 is at a cell at time t exactly when head 1 is there at time $t+1$, so its hops are the shifted hops of head 1 and its turns are the turns of head 1 minus one. Hence $\Phi_2(t) = \Phi_1(t+1)$ and $G(t) \equiv \Phi_1(t) + \Phi_1(t+1)$ up to a constant, and both identities follow by expanding $\Phi_1(t \pm 1) - \Phi_1(t)$. Each time is the earlier and the later endpoint of exactly one hop of each head. □

Remark 5.15 (Numerical: Conjecture A′ needs a misaligned shift)**.** With $B(4,0)$ on both heads at the aligned shift and $q=256$, exact evaluations on covers gave costs 0.080, 0.0996 and 0.0935 times n_h at $m=6$, 9 and 10, inside the budget $n_\mathrm{h}/8$, with 6 of 7, 8 of 10 and 9 of 11 levels doubly 0.1-served; Proposition 5.14 explains this, since at the aligned shift both heads see their single-head errors, doubled, on all but a vanishing fraction of hops. This indicates that Conjecture A′ cannot hold with the aligned shift, so that its choice of a misaligned shift is essential. At misaligned shifts (the golden shift, and a shift whose dyadic phases alternate near 1/3 and 2/3 at every level) the same labels collapse to the finest few levels at costs 0.22 to 0.24 times n_h.

5.4 Blindness of fine classes

Lemma 5.16 (Turn-free runs)**.** Not formalized; a complete proof is given below.

Let $m \ge 1$. Over one period, the gaps between consecutive turns of U have the following lengths: for each level $1 \le i \le m$, 2^{i+2} gaps of length R_i, except that two of the gaps of level m have length $R_m + R_0$ instead; and two gaps of length R_0.

Proof (not formalized). We find the gap that follows each turn. After an apex of level $i < m$ comes the back segment, R_i steps, and then the apex of the V attached at its end; after an apex of level m, a plain back stroke of R_m steps and then the valley. After a valley of level $i < m$ comes the forward segment, R_i steps, and then an apex. After a tip comes the segment of a level-0 stroke, R_0 steps, and then an apex. After a valley ν of level m, the forward stroke takes R_m steps and the V ends at $\nu + R_m$, together with every stroke of which it is the last part. If some enclosing stroke is a back stroke, the deepest such ends at $\nu + R_m$ with the valley of its V, a turn. If not, the level-0 stroke ends at $\nu + R_m$; this happens for one level-m V per level-0 stroke. If the next level-0 stroke is the second or fourth of its period, $\nu + R_m$ is a tip; otherwise the directions continue (from the second to the third stroke, and from the fourth to the first), the next segment adds R_0 steps, and the next turn is at $\nu + R_m + R_0$; this happens twice per period. Each level i has 2^{i+2} turns per period, and the lengths add up to $4m\Lambda + 4\Lambda = P$. □

Proposition 5.17 (Blindness)**.** Not formalized; a complete proof is given below.

Let $m \geq 1$, let G be turn-supported, let (p, u) be a hop of head h' of length $\ell = u - p$, and let the shift ξ be uniformly random in $\mathbb{Z}_P$. If $(p, u]$ contains no turn of the other head, then $F = 0$. The probability of this event is

$$\frac{1}{P} \sum_{gaps\ L} (L - \ell)^+ \ \geq \ \frac{1}{m+1} \sum_{i=0}^{m} \Big(1 - \frac{\ell}{R_i}\Big)^+,$$

the sum over the gaps of Lemma 5.16. For ξ uniform over the odd residues the bound decreases by at most $2/((m+1)R_m)$.

Proof (not formalized). F is the sum of the other head's jumps over the window, an empty sum. In the other head's own time, where it is U, the window $(p, u]$ is a window $(a', a' + \ell]$ with a' uniform in $\mathbb{Z}_P$: a hop of head 2, given as a hop (a, b) of U in the own time of head 2, occupies the window $(a - \xi, b - \xi]$ of head 1; and for a hop (a, b) of head 1, the turns of head 2 are those of U shifted by $-\xi$, so the window is $(a + \xi, b + \xi]$ in the time of U. This window contains no turn exactly when $\tau_j \leq a'$ and $a' + \ell < \tau_{j+1}$ for consecutive turns $\tau_j < \tau_{j+1}$, which gives $(L_j - \ell)^+$ values of a' for the gap of length $L_j = \tau_{j+1} - \tau_j$. Insert Lemma 5.16, using $(x + y - \ell)^+ \geq (x - \ell)^+ + (y - \ell)^+$ for $x, y, \ell \geq 0$ for the two long gaps, and $2^{i+2}/P = 1/((m+1)R_i)$, $4/P = 1/((m+1)R_0)$. For odd shifts, each interval of admissible values of a' contains at least $((L_j - \ell)^+ - 1)/2$ values of each parity, there are fewer than 2^{m+3} gaps, and $2^{m+3}/P = 2/((m+1)R_m)$. □

Corollary 5.18 (Fine classes are blind)**.** Not formalized; a complete proof is given below.

Let $1 \leq k \leq m$, $g = m - k$, and let ξ be uniform in $\mathbb{Z}_P$. On average over ξ, at least a fraction

$$\frac{k - 2 - \lceil \log_2(g+2) \rceil}{m+1}$$

of the hops of each class (h', k) contain no turn of the other head. Consequently, for every turn-supported label, the expected fraction of the hops of the class with $F \neq 0$ is at most

$$\frac{g + 3 + \lceil \log_2(g+2) \rceil}{m+1},$$

which is $O(\log m/m)$ for the classes with $g = O(\log m)$.

Proof (not formalized). By Lemma 5.1(b), the hops of the class are, for each V of level k, its apex hops $(\alpha-d, \alpha+d)$, $1 \le d \le R_k$, and its valley hops at the cells $v+\sigma x$, $1 \le x \le R_k$; the boundary hops among them obey the same length bounds as the others. An apex hop has length $2d \le 2R_k = R_{k-1}$, so by Proposition 5.17 it is blind with probability at least $\frac{1}{m+1}\sum_{i=0}^{k-1}(1-2^{i-k+1}) \ge \frac{k-2}{m+1}$. For $k \le l \le m$ and $R_{l+1} < x \le R_l$, the valley hop at $v+\sigma x$ is $(\nu-(m-l)R_l - x, \nu + x)$: for $x < R_l$ by Lemma 5.1(b), and for the boundary hop $x = R_l$ because the cell $v + \sigma R_l$ is visited at the valley time $\nu - n_l R_l$ of C_l (at the apex $\alpha = \nu - n_k R_k$ if $l = k$), left at the next step, and not visited again before $\nu + R_l$. So its length is at most $(m-l+2)R_l \le R_{l-a}$, where $a = \lceil \log_2(m-l+2) \rceil$; if $l \ge a$ it is blind with probability at least $\frac{1}{m+1}\sum_{i=0}^{l-a}(1-2^{i-l+a}) \ge \frac{l-a-1}{m+1} \ge \frac{k-1-\lceil \log_2(g+2) \rceil}{m+1}$, using $l \ge k$ and $m-l \le g$; if $l < a$ the claimed bound is negative. The second statement follows, because $F \ne 0$ needs a turn of the other head in the window. □

Why it matters. By Markov's inequality, for each class with $g = O(\log m)$ the set of shifts at which a fixed fraction $c > 0$ of the class has $F \ne 0$ has density $O(\log m/(cm))$: for all large m, at all but a vanishing density of shifts, these classes see the other head on less than a fraction c of their hops, for every label that changes only at turns. On their blind hops the error is the single-head error, so a head can serve them as if alone, for instance by a block label. Heuristically, then, at typical shifts no argument that charges errors to the other head's increments can bound the number of doubly served fine levels by $C \log_2 q + C' + O(\log m)$, and if Conjecture A′ holds at a typical shift, it holds because serving the coarse levels is expensive. This is not a proof: the corollary bounds an average over shifts, and the conjecture may choose an atypical shift. The measurements of Remark 5.19 agree with this picture.

Remark 5.19 (Numerical: block pairs at a misaligned shift)**.** With $B(4,0)$ on both heads at a misaligned shift and $q = 256$, the number of doubly 0.1-served levels was $4, 4, 4, 5, 5, 8, 8$ at $m = 10, 15, 20, 25, 30, 40, 50$, all in the blind fine region, at costs of 0.23 to 0.29 times n_{h}, above the budget; the value at $m = 10$ is exact on a cover, the others are sampled.

5.5 Visibility of coarse classes

The coarse classes see the other head's cancellations. In the other head's own time an apex hop of half-width d is a window $(c_0 - d, c_0 + d]$ around a centre c_0, and its other-head increment is $F(d) = \Phi_h(c_0 + d) - \Phi_h(c_0 - d)$.

Lemma 5.20 (Straddle involution)**.** Not formalized; a complete proof is given below.

Let head h carry $B(a, K)$, let c_0 be an integer, $k \ge 1$, and let V be a V of head h of a canceller level κ, with apex α_V, valley ν_V and stroke length $\Pi_\kappa = \nu_V - \alpha_V = n_\kappa R_\kappa$. Let $\mathcal{D}$ be the set of $d \in [1, R_k - 1]$ with $c_0 - d \in [\alpha_V, \nu_V)$, $c_0 + d \in [\nu_V, \nu_V + \Pi_\kappa)$ and $\Pi_\kappa - d \in [1, R_k - 1]$. Then $d \mapsto \Pi_\kappa - d$ is an involution of $\mathcal{D}$, and $F(d) + F(\Pi_\kappa - d) = -2Q - 2$ for every $d \in \mathcal{D}$. Hence at least half of the elements of $\mathcal{D}$ have $F(d) \le -Q - 1$.

Proof (not formalized). For $d \in \mathcal{D}$ put $o_p = c_0 - d - \alpha_V$ and $o_u = c_0 + d - \nu_V$, both in $[0, \Pi_\kappa)$. For $d' = \Pi_\kappa - d$ we get $c_0 - d' - \alpha_V = o_u$ and $c_0 + d' - \nu_V = o_p$, so $d' \in \mathcal{D}$ and the map swaps o_p and o_u. By Lemma 5.9, for a time x in the back stroke of V, $\Phi_h(x) = C_V + c_{\mathrm{b}}(V) + Y(x - \alpha_V)$, and for x in its forward stroke, $\Phi_h(x) = C_V + c_{\mathrm{f}}(V) + Y(x - \nu_V)$. Here C_V collects $\Phi_h(\alpha_{V_1} - 1)$ and the contributions of the V's containing V, the same for all times in V, and $Y(o)$ is the sum of $c_\delta(W)$ over the V's W inside a stroke of level κ that contain its time at offset o; by Lemma 5.12 these contributions depend only on levels, so Y is the same function for both strokes of V. Hence $F(d) = c_{\mathrm{f}}(V) - c_{\mathrm{b}}(V) + Y(o_u) - Y(o_p)$ and $F(\Pi_\kappa - d) = c_{\mathrm{f}}(V) - c_{\mathrm{b}}(V) + Y(o_p) - Y(o_u)$, and their sum is $2(c_{\mathrm{f}}(V) - c_{\mathrm{b}}(V)) = 2(-Q - 1)$ by Lemma 5.12. □

Theorem 5.21 (Theorem V: visibility of coarse classes on average over shifts)**.** Not formalized; a complete proof is given below.

Let head h carry $B(a,K)$ and let ξ be uniform in $\mathbb{Z}_P$. For every class (h',k) of the other head with $1 \le k \le m$ and $g = m-k \ge \lceil \log_2(g+1) \rceil + a$, the expected fraction of its apex hops with $d < R_k$ that have $F \le -Q-1$ is at least

$$\frac{g - \lceil \log_2(g+1) \rceil - 2a}{2^{a+4}(m+1)}.$$

Proof (not formalized). Let $j_1 = \min\{j \ge 0 : g+1-j \le 2^j\}$; then $1 \le j_1 \le \lceil \log_2(g+1) \rceil$, since $g \ge 1$. Let $\kappa = k + j$ be the first canceller level with $j \ge j_1$; cancellers recur every $a+1$ levels, so $j \le j_1 + a$, and $\kappa \le m$ by the hypothesis on g. Since $j \ge j_1$, $\Pi_\kappa = n_\kappa R_\kappa = (g+1-j)R_k 2^{-j} \le R_k$, and Π_κ is even.

Fix a V of level k of head h' and its apex hops. As ξ is uniform, the centre c_0 of their windows in the own time of head h is uniform modulo P. A window of half-width d *straddles* a V of level κ of head h if its left end lies in the back stroke and its right end in the forward stroke; then $d < \Pi_\kappa$, and c_0 lies in the span of that V, so for fixed c_0 at most one V of level κ is straddled, and for $d \le \Pi_\kappa - 1$ the partner $\Pi_\kappa - d$ lies in $[1, R_k - 1]$. So, for fixed c_0, the straddling half-widths form the set $\mathcal{D}$ of Lemma 5.20 for the straddled V. For a fixed d, the centres that straddle a given V of level κ form an interval of length $2\min(d, \Pi_\kappa - d)$, and there are $2^{\kappa+1}$ such V's per period; so

$$\mathbb{E}|\mathcal{D}| = \sum_{d=1}^{\Pi_\kappa - 1} \frac{2^{\kappa+2}\min(d, \Pi_\kappa - d)}{P} = \frac{2^{\kappa+2}}{P} \cdot \frac{\Pi_\kappa^2}{4} = \frac{2^\kappa \Pi_\kappa^2}{P}.$$

By Lemma 5.20, at least $|\mathcal{D}|/2$ of the $R_k - 1$ apex hops with $d < R_k$ have $F \le -Q-1$, so the expected fraction is at least

$$\frac{2^{\kappa-1}\Pi_\kappa^2}{PR_k} = \frac{n_\kappa^2 R_\kappa}{8(m+1)R_k} = \frac{(g+1-j)^2 2^{-j}}{8(m+1)},$$

using $2^\kappa = \Lambda / R_\kappa$ and $P = 4(m+1)\Lambda$. By the minimality of j_1, $g + 2 - j_1 > 2^{j_1 - 1}$, so with $G_1 = g + 2 - j_1$ we have $2^j \le 2^{j_1 + a} < 2^{a+1} G_1$ and $(g+1-j)^2 \ge (G_1 - 1 - a)^2 \ge G_1^2 - 2(a+1)G_1$, where $G_1 - 1 - a = g + 1 - j_1 - a \ge 1$. Hence $(g+1-j)^2 2^{-j} > (G_1 - 2(a+1))/2^{a+1} = (g - j_1 - 2a)/2^{a+1}$, and dividing by $8(m+1)$ and using $j_1 \le \lceil \log_2(g+1) \rceil$ gives the bound. □

Why it matters. Blindness bounds the visible fraction of a class from above by $(g + 3 + \lceil \log_2(g+2) \rceil)/(m+1)$, and Theorem V bounds it from below, for block labels, by about $g/(2^{a+4}(m+1))$: both are of order $(m-k)/(m+1)$ up to logarithmic factors and constants depending on a. So the coarse classes, a positive fraction of all levels, see the other head's cancellations at full size on a positive fraction of their apex hops, on average over shifts (see also Remark 5.22). The next proposition says what this would give for block pairs, but it needs more than Theorem V provides. First, Conjecture A′ fixes one shift for all classes at once. Second, the proposition needs the cancellation in a window modulo q, $-q/2 < F \le -Q_h/2$, while Theorem V gives only the integer event $F \le -Q-1$, with no lower bound on F: by the proof of Lemma 5.20, $F = -Q - 1 + Y(o_u) - Y(o_p)$, and Y sums contributions of up to $m - \kappa$ nested levels, so $|F|$ may grow linearly in m, and the event is guaranteed to imply the window only for q of order $(m-\kappa)Q$ or larger.

Remark 5.22 (Numerical: visible fractions)**.** Measured fractions of apex hops with $F \le -Q-1$ were ten to a hundred times larger than the bound of Theorem V.

Proposition 5.23 (Visibility at a fixed shift bounds the served levels of block pairs)**.** Not formalized; a complete proof is given below.

Let the heads carry $B(a_1, K_1)$ and $B(a_2, K_2)$ with $a_1, a_2 \geq 1$, at a fixed odd shift, on the r-fold cover with r a multiple of q, so that the two potentials define a label there modulo q, and let Q_1, Q_2 be their block masses. Let $0 < \beta < 1$ and $\theta_0 > 0$, and suppose that for each head h' and each class (h', k) with $m - k \geq \beta(m+1)$, a fraction at least $c(\beta) > 8\theta_0$ of its apex hops with $d < R_k$ have $-q/2 < F \leq -Q_h/2$, where h is the other head. Then at most $\beta(m+1) + 2$ levels are doubly θ_0-served.

Proof (not formalized). Since $Q_h = (2^{a_h} - 1)(2K_h + 2) \geq 2K_h + 2$, the inequalities $K_1 \geq Q_2/2$ and $K_2 \geq Q_1/2$ cannot both hold: they would give $Q_1 \geq 2K_1 + 2 \geq Q_2 + 2$ and $Q_2 \geq Q_1 + 2$. So some head h' has $K_{h'} < Q_h/2$ for the other head h. Let $k \geq 1$ with $m - k \geq \beta(m+1)$. By Lemma 5.10(a) an apex hop with $d < R_k$ of a V of level k of head h' has error $(A + F) \bmod q$ with $A \in \{0, K_{h'}\}$; on the hops of the hypothesis, $-q/2 < A + F < 0$, so the error lies in $[q/2, q)$. The apex hops with $d < R_k$ are a fraction $(R_k - 1)/(2R_k) \geq 1/4$ of the class, so the class has total error at least $c(\beta) \cdot \frac{1}{4} \cdot \frac{q}{2}$ times its size, more than $\theta_0 q$ times its size, and level k is not θ_0-served for head h'. The levels $k \geq 1$ with $m - k < \beta(m+1)$, together with level 0, number at most $\beta(m+1) + 2$. □

5.6 A safe form of Conjecture A′

Blindness suggests, heuristically, that cheap labels may serve a number of fine levels that grows with m; in the measurements of Remark 5.19 the block pairs serve between 4 and 8 levels for $10 \leq m \leq 50$, above the budget. The following form of the conjecture allows up to a fixed fraction of the levels to be served and keeps its consequence.

Conjecture 5.24 (Conjecture A″)**.** *There are $\theta_0 > 0$, $\eta < 1$ and constants C, C' such that for all s_0 and m some odd shift ξ has the following property: for all r and q, every label of cost at most $n_h/8$ on the r-fold cover of the pair $x_1 = U(s_0, m, 1)$, $x_2 = x_1(\cdot + \xi)$, has at most $C \log_2 q + C' + \eta(m+1)$ doubly θ_0-served levels.*

If Conjecture A″ holds and a label of cost at most εn_h with $\varepsilon \leq 1/8$ and $\varepsilon < (1 - \eta)\theta_0/2$ exists on such a cover, then, as in Section 2.5, it doubly serves more than $(m+1)(1 - 2\varepsilon/\theta_0)$ levels, so

$$\log_2 q \;>\; \frac{(1 - \eta - 2\varepsilon/\theta_0)(m+1) - C'}{C}.$$

With its walk-DAG analogue this makes the pairs depth-robust as $m \to \infty$, and TCP(2) would fail; only the admissible range of ε shrinks, from $\theta_0/2$ to $(1 - \eta)\theta_0/2$. Proposition 5.23 proves Conjecture A″ for pairs of block labels, with $\eta = \beta + O(1/m)$ and no logarithmic term, from visibility in a window modulo q at a fixed shift. Theorem V does not supply this hypothesis, even on average over shifts: it gives the integer event $F \leq -Q - 1$ with no lower bound on F, hence the window only for q of order $(m - \kappa)Q$, and only on average over shifts. Two steps remain open: a bound on $|F|$ on the straddling hops, or a version of Theorem V that counts the window directly; and, at one fixed shift for all coarse classes, equidistribution of a class's apex centres relative to the other head's canceller valleys of level about $k + \log_2(m - k)$, a mixed-radix carry problem that we could not solve.

6 Consequences for transposers

Theorem D of the fourth paper (Theorem 2.5) excludes oblivious linear-time transposers under two hypotheses, depth reduction and the crux. For machines whose work heads enter every cell

boundedly often, Theorem F supplies the first: the crux alone excludes them (Corollary F). *Directed Recovery Structures in Tape Computations* shows that the cone relations of every linear-time schedule are paired-distance separated after paid trimming, and asks whether all codes on such relations have rate at most $\frac{1}{2} + o(1)$ [14, Question 9.1]; an affirmative answer excludes bounded-entry transposers, oblivious (Corollary F′) or of low adaptivity (Corollary F″). Theorem M gives the analogous statement for machines with one unrestricted work head (Corollary 6.6).

6.1 Corollary F

Corollary 6.1 (Corollary F: the crux excludes bounded-entry oblivious transposers)**.** Not formalized; a complete proof is given below.

Let $k \geq 1$, $\beta \geq 2$ and $V \geq 1$ be integers, $C \geq 1$ and $0 < \varepsilon \leq 1$, and put

$$D_1 = D_{\mathrm{F}}(k, 2V-1, \varepsilon/(4\beta C)), \qquad P = \lceil 4\beta C/\varepsilon \rceil, \qquad D = D_1 P.$$

Suppose that $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ *holds for some $\rho > \varepsilon$. Then for every machine M with k work tapes and value width at most β there is d_0 such that for no $d \geq d_0$ do the following hold together at $L = d^2 b(d)$: M is oblivious at length L with halting time $T \leq C(L+1)$; every work-tape cell is entered at most V times on $[0, T)$; and M transposes all $d \times d$ matrices with $b(d)$-bit entries.*

Proof (not formalized). Let d_0 be given by the crux for $\eta = (\rho - \varepsilon)/2$, and suppose the three conditions hold at some $d \geq d_0$. By Corollary 3.2 the long hops of the work-head walks have owner-load at most $2V - 1$, so Theorem 3.6, at $\varepsilon' = \varepsilon/(4\beta C) \in (0, 1/8]$, gives a cut of the walk DAG with at most $\varepsilon' T$ edges and depth less than D_1. By Lemma 2.6 the dependency DAG has a genie Γ of cost at most εL and full depth less than D. As in the proof of Theorem D, M appends exactly L output bits, $I_\Gamma \in \mathcal{S}_{k,C,\beta,D}$, and the revealed word G_Γ is an index code for I_Γ with at most εL bits, by the cone lemma [8, Corollary 4.5] (a written proof there), while the crux requires at least $(\rho - \eta)L > \varepsilon L$ bits. The input head needs no hypothesis: it adds no edge to the dependency DAG. □

So the crux alone, at the depth D and a rate above ε, excludes oblivious linear-time transposers whose work heads enter every cell $O(1)$ times, whatever the coupling, drift and scales of the heads and whatever the input head does. Here $\log_2 D \approx 512k^3(2V-1)^3\beta C/\varepsilon$, so the crux is needed at a rate exceeding about $512k^3(2V-1)^3\beta C/\log_2 D$, the analogue of the rate $456k^3(1+A)\beta C/\log_2 D$ needed for home-based heads [8, Remark 4.15].

Remark 6.2 (Which machines are new)**.** Bounded entry allows unboundedly many reversals. A machine whose work heads make at most r reversals each enters every cell at most $r+1$ times, since each head's motion splits into $r+1$ weakly monotone pieces, each of which visits a cell in one run; a bound on the number of visits counted in time steps does not cover such machines, because a head may wait on a cell. Such a head has at most $r+1$ legs, each contributing at most T to its leg area about its start cell, so it satisfies the area hypothesis of the fourth paper's Corollary B [8, Corollary 4.23] with $A = r+1$, and that corollary, like Corollary F, puts no condition on the input head. Relative to Corollary B, Corollary F is therefore new exactly for bounded-entry heads whose leg area is superlinear about every home; such heads reverse unboundedly often. Conversely, home-based heads may enter cells near home unboundedly often, so the two corollaries are incomparable. Relative to the unconditional bounds, Corollary F also allows the input head to move freely: machines whose heads, the input head included, make $o(\log d)$ reversals in all cannot transpose at all, oblivious or not, by the first paper's pass bound, Lean-verified there [9, Theorems 5.9 and 5.12], and conservative reversal-bounded machines are excluded by Kirchherr's bound [33].

6.2 Corollaries F′ and F″

Question 9.1 of *Directed Recovery Structures* asks: for a fixed degree bound Δ and recovery relations N_j on finite sets V_j with $|V_j| \to \infty$, whose incidence graphs have maximum degree at most Δ and satisfy the paired-distance condition, must every sequence of storage codes have rate at most $\frac{1}{2} + o(1)$, for every alphabet? [14, Question 9.1] The incidence graph has vertices L_p and R_p for every coordinate p and an edge $L_p R_q$ whenever $p \in N(q)$; the paired-distance condition says that, for every fixed r, only $o(|V_j|)$ coordinates have L_p and R_p at distance at most r [14, Definition 2.3]. The same paper proves that the cone relations $N_\Gamma(q) = I_\Gamma(\mathrm{tpos}(q))$ of every schedule with $T \le c(L+1)$, k work tapes and a genie of fixed full depth satisfy this condition after the paid deletion of a small fixed fraction of coordinates of high reuse [14, Theorem 3.3 and Corollary 3.8], both partly formalized there, with written proofs of the native realization and transport estimates and of the cone-size premise; this holds also for a single frozen trace of an adaptive machine, as argued in prose in [14, Remark 3.9]. It proves, with a written proof that is not formalized, that an affirmative answer to Question 9.1 for every Δ gives rate at most $\frac{1}{2} + o(1)$ for all codes on these cone relations, uniformly over the schedules and genies [14, Proposition 9.2]; the proof of that proposition uses the schedule only through the cone bound, the paid trimming and the paired-distance theorem.

Corollary 6.3 (Corollary F′: Question 9.1 excludes bounded-entry oblivious transposers)**.** Not formalized; a complete proof is given below.

Suppose that Question 9.1 of [14] has an affirmative answer for every fixed Δ; the binary alphabet suffices. Then for all integers $k, V \ge 1$ and every $c > 0$, no machine with k work tapes is, for infinitely many d, oblivious at length $L = d^2 b(d)$, a transposer of all $d \times d$ matrices within $c(L+1)$ steps, and such that every work-tape cell is entered at most V times.

Proof (not formalized). Let M be such a machine, $\beta = \beta(M)$, $C = \max(c, 1)$ and $\varepsilon = 1/4$. At each such d, as in the proof of Corollary F, Theorem 3.6 and Lemma 2.6 give a genie Γ of cost at most $L/4$ and full depth at most a constant D depending only on k, V, β and C. The revealed word is an index code for I_Γ, so its fibres are binary storage codes for the cone relation N_Γ [8, Remark 4.9(b)], and some fibre has at least $2^{3L/4}$ words. By [14, Proposition 9.2], applied with k work tapes, linear-time constant c and cone depth D, every binary storage code on these relations has at most $2^{(1/2+o(1))L}$ words. This is a contradiction for large d. □

For home-based schedules the same conclusion is the remark of *Directed Recovery Structures* that every class of schedules with arbitrarily cheap genies of fixed depth would be excluded by the all-code half bound [14, Section 9.1], combined with the fourth paper's Corollary B; the bounded-entry class is supplied by Theorem F. The same argument extends to adaptive machines whose traces have entropy at most ηL for a fixed $\eta < \frac{1}{2}$, provided that every trace of probability at least $2^{-\eta L}$ has bounded entry.

Corollary 6.4 (Corollary F″: low adaptivity)**.** Not formalized; a complete proof is given below.

Let $k, V \ge 1$ be integers, $c > 0$ and $0 \le \eta < 1/2$. Suppose that Question 9.1 of [14] has an affirmative answer for every fixed Δ. Then for every machine M with k work tapes there is L_0 such that at no length $L = d^2 b(d) \ge L_0$ does M have the following three properties:

(i) on every input X it halts within $c(L+1)$ steps with output X^{T};

(ii) every trace of probability at least $2^{-\eta L}$, for a uniformly random input, enters every work-tape cell at most V times (for instance, every trace does);

(iii) the trace $\tau(X)$ has Shannon entropy $H(\tau(X)) \le \eta L$ for a uniformly random input (for instance, at most $2^{\eta L}$ distinct traces occur).

For $\eta = 0$ this is Corollary F′. The proof uses only one trace: (ii) and (iii) may be replaced by the single condition that some trace of probability at least $2^{-\eta L}$ enters every work-tape cell at most V times.

Proof (not formalized). Fix $\varepsilon \in (0,1]$ with $\eta+\varepsilon < \frac{1}{2}$, and let $D(\varepsilon)$ be the full depth that Theorem 3.6 and Lemma 2.6 give, as in the proof of Corollary F′, for k, V, $\beta(M)$, c and ε. Suppose that M has the three properties at a length L. Since $H(\tau(X)) \ge -\log_2 \max_\tau \Pr[\tau(X)=\tau]$, some trace τ_0 has probability at least $2^{-\eta L}$; let W be the set of inputs that follow it, so $|W| \ge 2^{(1-\eta)L}$. All runs on W share the dependency DAG H_{τ_0} of the trace (Section 2.4), and the local recursion and the cone lemma of the fourth paper [8, Lemma 4.2, Lemma 4.4 and Corollary 4.5], which have written proofs there, hold for all pairs of inputs in W, because their proofs use only the common schedule. By (ii), τ_0 has bounded entry, so Theorem 3.6 and Lemma 2.6, applied to H_{τ_0}, give a genie Γ of cost at most εL and full depth at most $D(\varepsilon)$. For a value g of the revealed word, the set W_g of inputs in W with revealed word g is a storage code for the cone relation of τ_0: two of its inputs that agree on $N_\Gamma(q)$ produce the same output bit at position $\mathrm{tpos}(q)$, which by correctness is their bit q. Some g has $|W_g| \ge 2^{(1-\eta-\varepsilon)L}$. The paired-distance theorem and its corollaries hold for one frozen trace [14, Remark 3.9], and the proof of [14, Proposition 9.2] uses the schedule only through them, so it bounds $\log_2|W_g|$ by $(\frac{1}{2}+o(1))L$, uniformly in the trace, with $o(1)$ depending only on k, c and $D(\varepsilon)$. Since $\eta+\varepsilon < \frac{1}{2}$, this is a contradiction for all $L \ge L_0$, for a suitable L_0. □

The trace's information is paid for once, in the choice of τ_0. This is the kind of paid treatment of adaptivity that the fourth paper anticipated, with a coding bound for codes that are correct only on a large set of inputs [8, Remark 4.24], and that *Directed Recovery Structures* leaves open in general [14, Section 9.3]; here it is available only for traces of entropy below $L/2$.

Remark 6.5 (A weak form of Question 9.1 suffices)**.** Corollary F′ does not need rate $\frac{1}{2}$. The paired-distance condition is a property of sequences of relations. Suppose that along every sequence of relations of incidence degree at most Δ that satisfies it, all codes have rate at most $1-\rho(\Delta)+o(1)$. The argument of Corollary F′, with the paid trimming of [14, Corollary 3.8] at the threshold $\lceil B/\delta \rceil$, where $B = \sum_{i \le D(\varepsilon)} (k+1)^i$ bounds the cone sizes, then works as soon as $(1-\delta)\rho(\Delta) > \varepsilon$ for some $\delta \in (0,1)$ and $\varepsilon \in (0,1]$ with $\Delta = \lceil B/\delta \rceil$. Since $D(\varepsilon)$ is exponential in $1/\varepsilon$, $\log\log\Delta$ is of order $1/\varepsilon$, and a deficit with $\rho(\Delta)\log\log\Delta \to \infty$ suffices. For Corollary F″ the deficit must exceed η as well. *Directed Recovery Structures* makes the same qualitative point: a deficit works if the cost of the genie can be chosen below it before the depth is fixed [14, Section 9.1].

6.3 One unrestricted head

Corollary 6.6 (Machine form of Theorem M)**.** Not formalized; a complete proof is given below.

Let $k \ge 2$, $\beta \ge 2$, $\ell \ge 1$, $q \ge 1$ and $h \ge 0$ be integers, $C \ge 1$ and $0 < \varepsilon \le 1$, and put $D = D_{\mathrm{M}}(k-1, \ell, q, h, \varepsilon/(4\beta C)) \cdot \lceil 4\beta C/\varepsilon \rceil$. Suppose that $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ holds for some $\rho > \varepsilon$. Then for every machine M with k work tapes and value width at most β there is d_0 such that for no $d \ge d_0$ is M oblivious at length $L = d^2 b(d)$ with halting time at most $C(L+1)$, a transposer of all $d \times d$ matrices, and such that all its work heads but one have long hops of owner-load at most ℓ whose family bands are covered by q windows of width h. The same holds with Question 9.1, in the form of Corollary F′.

Proof (not formalized). The proof of Corollary F uses Theorem F only through a cut of the walk DAG with at most $\varepsilon T/(4\beta C)$ edges and depth less than D_1. Theorem 4.2, applied with the unrestricted work head as x_0 and the long hops of the other $k-1$ work heads as H', at $\varepsilon/(4\beta C)$, supplies such a cut with $D_1 = D_{\mathrm{M}}(k-1, \ell, q, h, \varepsilon/(4\beta C))$. □

Corollary F also holds with an owner-load hypothesis on all work heads instead of bounded entry, with $D_1 = D_{\mathrm{F}}(k, \ell, \varepsilon/(4\beta C))$, since its proof uses only the owner-load. Together the two corollaries say the following. Let $0 < \varepsilon \le 1$ and $\rho > \varepsilon$, and let M be oblivious with $k \ge 2$ work tapes and value width at most β, transposing within $C(L+1)$ steps. For fixed ℓ, q and h, if $\mathrm{Crux}(\mathcal{S}_{k,C,\beta,D}, \rho)$ holds at the depth $D = D_{\mathrm{F}}(k, \ell, \varepsilon/(4\beta C)) \cdot \lceil 4\beta C/\varepsilon \rceil$ and at the depth D of Corollary 6.6, then at all large lengths M has at least two work heads of owner-load greater than ℓ, or exactly one, while the family bands of the other work heads are not covered by q windows of width h. Which of the two a transposer would use is open. The first paper's pass bound forces $\Omega(\log d)$ reversals [9, Theorem 5.12], but one head of owner-load one and a single scale supplies them: a zigzag of amplitude about $L/\log d$ has one family band, up to the families truncated at the ends, and makes $\log d$ reversals of amplitude larger than any smaller scale. So the pass bound does not force the case of many scales.

7 Linear machines: the linear crux as rigidity

For machines whose data path is linear over a field, Theorem D has a linear analogue, Theorem $\mathrm{D}_{\mathrm{lin}}$ (Theorem 7.5): *linear depth-TCP* together with a *linear crux* excludes oblivious linear-time transposers. Linear depth-TCP weakens the structural hypothesis, replacing depth by *linear* depth, in which long paths may cancel; the linear crux is required for a larger class of systems, the ball systems of the schedules, and it can be stated exactly as relative rigidity of the transposition matrix with respect to the dependency pattern of the schedule. Relative rigidity alone excludes no machine.

7.1 Linear data paths and path sums

Let F be a field; input bits are read as $0, 1 \in F$. Let M be oblivious at length L, with dependency DAG H_M and step values $v_t(w)$ as in Section 2.4, and write $\mathrm{out}(t)$ for the set of defined out-neighbours $t-1$, $\mathrm{prev}_1(t), \dots, \mathrm{prev}_k(t)$ of t in H_M.

Definition 7.1 (Linear data path)**.** M has an *F-linear data path* at length L if there are maps enc_t from the possible values of step t to F^m, vectors $c_t, a_t \in F^m$ and matrices $A_{t,s} \in F^{m \times m}$, none depending on the input, such that for every $w \in \{0,1\}^L$

$$\mathrm{enc}_t(v_t(w)) = c_t + \sum_{s \in \mathrm{out}(t)} A_{t,s}\, \mathrm{enc}_s(v_s(w)) + a_t\, w_{x_0(t)},$$

where the last term is present only if the input head reads a position $x_0(t) \in [0, L)$, and such that output bit ℓ, appended at step t_ℓ, equals $\varphi_\ell(\mathrm{enc}_{t_\ell}(v_{t_\ell}(w)))$ for a fixed affine functional φ_ℓ with linear part $\varphi_\ell^{\mathrm{lin}}$. Such a *representation* need not be unique.

Write v_t for $\mathrm{enc}_t(v_t)$. For a genie Γ, expanding the recursion along paths and stopping at steps of Γ gives

$$v_t = c'_t + \sum_{s \in \Gamma} T_{t,s} v_s + \sum_j C_{t,j} w_j.$$

Here, for a path $\pi = (t = t_0 \to \cdots \to t_n)$ of H_M with weight $W(\pi) = A_{t_0,t_1} \cdots A_{t_{n-1},t_n}$, $T_{t,s}$ is the sum of $W(\pi)$ over the paths from t to $s \in \Gamma$ with $n \ge 1$ whose interior avoids Γ; $C_{t,j}$ is the sum of

$W(\pi)a_{t_n}$ over the Γ-free paths whose last step reads position j; and c'_t is the sum of $W(\pi)c_{t_n}$ over all Γ-free paths from t. The sums are finite because H_M is acyclic. In matrix form they are the blocks of $(I - A_\Gamma)^{-1}(A - A_\Gamma)$ and $(I - A_\Gamma)^{-1}B$, where A_Γ is the block matrix A with the columns of Γ set to zero and B is the input matrix: the transfer matrices of the linear network cut at Γ, in the sense of Koetter and Médard [34].

Definition 7.2 (Linear depth)**.** Let $I^D_\Gamma(\ell)$ be the set of input positions read at steps reachable from t_ℓ by Γ-free paths of length at most D, and $B^D(\ell) \supseteq I^D_\Gamma(\ell)$ the set of positions read at steps reachable from t_ℓ by directed paths of length at most D in H_M, so that $B^D(\ell) = I^D_\emptyset(\ell)$. Output step t_ℓ has *linear depth at most D for Γ* if $\varphi^{\mathrm{lin}}_\ell C_{t_\ell,j} = 0$ for every $j \notin I^D_\Gamma(\ell)$.

For such j every Γ-free path to a read of j is longer than D, so the condition says that the long contributions to inputs outside $I^D_\Gamma(\ell)$ cancel; output bit ℓ is then an affine function of the genie values and of the bits w_j, $j \in I^D_\Gamma(\ell)$. Structural depth at most D implies linear depth at most D at every output step. Linear depth is not monotone in Γ: adding a step to Γ can break a cancellation between a path through it and a path that avoids it.

7.2 Rigidity relative to the schedule

Let P_{tpos} be the $L \times L$ matrix whose row ℓ is the unit vector $e_{\mathrm{tpos}(\ell)}$. The *minrank* of a system $(I(\ell))_\ell$ over F is the least rank of a matrix R such that row ℓ of $P_{\mathrm{tpos}} - R$ is supported in $I(\ell)$ for every ℓ; it is the least length of an F-linear index code for the system, decodable on all of F^L with linear decoders [26, 3].

Lemma 7.3 (Per-machine rigidity)**.** Not formalized; a complete proof is given below.

Suppose that M has an F-linear data path at length L (with some representation), transposes all $d \times d$ matrices, and has a genie Γ under which every output step has linear depth at most D. Then

$$P_{\mathrm{tpos}} = R + S, \qquad \operatorname{rank} R \le \beta|\Gamma|, \qquad \operatorname{supp}(\textit{row } \ell \textit{ of } S) \subseteq I^D_\Gamma(\ell) \subseteq B^D(\ell);$$

equivalently, the system $(I^D_\Gamma(\ell))_\ell$ has minrank at most $\beta|\Gamma|$ over F.

Proof (not formalized). Define $V_t(w)$ for all $w \in F^L$ by the recursion of the data path. By induction $V_t(w) = v_t(w)$ for $w \in \{0,1\}^L$, and the path-sum expansion holds for V identically in $w \in F^L$, being the unrolled recursion. As M transposes, it appends exactly L bits, and the affine function $\varphi_\ell(V_{t_\ell}(w))$ equals $w_{\mathrm{tpos}(\ell)}$ on $\{0,1\}^L$, hence on all of F^L, because an affine function is determined by its values at 0 and at the unit vectors. By linear depth,

$$w_{\mathrm{tpos}(\ell)} = \varphi_\ell(c'_{t_\ell}) + \sum_{s\in\Gamma} \varphi^{\mathrm{lin}}_\ell T_{t_\ell,s} V_s(w) + \sum_{j\in I^D_\Gamma(\ell)} \varphi^{\mathrm{lin}}_\ell C_{t_\ell,j} w_j \qquad (w \in F^L).$$

So $E(w) = (V_s(w) - V_s(0))_{s\in\Gamma}$ is an F-linear index code for $(I^D_\Gamma(\ell))_\ell$, decodable on all of F^L with linear decoders. On $\{0,1\}^L$ it takes at most $2^{\beta|\Gamma|}$ values, since each v_s takes at most 2^β values. A linear map of rank ρ takes at least 2^ρ values on $\{0,1\}^L$: it is injective on the binary vectors supported on ρ coordinates whose columns are independent, because differences of such vectors have coefficients in $\{-1,0,1\}$. So $\operatorname{rank} E \le \beta|\Gamma|$, and the linear parts of the displayed identity, stacked over ℓ, give the decomposition, with the rows of R in the row space of E. □

Definition 7.4 (Linear depth-TCP and the linear crux)**.** *Linear depth-TCP$(k, C, \beta, \varepsilon, D; F)$* says: for every machine M with k work tapes and value width at most β there is d_0 such that at every $d \ge d_0$ at which M is oblivious at length L, has an F-linear data path and halting time at most

$C(L+1)$, there are a representation of its data path and a genie Γ with $\beta|\Gamma| \le \varepsilon L$ under which every output step has linear depth at most D. Let $\mathcal{S}^{\mathrm{ball}}_{k,C,\beta,D}$ be the class of the ball systems $(B^D(\ell))_{\ell\in[0,L)}$, at the lengths $L = d^2b(d)$, of the oblivious machines with k work tapes, value width at most β and halting time at most $C(L+1)$ that append exactly L bits, as for $\mathcal{S}_{k,C,\beta,D}$; no linear data path is required. $\mathrm{LinCrux}_F(\mathcal{S}^{\mathrm{ball}}_{k,C,\beta,D}, \rho)$ says: for every $\eta > 0$ there is d_0 such that every system of $\mathcal{S}^{\mathrm{ball}}_{k,C,\beta,D}$ at a size $d \ge d_0$ has minrank at least $(\rho-\eta)L$ over F.

Theorem 7.5 (Theorem $\mathrm{D}_{\mathrm{lin}}$)**.** Not formalized; a complete proof is given below.

Let $k \ge 1$, $D \ge 0$ and $\beta \ge 2$ be integers, $C > 0$, $0 < \varepsilon < \rho$ and F a field. If linear depth-TCP$(k, C, \beta, \varepsilon, D; F)$ and $\mathrm{LinCrux}_F(\mathcal{S}^{\mathrm{ball}}_{k,C,\beta,D}, \rho)$ hold, then for every machine M with k work tapes and value width at most β there is d_0 such that for no $d \ge d_0$ is M oblivious at length L with an F-linear data path, halting time at most $C(L+1)$, and a transposer of all $d \times d$ matrices.

Proof (not formalized). Let $\eta = (\rho-\varepsilon)/2$ and let d_0 be the larger of the thresholds of the two hypotheses. If M were such a transposer at some $d \ge d_0$, linear depth-TCP would give a genie Γ with $\beta|\Gamma| \le \varepsilon L$ under which every output step has linear depth at most D, and by Lemma 7.3 the system $(I^D_\Gamma(\ell))_\ell$, and hence the larger system $(B^D(\ell))_\ell \in \mathcal{S}^{\mathrm{ball}}_{k,C,\beta,D}$, would have minrank at most $\varepsilon L < (\rho-\eta)L$. □

This is the proof of Theorem D with the cone lemma replaced by the path-sum expansion, which is exact for linear data paths.

Why it matters. Read as a matrix statement, $\mathrm{LinCrux}_F(\mathcal{S}^{\mathrm{ball}}, \rho)$ says that P_{tpos} is ρL-rigid relative to the D-ball pattern of every linear-time schedule: it is not the sum of a matrix of rank below ρL and a matrix supported on that pattern. So, for linear machines, linear depth-TCP together with a linear crux excludes oblivious linear-time transposers, and the linear crux can be stated exactly as relative rigidity of the transposition matrix; Lemma 7.3 applies only to a machine that has a cheap genie under which every output has linear depth at most D. This is a relative form of Valiant's route from depth reduction and rigidity to lower bounds for linear circuits [43]. The route from small depth to index coding also goes back to Riis's guessing games [41], and it was used in the language of min-rank by Jukna and Schnitger [31]. For static data structures, lower bounds imply rigidity [21], and for systematic linear data structures the two are equivalent [38]. Without the pattern the statement is false for permutation matrices: splitting the receivers into groups of $s+1$, each knowing the bits demanded by the others in its group, gives rank $\lceil L/(s+1)\rceil$ with s off-diagonal entries per row [8, Section 4.3]. Any proof must use the schedule's structure. On the construction side, a linear oblivious transposer at size d either has no genie of cost εL that makes every output linearly D-shallow, or its formal revealed map is a linear index code of rank at most εL for its own ball system.

Remark 7.6 (Scope)**.** (a) The genie plays no role in the crux class: since $I^D_\Gamma(\ell) \subseteq I^D_\emptyset(\ell) = B^D(\ell)$ and the empty genie is allowed, the linear crux for the systems $(I^D_\Gamma(\ell))_\ell$ over all genies is equivalent to the linear crux for the ball systems; and $\mathcal{S}^{\mathrm{ball}}$ dominates the fourth paper's class $\mathcal{S}_{k,C,\beta,D}$ pointwise. Neither Theorem $\mathrm{D}_{\mathrm{lin}}$ nor the linear form of Theorem D implies the other: $\mathrm{D}_{\mathrm{lin}}$ trades a weaker structural hypothesis for a larger crux class. For home-based linear machines $\mathrm{D}_{\mathrm{lin}}$ gives nothing new, since Corollary B supplies structural genies. (b) The trade-off between cost and depth survives: the machine that appends, at every step, the parity of all bits read so far forces at least $\lfloor (L-1)/(D+1)\rfloor$ revealed steps for linear depth D, as in the fourth paper's forced trade-off [8, Proposition 4.13]. (c) Linear depth is robust to dummy traffic on separate coordinates: copies of an input bit carried along deep paths of a dummy schedule on extra tapes and combined so that they cancel contribute

$W(\pi_1) + W(\pi_2) = 0$ to the path sums, whatever their lengths, provided the genie reveals single coordinates (fields, as in [8, Remark 4.6]) and does not cut one of the two paths; Lemma 7.3 and Theorem $\mathrm{D}_{\mathrm{lin}}$ hold verbatim for such coordinate genies, with cost equal to the total bit width of the revealed coordinates. With whole-step genies the robustness fails, and it fails for dummy traffic that is entangled with revealed coordinates, for instance a value masked by a pad and unmasked through a deep partner copy. (d) Linear depth-TCP quantifies over all linear machines, and the theorem uses it only for the transposer itself. It is not a statement about head schedules alone: it depends on the data path and on its representation.

8 The crux in special linear settings

The crux asks for a deficit below rate 1, and its undirected core asks whether storage codes on locally bipartite side information of bounded degree can beat rate $\frac{1}{2}$ [8, Question 6.1]. For all codes this is open beyond the classes of the companion paper, *Nested Elimination and Half-Rate Bounds for Storage Codes* [10], which proves the half bound for every storage code on triangle-free graphs of maximum degree three, among other classes, with the guessing number equal to the fractional matching number there. This section proves two linear cases in which a natural source of high-rate codes fails: codes lifted along towers of covers in characteristic 2 (Section 8.1), and scalar codes at degree four whose exceptional vertices are few or far apart (Section 8.2). By Section 7, linear codes are the relevant ones for linear machines.

8.1 Lifted codes along towers of covers

Let H be a finite connected graph with vertex set V, $|V| = N$, and let $M \in \mathbb{F}_2^{V\times V}$ have unit diagonal and off-diagonal support in the edges of H; the entries M_{uv} and M_{vu} may differ. Its kernel is a linear storage code on H [8, Lemma 6.14]. Let $\mathcal{T}$ be the universal cover of H, a tree, and $\pi = \pi_1(H)$, a free group acting freely on $\mathcal{T}$. A *finite cover* is $\mathcal{T}/\Lambda$ for a subgroup $\Lambda \le \pi$ of finite index; more generally, a finite π-set X defines a cover with vertex set $X \times V$. Choosing a spanning tree and a base vertex turns M into a matrix $\tilde{M} \in \mathrm{Mat}_V(\mathbb{F}_2[\pi])$, whose entry (u, v) is the sum of $M_{uv}g_e$ over the edges e from u to v, where $g_e \in \pi$ is the closed walk along the tree to u, across e and back along the tree, and $\tilde{M}_{vv} = 1$. The *lift* M_X is the image of $\tilde{M}$ under the permutation representation of π on $\mathbb{F}_2[X]$, applied entrywise: a matrix with unit diagonal and support in the edges of the cover, whose kernel is a linear storage code there, of rate $\dim\ker M_X/(N|X|)$. The *inner rank* $\operatorname{irk}\tilde{M}$ is the least ρ such that $\tilde{M} = PQ$ with $P \in \mathrm{Mat}_{N\times\rho}(\mathbb{F}_2[\pi])$ and $Q \in \mathrm{Mat}_{\rho\times N}(\mathbb{F}_2[\pi])$. The group algebra of a free group is a free ideal ring, and its inner rank equals the rank over its universal field of fractions, Cohn's free field [19]. The *universal rate* is

$$r_U(M) = (N - \operatorname{irk}\tilde{M})/N.$$

Let $\chi\colon \pi \to \mathbb{Z}/2$ be the parity of the length of closed walks, $\pi^+ = \ker\chi$, $\hat{\pi}$ the pro-2 completion of π and $\hat{\pi}^+$ the closure of π^+ in it. A cover $\mathcal{T}/\Lambda$ is bipartite if and only if $\Lambda \le \pi^+$.

Theorem 8.1 (The equivariant crux in characteristic two)**.** Not formalized; a complete proof is given below.

Let H be non-bipartite.

(a) Every finite cover X satisfies $\mathrm{rate}(M_X) \ge r_U(M)$.

(b) $r_U(M) \le \frac{1}{2}$.

(c) *Along every tower* $\pi = \Lambda_0 \geq \Lambda_1 \geq \cdots$ *in which each* Λ_{n+1} *has index* 2 *in* Λ_n*, the rate is non-increasing. More precisely, with* $N_n = 2^n N$ *and* M_n *the lift at level* n*, the excess* $e_n = \dim\ker M_n - N_n/2$ *satisfies* $e_{n+1} = 2e_n - r_n$*, where* $r_n \geq 0$ *is the rank of the* twist pairing $\ker M_n^{\mathsf{T}} \times \ker M_n \to \mathbb{F}_2$*,* $(y, x) \mapsto \sum(M_{uv} y_u x_v + M_{vu} y_v x_u)$*, the sum over the edges* uv *of level* n *that the double cover twists.*

(d) *Let* $C = \bigcap_n \overline{\Lambda_n} \leq \hat{\pi}$*. Some level of the tower is bipartite if and only if* $C \leq \hat{\pi}^+$*, and from that level on the rate is at most* $\frac{1}{2}$*.*

(e) *Let* $\iota\colon \pi \to \mathbf{F}$ *be an injective homomorphism into a free-by-*$\mathbb{Z}_2$ *pro-2 group* $\mathbf{F}$*, where* $\mathbb{Z}_2$ *denotes the 2-adic integers (for instance a finitely generated free pro-2 group), and let* $\mathbf{F} = \mathbf{N}_1 \geq \mathbf{N}_2 \geq \cdots$ *be open normal subgroups with trivial intersection. Then* $\operatorname{rate}(M_{\iota^{-1}(\mathbf{N}_j)}) \to r_U(M)$*.*

(f) *For the Clebsch graph and* $M = I + A$*,* $r_U(M) = 0$*.*

Proof (not formalized). (a) Write $\tilde{M} = PQ$ with inner dimension $\rho = \operatorname{irk}\tilde{M}$. The permutation representation is a ring homomorphism, so $M_X = P_X Q_X$ has rank at most $|X|\rho$ and nullity at least $|X|(N - \rho)$.

(b) Apply (a) to the bipartite double cover $H \times K_2 = \mathcal{T}/\pi^+$, on which every edge is twisted. With $M = I + B$, its lift is $\left(\begin{smallmatrix} I & B \\ B & I \end{smallmatrix}\right)$, and (x, y) lies in its kernel exactly when $x = By$ and $(I + B^2)y = 0$; so its nullity is the nullity of $M^2 = I + B^2$, at most N. Hence $2Nr_U(M) \leq N$. The bound is also contained in *Covers and Local Generation* [13, Proposition 3.2], a lower bound $\lceil N/2 \rceil$ for the inner rank over every field, whose separation hypothesis holds for graph templates and whose proof uses the specialization argument of (a).

(c) A double cover of level n is given by voltages in $\mathbb{Z}/2 = \{1, g\}$ on the edges, and $\mathbb{F}_2[\mathbb{Z}/2] \cong \mathbb{F}_2[t]/(t^2)$ with $g = 1 + t$. So the lift is $\varphi = M_n + tM_n^{\mathrm{tw}}$ acting on $(\mathbb{F}_2[t]/(t^2))^{V_n}$, where M_n^{tw} keeps the entries of M_n on twisted edges. Its kernel consists of the $a + tb$ with $M_n a = 0$ and $M_n b = M_n^{\mathrm{tw}} a$. Since $\operatorname{im} M_n = (\ker M_n^{\mathsf{T}})^{\perp}$, the admissible a form the kernel of $x \mapsto (y \mapsto y^{\mathsf{T}} M_n^{\mathrm{tw}} x)$ on $\ker M_n$, of dimension $\dim\ker M_n - r_n$, and each admits a coset of $\ker M_n$ of choices of b. So $\dim\ker M_{n+1} = 2\dim\ker M_n - r_n$, which is the recursion, and the rate does not increase. For symmetric checks, the adjacency checks $M = I + A$, the recursion is the lift identity of *Covers and Local Generation* [13, Lemma 9.1], for arbitrary binary lifts of finite connected graphs; the monotonicity of the normalized nullity under refinements with 2-group kernels in characteristic 2 is its Lemma 4.1, and it is also the index-2 case of the argument of Bergeron, Linnell, Lück and Sauer [5, Lemma 4.1 and Theorem 1.6] (see also Kionke [32, Lemma 2.6]). Part (c) gives the exact identity, with the twist pairing, for every unit-diagonal, possibly asymmetric, binary check, along arbitrary towers of index-2 subgroups.

(d) If every level were non-bipartite, the sets $\overline{\Lambda_n} \cap (\hat{\pi} \setminus \hat{\pi}^+)$ would form a decreasing sequence of nonempty compact sets, so $C \not\leq \hat{\pi}^+$. Conversely, a bipartite level gives $C \leq \overline{\Lambda_n} \leq \hat{\pi}^+$. Covers of bipartite graphs are bipartite, and every storage code on a bipartite graph has rate at most $\frac{1}{2}$ [8, Corollary 6.9].

(e) A free pro-2 group of positive rank maps onto $\mathbb{Z}_2$ with a free pro-2 kernel, so it is free-by-$\mathbb{Z}_2$. By Jaikin-Zapirain and Souza [30, Theorem 8.1], applied to the finitely generated subgroup $\iota(\pi) \cong \pi$, the normalized dimensions of the kernels converge to the dimension over the universal division ring of $\mathbb{F}_2[\iota(\pi)]$; in the case of a finitely generated free pro-2 group, which is all that Proposition 8.2 uses, this is Jaikin-Zapirain's [29, Proposition 3.5]. Since ι is injective, this division ring is Cohn's free field of $\mathbb{F}_2[\pi]$, and the limit is $r_U(M)$.

(f) The Clebsch graph contains the 4-cube, whose Gray code is a Hamiltonian cycle, so it has a spanning 2-factor, and $I + A$ has unit diagonal and nonzero scalars on all edges. For such checks,

Covers and Local Generation shows that the abelianization of $\tilde{M}$, over the Laurent polynomial ring in the chord generators, has nonzero determinant [13, Lemma 6.1], with a written proof there. A factorization $\tilde{M} = PQ$ through fewer than N columns would specialize to a factorization of this invertible matrix, so $\operatorname{irk} \tilde{M} = N = 16$, as in the proof of [13, Theorem 6.2], and $r_U(M) = 0$. As an independent computational check, the script `tools/clebsch_lift.py` constructs an explicit 16-sheeted lift X_0 of the Clebsch graph, by permutation voltages on the 25 edges outside a breadth-first spanning tree, and computes, by exact elimination over $\mathbb{F}_2$, that the lift of $I + A$ has nullity 10; by (a), $10 \geq 16(16 - \operatorname{irk} \tilde{M})$, so again $\operatorname{irk} \tilde{M} = 16$. □

Proposition 8.2 (Towers that never become bipartite)**.** Not formalized; a complete proof is given below. *There are towers of normal subgroups of* $\pi_1(\text{Clebsch})$ *with trivial intersection whose covers are non-bipartite at every level and have girth tending to infinity, and along which the rate of the lifted Clebsch code tends to* 0.

Proof (not formalized). The Clebsch graph has 16 vertices and 40 edges, so $\pi_1(\text{Clebsch})$ is free of rank 25. The kernel $K = \ker(F_9 \to \mathbb{Z}/3)$ of a surjection from the free group of rank 9 is free of rank $1 + 3 \cdot 8 = 25$ by Schreier's formula. It is dense in the pro-2 completion $\hat{F}_9$, since for every normal subgroup N of 2-power index the index of KN divides both 3 and a power of 2; and K embeds in $\hat{F}_9$, since free groups are residually 2. A character of K is continuous for the induced topology exactly when it is the restriction of a character of F_9, and restriction is injective on characters because F_9/K has order 3; so exactly 2^9 of the 2^{25} characters of K are continuous. Since $\operatorname{Aut}(F_{25})$ maps onto $\mathrm{GL}_{25}(\mathbb{F}_2)$, it acts transitively on nonzero characters, so there is an isomorphism $\varphi\colon \pi_1(\text{Clebsch}) \to K$ for which $\chi \circ \varphi^{-1}$ is not continuous. For a descending chain of open normal subgroups $U_j \leq \hat{F}_9$ with trivial intersection, put $\Lambda_j = \varphi^{-1}(K \cap U_j)$. These are normal with trivial intersection; $\Lambda_j \not\leq \pi^+$ for every j, since otherwise $\chi \circ \varphi^{-1}$ would vanish on $K \cap U_j$ and be continuous; and the girth tends to infinity, because only finitely many conjugacy classes have bounded translation length and each nontrivial element eventually leaves the normal subgroups Λ_j. By Theorem 8.1(e), with ι the composite of φ and the inclusion $K \subseteq \hat{F}_9$, and by Theorem 8.1(f), the rate tends to $r_U = 0$; the limit also follows from [13, Theorem 6.2]. □

Why it matters. A construction cannot obtain its coding gain from a fixed small code lifted along towers that become bipartite, nor, for binary linear codes, along towers pulled back injectively from free pro-2 groups: bipartite levels have rate at most $\frac{1}{2}$ for every code (Theorem 8.1(d)), lifting a binary check along double covers never raises the rate (Theorem 8.1(c)), and along towers of type (e) the rate tends to the universal rate, at most $\frac{1}{2}$, and to 0 for the Clebsch code, whose rate is $\frac{5}{8}$ on its own graph [8, Corollary 6.11]. The theorem leaves open: normal towers with trivial intersection and $C \not\leq \hat{\pi}^+$ that are not of type (e), such as 2-adic analytic towers, where a positive limit for the Clebsch code would contradict the mod-2 case of the Lück approximation conjecture in positive characteristic, which Jaikin-Zapirain and Souza discuss for chains with trivial intersection [30, Section 8]; non-normal towers; towers with short even cycles at positive density; and codes that are not lifts of a code on the base. Trivial intersection matters: for the Clebsch adjacency check, the cyclic covers of degree 2^j with the voltage on a single edge form a normal tower of binary lifts whose intersection is not trivial; odd cycles that avoid that edge lift to odd cycles, so no level is bipartite, and the dimension $N/2 + 2$ at every level [13, Theorem 9.5] makes the rate tend to $\frac{1}{2}$, not to $r_U = 0$.

Relation to earlier work. *Covers and Local Generation in Linear Storage Codes* [13] contains several parts of the theorem. Its lift identity [13, Lemma 9.1] is the recursion of (c) for symmetric binary checks, and its Lemma 4.1 gives the monotonicity in (c) for towers of normal subgroups. Its

Proposition 3.2 contains (b). Its Lemma 6.1 and Theorem 6.2, full inner rank and rate tending to 0 along every chain from a faithful embedding into a free-by-$\mathbb{Z}_p$ pro-p group, for every check with nonzero edge entries on a graph with a spanning 2-factor, give (f) without computation and the limit in Proposition 8.2. It also proves, under pro-p and separation hypotheses, an asymptotic half-rate bound for homogeneous checks on a fixed free-group voltage template in every characteristic p, uniformly over coefficient fields and values [13, Theorem 4.3], through Jaikin-Zapirain and Souza's Theorem 8.1, and the exact finite bound $N/2 + 2$ for the Clebsch adjacency code along every sequence of connected binary lifts [13, Theorem 9.2]. New here are the twist-pairing recursion of (c) for asymmetric binary checks along arbitrary towers of index-2 subgroups, part (d), and the towers of Proposition 8.2, which never become bipartite, have girth tending to infinity and are pulled back injectively from $\hat{F}_9$.

8.2 Scalar codes at degree four

Let F be a field and $M \in F^{V\times V}$ a matrix with unit diagonal, $N = |V|$. The *recovery set* of v is $S_v = \{u \neq v : M_{vu} \neq 0\}$, and we say that u *points to* v if $v \in S_u$. The matrix has *no recovery triangle* if $q \notin S_p$ whenever $p, q \in S_r$ and $p \neq q$; this holds whenever the off-diagonal support of M lies in the edges of a triangle-free graph. Following the fourth paper's Theorem S [8, Theorem 6.13], consider at a vertex v the conditions

($\star\star$) $|S_v| \leq 2$, or $|S_v| = 3$ and $r \in S_v$ whenever $v \in S_r$;

($\star 4$) $|S_v| \leq 3$, or $|S_v| = 4$ and $r \in S_v$ whenever $v \in S_r$.

If the support of M lies in a graph of maximum degree at most 4, every vertex satisfies ($\star 4$), since $|S_v| = 4$ forces S_v to be the whole neighbourhood. A vertex violating ($\star\star$) is *heavy*, and H is the set of heavy vertices. The *separation* conditions are: (S1) no vertex has two heavy vertices in its recovery set; (S2) no heavy vertex has a heavy vertex in its recovery set. If the support lies in a graph G and the heavy vertices are pairwise at distance at least 3 in G, both hold.

Theorem 8.3 (Theorem SH: separated heavy vertices)**.** Not formalized; a complete proof is given below.
Let M have unit diagonal and no recovery triangle, and let every vertex satisfy ($\star 4$). If $|H| \leq 2$, or if the heavy vertices satisfy (S1) and (S2), then $\dim\ker M \leq \lfloor N/2 \rfloor$*, over every field. In graph form: if G is triangle-free of maximum degree at most 4, the off-diagonal support of M lies in the edges of G, and the vertices violating ($\star\star$) number at most two or lie pairwise at distance at least 3 in G, then* $\operatorname{rank} M \geq \lceil N/2 \rceil$.

Theorem S of the fourth paper is the case $H = \emptyset$. Since $\dim\ker M^{\mathsf{T}} = \dim\ker M$ and the absence of recovery triangles is invariant under transposition, the same holds when M^{T} satisfies ($\star 4$) and its heavy vertices, defined through the sets $\{u \neq v : M_{uv} \neq 0\}$, satisfy the hypothesis instead.

Corollary 8.4 (Barrier and matching forms)**.** Not formalized; a complete proof is given below.
Let M have unit diagonal and no recovery triangle, let $U \subseteq V$, and let $V \setminus U$ be partitioned into blocks with $M_{uv} = 0$ for u, v in different blocks. If on every block B every vertex satisfies ($\star 4$) on B and the vertices violating ($\star\star$) on B are at most two or satisfy (S1) and (S2) on B, then $\dim\ker M \leq |U| + \sum_B \lfloor |B|/2 \rfloor$*. If the support of M lies in a triangle-free graph G of maximum degree at most 4, then*

$$\dim\ker M \;\leq\; \nu(G) + |H| - \max\bigl(\rho_3(H), \min(|H|, 2)\bigr),$$

where $\nu(G)$ is the matching number and $\rho_3(H)$ the largest size of a subset of H at pairwise distance at least 3 in G; the same holds with the heavy set of M^{T} in place of H.

Proof (not formalized). The kernel of M meets $\{x_U = 0\}$ in a subspace of codimension at most $|U|$, which restricted to $V \setminus U$ lies in $\ker M[V \setminus U, V \setminus U] = \bigoplus_B \ker M[B, B]$; apply Theorem 8.3 to each block. Heaviness and the conditions pass to subsets (see the heredity statement in the proof of Theorem 8.3 below), so for $W \subseteq V$ the classes of pairs $(G[W], M[W, W])$ whose heavy vertices on W number at most two, respectively lie pairwise at distance at least 3 in $G[W]$, are hereditary. By the Tutte–Berge formula [4], every graph G' has a set U' with $|U'| + \sum_B \lfloor |B|/2 \rfloor = \nu(G')$ over the components B of $G' - U'$, so every member of such a class satisfies $\dim\ker \le \nu$, as in the matching form of Theorem S [8, Theorem 6.16]. Take $U = H \setminus D$, where D is a set of $\min(|H|, 2)$ heavy vertices, or a set of $\rho_3(H)$ heavy vertices at pairwise distance at least 3: the heavy vertices of $M[V \setminus U, V \setminus U]$ lie in D, so $\dim\ker M \le |U| + \nu(G - U) \le |H| - |D| + \nu(G)$. □

Why it matters. The theorem extends the fourth paper's Theorem S, the case $H = \emptyset$, to matrices with few or scattered heavy vertices at degree four. A scalar linear code $\ker M$ beating $\lfloor N/2 \rfloor$ by εN on a triangle-free graph of maximum degree four needs $|H| - \rho_3(H) \ge \varepsilon N$, for M and for M^{T} alike: a linear number of heavy vertices that cannot be scattered, that is, dense clusters of vertices whose rows recover them from four neighbours or from three non-reciprocally. For all codes, guessing numbers of triangle-free graphs need not equal the lower bound given by fractional clique covers, as Cameron, Dang and Riis showed with the Higman–Sims graph [18]. This is the regime of the fourth paper's tree code, in which every interior vertex is heavy and no locally confined argument can work [8, Proposition 6.28]. For linear machines, Theorem 7.5 needs the linear crux, and the theorem says where a linear gain at degree four would have to live.

Proof of Theorem 8.3. Write $T_v = \{v\} \cup S_v$ for the support of the row z_v of M, $\mathcal{C} = \ker M$, and $\mathcal{C}^\perp$ for the row space. For a vertex set W, "on W" means that S_v is replaced by $S_v \cap W$ and only rows $r \in W$ are considered. The class $\mathcal{H}$ of pairs $(W, M[W, W])$ satisfying the hypotheses of Theorem 8.3 on W is *hereditary*: no recovery triangle and $(\star 4)$ pass to subsets, since a recovery set either keeps its size or loses elements, and the reciprocity condition only loses in-neighbours; a vertex heavy on $W' \subseteq W$ is heavy on W, for the same reason; so $|H| \le 2$, (S1) and (S2) on W imply them on W'. Let (V, M) be a counterexample in $\mathcal{H}$ with N minimal, and H its set of heavy vertices. We treat the separated case first, and the case $|H| \le 2$ at the end.

Lemma 8.5 (Axioms of a minimal counterexample)**.** Not formalized; a complete proof is given below.

(a) $N = 2m + 1$, $\dim \mathcal{C} = m + 1$ *and* $\operatorname{rank} M = m$.

(b) For every nonempty $X \subseteq V$, $\dim(\mathcal{C}^\perp \cap F^X) \le \lceil |X|/2 \rceil - 1$*; in particular, the rows supported in* X *span at most* $\lceil |X|/2 \rceil - 1$ *dimensions.*

(c) $|S_v| \ge 2$ *for every* v.

(d) For every nonempty $X \subsetneq V$*, the set* $\mathrm{Out}(X) = \{v \in X : S_v \not\subseteq X\}$ *has at least two elements if* $|X|$ *is even and at least one if* $|X|$ *is odd.*

Proof (not formalized). (a) Minimality gives $\dim\ker M[W, W] \le \lfloor |W|/2 \rfloor$ for $W \subsetneq V$. The vectors of $\mathcal{C}$ vanishing at a vertex v, with that coordinate dropped, lie in $\ker M[V - v, V - v]$, so $\dim \mathcal{C} \le \lfloor (N - 1)/2 \rfloor + 1$, which exceeds $\lfloor N/2 \rfloor$ only if $N = 2m + 1$ and $\dim \mathcal{C} = m + 1$. (b) For $X = V$ this is $\operatorname{rank} M = m = \lceil N/2 \rceil - 1$. Otherwise $\mathcal{C} \cap F^{V \setminus X} \subseteq \ker M[V \setminus X, V \setminus X]$, so with $\rho(X) = \dim \mathcal{C}|_X$ we get $m + 1 - \rho(X) \le \lfloor (N - |X|)/2 \rfloor$, that is, $\rho(X) \ge \lfloor |X|/2 \rfloor + 1$; and $\mathcal{C}^\perp \cap F^X$ is the annihilator of $\mathcal{C}|_X$ in F^X, of dimension $|X| - \rho(X)$. (c) $z_v \ne 0$ lies in $\mathcal{C}^\perp \cap F^{T_v}$, which is zero if $|T_v| \le 2$. (d) Let $I = X \setminus \mathrm{Out}(X)$, and suppose $I \ne \emptyset$. The rows z_u, $u \in I$, are supported in X, so by (b) their

rank is at most $\lceil |X|/2 \rceil - 1$; restricted to the columns I they form $M[I, I]$, of rank at least $\lceil |I|/2 \rceil$ by minimality. So $\lceil |I|/2 \rceil \le \lceil |X|/2 \rceil - 1$, which gives the claim. □

Explorations. A *run* from a vertex v_0 is a sequence of moves $(r_1, \pi_1), \ldots, (r_t, \pi_t)$ with $X_0 = \{v_0\}$, $r_i \in X_{i-1}$, $\pi_i = S_{r_i} \setminus X_{i-1} \neq \emptyset$, $X_i = X_{i-1} \cup \pi_i$ and $R_i = \{z_{r_1}, \ldots, z_{r_i}\}$; necessarily $r_1 = v_0$. The row z_{r_j} is supported in X_j and nonzero on all of π_j, so R_i is linearly independent. For $u \in X$ the *pending set* is $P_X(u) = S_u \setminus X$, and u is *closed* at X if $P_X(u) = \emptyset$; a mover is closed after its move. Every $u \in X_i$ other than v_0 entered as a member of a block $\pi_j \subseteq S_{r_j}$, and r_j is its *parent*. Put $\Phi(X_i, R_i) = \lceil |X_i|/2 \rceil - 1 - |R_i|$; then $\Phi \ge 0$ by Lemma 8.5(b), and $\Phi(X_0, \emptyset) = 0$. A move with a block of size j changes Φ by $-1, 0, 0, +1$ for $j = 1, 2, 3, 4$ at odd $|X|$, and by $0, 0, +1, +1$ at even $|X|$. A run is *neutral* if all its moves leave Φ unchanged, and its states are then *good* ($\Phi = 0$). At a good state, $|R_i| = \lceil |X_i|/2 \rceil - 1 \ge \dim(\mathcal{C}^\perp \cap F^{X_i})$, so every row supported in X_i lies in $\operatorname{span} R_i$; and a move with Φ-change -1, in particular a single pending vertex at an odd good state, is a contradiction. A heavy vertex g is *non-closed* at X if $g \notin X$ or $P_X(g) \neq \emptyset$; the set $\mathrm{NC}(X)$ of these vertices only shrinks as X grows.

Lemma 8.6 (Pending sizes, immediate closure, a Hall count)**.** Not formalized; a complete proof is given below.

(a) In any run, v_0 is closed after the first move, a light vertex $u \neq v_0$ has $|P_X(u)| \le 2$, and a heavy vertex $u \neq v_0$ has $|P_X(u)| \le 3$.

*(b) Let $w \in \pi_i$ with $|\pi_i| \ge 2$, and suppose $z_w \in \operatorname{span} R_t$ for some $t \ge i$. Then $\pi_j \subseteq S_w$ for some j with $i < j \le t$. Consequently (*immediate closure*): at a good state X_t whose last block has size at least 2, every member w of the last block has $P_{X_t}(w) \neq \emptyset$; and a neutral run cannot end at $X_t = V$ with a last block of size at least 2.*

(c) If $I \subseteq V$ is nonempty and no vertex of I points to another, then $|I \cup \bigcup_{v \in I} S_v| \ge 2|I| + 1$.

Proof (not formalized). (a) If $|S_u| = 3$ and u is light, ($\star\star$) puts its parent, which points to u, into S_u; if $|S_u| = 4$, ($\star 4$) does the same; the parent lies in X. (b) Write $z_w = \sum_j \lambda_j z_{r_j}$ and let j^* be the largest j with $\lambda_j \neq 0$. The rows z_{r_j}, $j < j^*$, are supported in X_{j^*-1}, which is disjoint from π_{j^*}, so z_w agrees with $\lambda_{j^*} z_{r_{j^*}}$ on π_{j^*} and $\pi_{j^*} \subseteq T_w$. If $j^* < i$, then $w \notin X_{j^*}$ and $z_w(w) = 0$, contradicting $z_w(w) = 1$. If $j^* = i$, some $p \in \pi_i - w$ lies in S_w, while $p, w \in S_{r_i}$: a recovery triangle. So $j^* > i$, $w \in X_{j^*-1}$, and $\pi_{j^*} \subseteq S_w$. For immediate closure, a member w of the last block with $T_w \subseteq X_t$ would have $z_w \in \operatorname{span} R_t$ and no later block. (c) The rows z_v, $v \in I$, are independent, since $z_v(v) = 1$ and $z_v(u) = 0$ for $u \in I - v$, and they are supported in $I \cup \bigcup_{v \in I} S_v$; apply Lemma 8.5(b). □

Lemma 8.7 (The odd-state lemma)**.** Not formalized; a complete proof is given below.

Assume (S1) and (S2). For every $t \ge 0$: if a neutral run with at least one move reaches an odd good state $X \neq V$ with $|\mathrm{NC}(X)| \le t$, there is a contradiction.

Proof (not formalized). By induction on t; the cases that invoke the hypothesis for $t - 1$ do not arise when $t = 0$. Extend the run by neutral moves only, so that all states are good.

Phase O. At the current odd state $X' \neq V$: (O1) a vertex with exactly one pending vertex gives a contradiction; (O2) otherwise, if a light vertex u has $|P(u)| = 2$, make that move, which keeps the state odd, and if it yields V, its last block is a pair, contradicting immediate closure; (O3) otherwise, if a heavy vertex g has $|P(g)| = 2$, make that move: the new state is odd with g closed, so either it is V, contradicting immediate closure, or it has at most $t - 1$ non-closed heavy vertices and the induction hypothesis gives a contradiction; (O4) otherwise every vertex of $\mathrm{Out}(X')$ is heavy

with exactly three pending vertices, by Lemma 8.6(a) and because v_0 is closed. By Lemma 8.5(d), $\mathrm{Out}(X') \neq \emptyset$, so $t \geq 1$; go to Phase E with $Z = X'$ and some $h \in \mathrm{Out}(Z)$. Every move enlarges the state, so Phase O ends.

Phase E. Let $P = P_Z(h) = \{t_1, t_2, t_3\}$. The move (h, P) is neutral at the odd state Z; $X_0 = Z \cup P$ is an even good state, h is closed, $|\mathrm{NC}(X_0)| \leq t - 1$, and $R_{X_0} = R_Z \cup \{z_h\}$ with every row of R_Z supported in Z. No t_i points to another, since M has no recovery triangle, and all t_i are light by (S2). Below, a *state* is a good state reached from X_0 by a neutral run, and $Y = V \setminus X$; $|Y|$ is odd at even states.

(F1) No early singles. At an even state with $|Y| \geq 3$, no vertex has exactly one pending vertex: that move is neutral and leads to an odd good state $\neq V$ with at most $t - 1$ non-closed heavy vertices, contradicting the induction hypothesis.

(F2) Pair coincidence. At an even state with $|Y| \geq 3$, any two pending sets of size exactly 2 are equal. We argue by induction on $|Y|$. Let $q_1 = P_X(u_1) \neq q_2 = P_X(u_2)$ have size 2. If $|Y| = 3$, then $Y = \{p, q, s\}$ with $q_1 = \{p, q\}$ and $q_2 = \{p, s\}$. Make the move (u_1, q_1); at the even proper state $V - s$, Lemma 8.5(d) gives a vertex with pending set $\{s\}$; add s. This neutral run ends at V, where $\operatorname{span} R = \mathcal{C}^\perp \ni z_p$, so by Lemma 8.6(b) with $w = p$ the block $\{s\}$ lies in S_p; but $p, s \in S_{u_2}$, a recovery triangle. If $|Y| \geq 5$ and $|q_1 \cap q_2| = 1$, the move (u_1, q_1) leaves u_2 with a single pending vertex at an even state with $|Y| - 2 \geq 3$, contradicting (F1). So $q_1 \cap q_2 = \emptyset$. Let $X_2 = X \cup q_2$, after the move of u_2, and $X_{12} = X \cup q_1 \cup q_2$, after the moves of u_1 and then u_2; both have last block q_2. Let $c \in q_2$. By immediate closure at X_2, $P_{X_2}(c) \neq \emptyset$, and by (F1) at X_2 it does not have size 1. If it has size 2, the inductive hypothesis at X_2, applied to c and u_1, whose pending set there is still q_1, gives $P_{X_2}(c) = q_1$; then $P_{X_{12}}(c) = \emptyset$, contradicting immediate closure at X_{12}. So $|P_{X_2}(c)| \geq 3$ and c is heavy, by Lemma 8.6(a). Then both members of q_2 are heavy and lie in S_{u_2}, contradicting (S1).

(F3) The final count. By immediate closure at X_0, every t_i has $P_{X_0}(t_i) \neq \emptyset$. If $|V \setminus X_0| \geq 3$, then by (F1) and Lemma 8.6(a) $|P_{X_0}(t_i)| = 2$, and by (F2) all three pending pairs equal one pair p. A light t_i with $|S_{t_i}| = 3$ has $h \in S_{t_i}$ by $(\star\star)$, since h points to t_i, so $S_{t_i} = \{h\} \cup p$; otherwise $S_{t_i} = p$. Then $|P \cup \bigcup_i S_{t_i}| \leq 6 < 7$, contradicting Lemma 8.6(c) with $I = P$. If $V \setminus X_0 = \{w\}$, every t_i points to w. Make the move $(t_1, \{w\})$, reaching V. Then $z_{t_2} \in \operatorname{span} R$, and z_{t_1} is the only row of R that is nonzero at w, so $z_{t_2} = \lambda z_{t_1} + y$ with $\lambda = M_{t_2 w}/M_{t_1 w} \neq 0$ and $y \in \operatorname{span} R_{X_0}$. Now $y(t_2) = 1$ and $y(t_3) = 0$, because t_1 points to neither. Every row of R_Z vanishes on P, so $y|_P = \mu z_h|_P$ for some μ; $z_h(t_3) \neq 0$ forces $\mu = 0$, and then $y(t_2) = 0$, a contradiction. □

Since $\mathrm{NC}(X) \subseteq H$, Lemma 8.7 with $t = |H|$ says that in a minimal counterexample no neutral run with at least one move reaches an odd good state other than V.

The start. By Lemma 8.5(c), $|S_v| \geq 2$ for all v; light vertices have $|S_v| \leq 3$ and heavy ones $|S_v| \in \{3, 4\}$. *Case 1: some v_0 has $|S_{v_0}| = 2$.* The move (v_0, S_{v_0}) is neutral and gives the odd good state T_{v_0}. If $T_{v_0} = V$, immediate closure gives a contradiction; otherwise Lemma 8.7 does. *Case 2: no $|S_v| = 2$, and some heavy h has $|S_h| = 3$.* The move (h, S_h) is neutral at $|X| = 1$ and gives the even good state $X_0 = T_h$ with h closed. This is Phase E with $Z = \{h\}$, $P = S_h$ and $R_Z = \emptyset$: the t_i are light by (S2), (F1) holds by Lemma 8.7 with $t = |H|$, and (F2) and (F3) give a contradiction. *Case 3: every light v has $|S_v| = 3$ and every heavy v has $|S_v| = 4$.* By $(\star\star)$ and $(\star 4)$ every in-neighbour of v lies in S_v; summing, in-sets and recovery sets have the same total size, so they coincide and the relation is symmetric. Its graph Γ is triangle-free, with light vertices of degree 3 and heavy ones of degree 4. The degree sum $3N + |H|$ is even and N is odd, so $|H|$ is odd and $H \neq \emptyset$. Pick $h \in H$ and $v_0 \in S_h$. Then v_0 is light by (S2), $S_{v_0} = \{h, s_1, s_2\}$, and s_1, s_2 are light by (S1), since v_0 points to h. The move (v_0, S_{v_0}) is neutral at $|X| = 1$, and $X_0 = \{v_0, h, s_1, s_2\}$ is an even good state. (F1) and (F2) hold at the even states reached from X_0, with the same proofs, using Lemma 8.7 with $t = |H|$. The vertex h and its four neighbours give $N \geq 5$. If $N = 5$, the

neighbours of h are pairwise non-adjacent and would have degree 1. If $N = 7$, each neighbour of h needs its two further neighbours among the remaining two vertices, which then have degree at least 4, so a neighbour of h points to two heavy vertices, contradicting (S1). So $N \geq 9$. Since Γ is triangle-free, $S_{s_i} \cap X_0 = \{v_0\}$, so $P_{X_0}(s_i)$ has size 2, and by (F2) both pending sets equal one pair p_1, as $|V \setminus X_0| \geq 5$. Make the move (s_1, p_1); $X_1 = X_0 \cup p_1$ is an even good state with $|V \setminus X_1| \geq 3$. By (S1) some $a \in p_1$ is light. As $a \in S_{s_1} \cap S_{s_2}$, symmetry gives $S_a = \{s_1, s_2, x\}$, so $P_{X_1}(a) \subseteq \{x\}$; by immediate closure it is nonempty, and a has a single pending vertex, contradicting (F1).

All cases are contradictory, which proves Theorem 8.3 in the separated case.

At most two heavy vertices. The class of matrices satisfying the hypotheses with $|H| \leq 2$ is hereditary; take a minimal counterexample. Lemmas 8.5 and 8.6 hold, and we prove the odd-state lemma for $t \leq 2$ with two changes. In (F2), the members c of q_2 with $|P_{X_2}(c)| \geq 3$ are heavy and non-closed at X_2, and $\mathrm{NC}(X_2) \subseteq \mathrm{NC}(X_0)$ has at most $t - 1 \leq 1$ element, so not both members of q_2 can be of this kind: (F2) holds without (S1). In (F3), P contains at most one heavy vertex, since a heavy member of P is non-closed at X_0 by immediate closure. If it contains none, (F3) applies verbatim. Otherwise let $t_3 = g$ be heavy and t_1, t_2 light. If $|V \setminus X_0| \geq 5$, then as in (F3) t_1 and t_2 have the same pending pair $p_1 = \{a, b\}$; make the move (t_1, p_1), reaching an even good state X_1 with $|V \setminus X_1| \geq 3$. The vertices a, b are light, since g is the only non-closed heavy vertex. By immediate closure and (F1) at X_1 each of them has a pending pair, the same pair p_2 by (F2). Both t_1 and t_2 point to a, so if $|S_a| = 3$, ($\star\star$) would give $S_a \supseteq \{t_1, t_2\} \cup p_2$, four elements; hence $S_a = p_2$, and likewise $S_b = p_2$, and Lemma 8.6(c) with $I = \{a, b\}$ is contradicted, as $|\{a, b\} \cup p_2| = 4 < 5$. If $V \setminus X_0 = \{a, b, w\}$ with $p_1 = \{a, b\}$ the common pending pair of t_1 and t_2, make the move (t_1, p_1) and then add w, using Lemma 8.5(d) at $V - w$; t_2 does not point to w, and the last row is the only row nonzero at w, so in the expansion of z_{t_2} that row has coefficient 0; the rows of R_{X_0} vanish on p_1, so z_{t_2} agrees on p_1 with λz_{t_1} for some $\lambda \neq 0$, and $y = z_{t_2} - \lambda z_{t_1} \in \operatorname{span} R_{X_0}$ has $y(t_2) = 1$ and $y(g) = 0$, while $y|_P = \mu z_h|_P$ forces $\mu = 0$: a contradiction. If $V \setminus X_0 = \{w\}$, argue as in (F3), with $t_3 = g$. The starts are as before: Cases 1 and 2 use the modified (F2) and (F3), and in Case 3 the relation is symmetric and $|H|$ is odd, so $|H| = 1$, where (S1) and (S2) hold. This completes the proof of Theorem 8.3.

Remark 8.8 (Where the argument stops)**.** Within the odd-state lemma, (S1) is used only at the end of (F2) and (S2) only to make the members of a triple block light; the starting cases use them as well. Without them the argument fails exactly at two configurations: two vertices whose pending pairs are disjoint and consist of four heavy vertices, and a triple block of a heavy vertex that contains a heavy vertex. So a minimal counterexample to the general degree-four question has heavy vertices at distance at most two, and the contradiction tree of the argument must break at a cluster of heavy vertices.

Remark 8.9 (Numerical: small cases)**.** Exact searches over $\mathbb{F}_2$ for all digraphs satisfying the hypotheses of Theorem 8.3 with at most nine vertices, with a positive control that allows recovery triangles, found no counterexample; such small cases cannot test the asymptotic question.

9 What a linear-time transposer would need

Our aim is to decide (T), and we would prefer a construction. This section collects what the results above require of a linear-time transposer, oblivious unless said otherwise. It contains no new theorems: every claim is one of the results above or of earlier papers of the series, or is labelled as a heuristic or as resting on a proof sketch.

The interface. For an oblivious transposer M with k work tapes and running time $C(L+1)$, and every $\varepsilon \in (0,1]$ and $D \geq 0$, Theorem D says: either the dependency DAG has no genie of cost at most εL and full depth at most D, or the revealed word of such a genie is an index code with at most εL bits for the machine's own cone side information, and some fibre is a binary storage code of rate at least $1-\varepsilon$ on the cone relation [8, Theorem 4.10 and Remark 4.9(b)]. After a paid trimming of a small fixed fraction of coordinates, that relation is paired-distance separated [14, Theorem 3.3 and Corollary 3.8]. So a transposer must either defeat cheap depth reduction or carry, on a paired-distance separated relation of bounded degree, a code of rate at least $1-\varepsilon$, a counterexample to Question 9.1 of [14] far beyond rate $\frac{1}{2}$.

Requirements on head motion. In addition to the earlier requirements, $\Omega(\log d)$ passes [9, Theorems 5.9 and 5.12], curved macroscopic motion [8, Theorems 3.10 and 3.12], and heads that are not all home-based unless the crux fails [8, Corollary 4.23], an oblivious transposer must satisfy the following; (R1) assumes the crux at the depths of Corollary F and of Corollary 6.6.

(R1) *Unbounded nesting.* Some work heads must have unboundedly many simultaneously open nested structures with distinct turning points, hence cells entered unboundedly often (Lemma 3.1, Corollary F); either two such heads, or one such head together with heads whose families occupy unboundedly many separated scales (Corollary 6.6). Unboundedly many reversals alone do not suffice.

(R2) *A deep part of vanishing capacity.* Whatever depth-robustness the schedule has lives on the cells entered more than $\lceil 1/\varepsilon \rceil$ times, fewer than $k\varepsilon T$ of them; the rest of the walk DAG is cut to depth $2^{O(k^3/\varepsilon^4)}$ by εT edges (Corollary 3.8). The deep part may relay information between shallow stretches, and must be allowed to (Proposition 3.9). On the deep-nesting family, under the walk-DAG form of Conjecture A′, its size must be of order $T/\log D$ when the rest is cut at depth D (Proposition 5.8), and depth-robust schedules carry a mean time-cut capacity that grows with the depth they force (Remark 3.10, proof sketch only).

(R3) *Not a turn counter.* Families of walks without pauses with $(\mathcal{N}/T)\cdot H$ bounded, where $\mathcal{N}$ is the number of turns and H the maximal number of hops spanning a time, are depth-reducible (Proposition 5.7); so the deep part must combine many turns with many hops open at once, as the deep-nesting pairs do.

Requirements on the codes. The fibre code on the cone relation must avoid every class on which rate $\frac{1}{2}$ is proved, on all but a small fraction of its coordinates: triangle-free side information of maximum degree three and the further directed classes of the companion paper [10], symmetric side information of circular chromatic number below three [12, Corollary 3.2], hyperfinite incidence graphs and strongly connected components with at most two internal inputs [14, Corollary 4.4 and Theorem 5.1], and, for linear codes, bounded complex coefficients [8, Theorem 6.17]. This paper adds the following for linear codes; parts of (R4) were known before [13].

(R4) *No fixed binary lift along a controlled tower.* Along a tower of covers that becomes bipartite, every code has rate at most $\frac{1}{2}$ from that level on [8, Corollary 6.9]. A fixed small binary linear code lifted along a tower of double covers never gains rate, and along a tower pulled back injectively from a chain of open normal subgroups with trivial intersection of a free-by-$\mathbb{Z}_2$ pro-2 group its rate tends to its universal rate, at most $\frac{1}{2}$; for the Clebsch code the limit is 0 (Theorem 8.1 and Proposition 8.2; compare [13, Lemma 9.1 and Theorem 6.2]).

(R5) *Clustered heavy vertices at degree four.* A scalar linear code $\ker M$ beating $\lfloor N/2 \rfloor$ by εN on triangle-free side information of maximum degree four needs a matrix M, and likewise M^{T}, with a linear number of heavy vertices that cannot be scattered (Corollary 8.4).

(R6) *Non-rigidity, for linear machines.* The transposition matrix must be the sum of a matrix of rank at most εL and a matrix supported on the depth-D ball pattern of the schedule, for a genie under which every output is linearly D-shallow (Lemma 7.3); otherwise the machine has no such genie.

Coding gaps. By the first paper, every linear-time transposer yields a causal code for complete exchange on the block network of its run that beats multicommodity flow by a factor $\Omega(\log d)$, on all inputs if the machine is oblivious [9, Theorems 8.8, 9.2 and 9.9], Lean-verified there; for oblivious algorithms the connection goes back to Adler, Harvey, Jain, Kleinberg and Lehman [1, Theorem 6.2]. Coding gains over routing in undirected networks exist, and the known gains grow with the size of the network, as $(\log n)^{\varepsilon}$ in [7] and polylogarithmically in [44]. Requirements (R1) and (R2) suggest where a gain of order $\log d$ would have to be produced: in a deep part of small capacity, visited over and over, while the shallow part only moves data. This is a heuristic. A conditional theorem of Brende and GPT-6 Astra shows that the shape is not empty for computations in general: if TCP(k) fails for some k, one fixed native bijection with a linear-time native inverse and logarithmic work span needs genie entropy above δL at every fixed depth [17, Theorem H].

Mechanisms that fail. The following cannot be the whole of an oblivious transposer for which the crux holds at the depths of Corollaries F and B and of Corollary 6.6: home-based work heads (Corollary B); work heads that enter every cell boundedly often, in any coupling and at any scales (Corollary F); one arbitrary head together with bounded heads at boundedly many scales (Corollary 6.6). And the following cannot supply its codes on more than a small fraction of coordinates: the classes listed above, fixed binary lifts along controlled towers for binary linear codes, and, for scalar linear codes at degree four, scattered heavy vertices.

10 Open problems

For oblivious machines the series reduces ¬(T) to depth reduction and a coding bound (Theorem 2.5). This paper settles depth reduction for bounded owner-load and for one arbitrary head plus bounded heads at boundedly many scales, and reduces the coding side, for bounded-entry machines, to Question 9.1 of *Directed Recovery Structures*. The first item below is the question itself. For the others, by Theorem 2.5 and the proof of Corollary F′, TCP(k) for every k together with an affirmative answer to Problem 10.3 would exclude oblivious linear-time transposers, while a depth-robust family of walks (Problems 10.1 and 10.2) or a negative answer to Problem 10.3 would show where a construction has to work.

1. *Decide (T).* Construct a machine that transposes $d \times d$ matrices with $b(d)$-bit entries in time $O(L)$, or prove that none exists. The first paper's frozen checker accepts either answer [9].

2. *Depth reduction at unbounded owner-load.* By Corollary 4.4, a depth-robust family of walks has two heads of unbounded owner-load, or one such head together with bounded heads whose families occupy unboundedly many separated scales.

   Problem 10.1 (Two remaining cases of TCP)**.** *(a) Are the deep-nesting pairs depth-robust as $m \to \infty$, for some sequence of odd shifts? Conjecture A″ (Conjecture 5.24), with its walk-DAG*

analogue, would give depth $2^{\Omega(m)}$ *at fixed cost and a negative answer to* TCP(2)*; the counter gives depth* $2^{m+O(1)}/\varepsilon$ *at every shift (Corollary 5.5). (b) Is TCP true for one arbitrary walk together with walks of owner-load one whose families occupy unboundedly many separated scales, for instance the pair of Remark 4.7?*

Conjecture A″ for block pairs would follow from visibility of the coarse classes in a window modulo q at a fixed shift (Proposition 5.23). Theorem V gives less, even on average over shifts: the integer event $F \le -Q-1$, with no lower bound on F, which implies the window only for q of order $(m-\kappa)Q$, with κ the canceller level of Lemma 5.20. Two points are open: a bound on $|F|$ on the straddling hops, or a version of Theorem V that counts the window directly; and, at one fixed shift, equidistribution of the apex centres of one head relative to the canceller valleys of the other. A proof for all labels must use cost, since fine levels are blind (Corollary 5.18), and packing certificates cannot do it [8, Proposition 5.14].

3. *Three stacks.* SR(2) holds by planarity (Proposition 4.6), and SR($2k$) implies TCP(k) (Proposition 4.5).

   Problem 10.2 (Three-stack DAGs)**.** *Is every family of* 3*-stack DAGs with a spine depth-reducible, that is, does* SR(3) *hold? A depth-robust family of* 3*-stack or* 4*-stack DAGs, each realized as the walk DAG of two walks, would refute* TCP(2)*.*

4. *The coding bound.* By Corollaries F′ and F″, an affirmative answer to the following question excludes oblivious bounded-entry transposers, and adaptive ones whose traces have entropy at most ηL for a fixed $\eta < \frac{1}{2}$ and whose traces of probability at least $2^{-\eta L}$ have bounded entry.

   Problem 10.3 (Question 9.1 of [14])**.** *Do all storage codes, over every alphabet, on recovery relations of bounded incidence degree that satisfy the paired-distance condition have rate at most* $\frac{1}{2}+o(1)$*? By Remark 6.5, Corollary F′ already follows from a deficit* $\rho(\Delta)$ *with* $\rho(\Delta)\log\log\Delta \to \infty$*.*

   The companion paper proves exact half bounds on several classes of relations without short paired paths and states the directed degree-three case as a conjecture [10]; *Directed Recovery Structures* proves the hyperfinite case and several others [14].

5. *Degree four.*

   Problem 10.4 (Triangle-free side information of degree four)**.** *Does* $2h(V) \le \sum_v h(\{v\})$ *hold for every polymatroid valid for a triangle-free graph of maximum degree* 4*, so that every storage code there has rate at most* $\frac{1}{2}$*? For scalar linear codes, does* $\dim\ker M \le \lfloor N/2 \rfloor$ *hold when the heavy vertices of Theorem 8.3 form clusters?*

   At degree three both are settled, for all codes by the companion paper [10] and for scalar linear codes by the fourth paper's Theorem S. At degree four, Theorem 8.3 reduces the scalar question to clusters of heavy vertices, the regime in which tree-local arguments are impossible [8, Proposition 6.28].

6. *The scratch.* If TCP fails at unbounded owner-load, the route of Theorem D needs input specific to transposition: fixed-depth genies, even charged by entropy, then fail for some easy oblivious bijections [17, Theorem H].

Problem 10.5 (The scratch question)**.** *Can an oblivious linear-time transposer use a set of fewer than εL work-tape cells, each visited unboundedly often, to supply the global information that the crux denies to the shallow part of its schedule (Corollary 3.8)?*

7. *Rigidity.* For linear machines, linear depth-TCP together with a linear crux excludes oblivious linear-time transposers (Theorem 7.5), and the linear crux can be stated exactly as the following question.

   Problem 10.6 (Relative rigidity of transposition)**.** *Is there $\rho > 0$ such that, for every linear-time schedule and every fixed D, the transposition matrix is not the sum of a matrix of rank below ρL and a matrix supported on the depth-D ball pattern of the schedule, for all large d?*

8. *Towers.* Do binary codes lifted along normal 2-adic analytic towers with trivial intersection that stay non-bipartite have asymptotic rate at most $\frac{1}{2}$? Without trivial intersection the answer is negative: two Clebsch graphs joined by two edges, the code $\ker((I+A)\oplus(I+A))$, which ignores the joining edges, and a voltage on one joining edge give connected non-bipartite covers of rate $\frac{5}{8}$ at every level. For the Clebsch adjacency code, connected binary-lift towers have dimension at most $N/2+2$ [13, Theorem 9.2], and along a normal tower with trivial intersection a positive limit would contradict the mod-2 case of the Lück approximation conjecture in positive characteristic [30, Section 8] (Section 8.1).

A Index of Lean declarations

Every declaration of this paper's library cited in the text, grouped by the Lean module that contains it, relative to the namespace `BoundedNesting`. All of them depend only on the axioms `propext`, `Classical.choice` and `Quot.sound` (Appendix B). The declarations of the fourth paper's library named in the text (`MotionDepthCodes.lemma_Zq` and `MotionDepthCodes.card_descents_eq`) are verified there [8].

`BoundedNesting.Clock` `Fam`, `Fam.Valid`, `Fam.Valid.band_at_end`, `Fam.Valid.band_of_mem`, `Fam.Valid.overlap`, `Fam.Valid.sum_d_le`, `Fam.Valid.unique_span`, `Fam.band`, `Fam.card_wrap_le`, `Fam.clock`, `Fam.clock_le_of_no_wrap`, `Fam.clock_lt_of_no_wrap`, `Fam.cnt`, `Fam.depth_le_of_lex`, `Fam.digit`, `Fam.group`, `Fam.occ`, `Fam.rk`, `flr`, `grp`, `lam`.

`BoundedNesting.ClockCost` `Fam.Valid.card_gchange_le`, `Fam.Valid.card_higher_le`, `Fam.Valid.card_incons_le`, `Fam.Valid.card_viol_calm_le`, `Fam.Valid.cost_sum_le`, `Fam.Valid.exists_charge`, `Fam.Valid.exists_cheap`, `Fam.Valid.sum_calm_growth_le`, `Fam.Valid.sum_gjumps`, `Fam.Valid.sum_higher_le`, `Fam.Valid.sum_jumps_lam`, `Fam.Valid.sum_jumps_pow`, `Fam.Valid.sum_old_le`, `Fam.Valid.sum_viol_calm_le`, `Fam.calm`, `Fam.card_desc_still_le`, `Fam.card_wrap_union`, `Fam.descents_subset`, `Fam.digit_lt`, `Fam.gchange`, `Fam.gjumps`, `Fam.higher`, `Fam.incons`, `Fam.still`, `Fam.sum_desc_still_le`, `Fam.violated_subset`, `card_grp_ne_le`, `exists_change`, `sum_floor_pow_le`.

`BoundedNesting.Families` `card_fam_le`, `disj_aux`, `fam`, `famData`, `famData_disj`, `famData_valid`, `famDepth`, `famDepth_le`, `famDepth_lt`, `famP`, `famP_ne`, `famU`, `fam_cell_injOn`, `fam_cell_mem`, `fam_long`, `fam_nested`, `fam_no_cross`, `fam_span`, `famd`, `side`, `sw`, `sw_above`, `vcode`, `vcode_lt`, `vh_eq`.

`BoundedNesting.Labels` `card_marked_add_le`, `depth_le_of_label`.

BoundedNesting.OneWalk IsHop, IsLine, card_cutHops_le, card_cutStates_le, cutHops, cutStates, exists_line_residue, exists_offset, hop_unique, hops, strip, stripCount, stripLabel, strip_cut, strip_cut_depth, strip_eq_of_adj.

BoundedNesting.Owner IsOwner, IsOwner.unique, IsUpperHop, IsWalk, card_crossings_le, crossings, entries, owner_entries.

BoundedNesting.TheoremF DF, OwnerLoadLE, ceil_le_two_pow_clog, cost_le_eps, hopOwner, hopOwner_spec, hopPairs, theoremF, theoremF_owner.

BoundedNesting.TurnCounter turnCounter_cost.

B Building and verifying the formalization

The Lean project is the directory `lean/` next to this paper. It uses Lean and Mathlib, both at version v4.34.1 (`lean-toolchain`, `lakefile.toml`, `lake-manifest.json`), and requires the library of the fourth paper [8] through a relative path; that library imports the libraries of the second and first papers and the repository's shared library `lean-machine`. Keep the repository's directory layout, so that these sibling directories stay in place.

Build. With `elan` installed, run in `lean/`:

```
lake exe cache get && lake build
```

The first command downloads the compiled Mathlib; the second compiles the library and the parts of the earlier libraries that it uses. Before the first build, fetch the second paper's pinned external dependency as its README describes (`python3 verification/fetch_openai_metric.py`, run in that paper's directory).

Axioms. The file `PaperAxioms.lean`, generated from the LaTeX sources of this paper by `tools/lean_index.py`, runs `#print axioms` on every declaration of this library named here (Appendix A):

```
lake env lean PaperAxioms.lean
```

Every declaration depends at most on `propext`, `Classical.choice` and `Quot.sound`. No Lean file of this library uses `sorry`, `admit` or `native_decide`, or declares an `axiom`. The script regenerates both the index of Appendix A and `PaperAxioms.lean`; with the option `-index-only` it writes only the index.

The computational check of Theorem 8.1(f). The script `tools/clebsch_lift.py`, which needs only Python 3, builds the 16-sheeted lift of the Clebsch graph described at its top and prints the rank and the nullity of the lift of $I + A$ over $\mathbb{F}_2$:

```
python3 tools/clebsch_lift.py
```

It reports rank 246 and nullity 10. The computation is exact elimination over $\mathbb{F}_2$ and is not checked in Lean.

Where things are. `Owner.lean` (walks, upper hops, owners, entries and crossings; the owner lemma), `Labels.lean` (cuts are labels), `OneWalk.lean` (hops, lines, strips, the strip label and Lemma W), `TurnCounter.lean` (the label arithmetic of the turn counter, from the fourth paper's Lemma Zq), `Clock.lean` (abstract family data, the family clock and the comparison of clocks), `ClockCost.lean` (the cost of the family clock, summed over offsets: the bounds of the proof of Theorem F), `Families.lean` (sides, families, spans, heights, depths and virtual heads of the hops of walks; the structure of families), and `TheoremF.lean` (owner-load, owners, the bound D_{F} and Theorem F).

References

[1] Micah Adler, Nicholas J. A. Harvey, Kamal Jain, Robert D. Kleinberg, and April Rasala Lehman. On the capacity of information networks. In *Proceedings of the Seventeenth Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2006)*, pages 241–250, 2006.

[2] Joël Alwen, Jeremiah Blocki, and Krzysztof Pietrzak. Depth-robust graphs and their cumulative memory complexity. In *Advances in Cryptology – EUROCRYPT 2017*, volume 10212 of *Lecture Notes in Computer Science*, pages 3–32. Springer, 2017. doi:10.1007/978-3-319-56617-7_1.

[3] Ziv Bar-Yossef, Yitzhak Birk, T. S. Jayram, and Tomer Kol. Index coding with side information. *IEEE Transactions on Information Theory*, 57(3):1479–1494, 2011.

[4] Claude Berge. Sur le couplage maximum d'un graphe. *Comptes Rendus de l'Académie des Sciences, Paris*, 247:258–259, 1958.

[5] Nicolas Bergeron, Peter Linnell, Wolfgang Lück, and Roman Sauer. On the growth of Betti numbers in p-adic analytic towers. *Groups, Geometry, and Dynamics*, 8(2):311–329, 2014. Preprint arXiv:1204.3298; Theorem 1.6 and Lemma 4.1 cited. doi:10.4171/GGD/227.

[6] Jean Bourgain and Amir Yehudayoff. Expansion in $\mathrm{SL}_2(\mathbb{R})$ and monotone expanders. *Geometric and Functional Analysis*, 23(1):1–41, 2013. doi:10.1007/s00039-012-0200-9.

[7] Mark Braverman and Zhongtian He. The multiple unicast conjecture is false. arXiv:2610.10108, 2026. URL: https://arxiv.org/abs/2610.10108.

[8] Hans Brende and Claude Opus 5.5 (Anthropic). Curved motion, depth reduction, and storage codes for matrix transposition on tapes. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes.

[9] Hans Brende and Claude Opus 5.5 (Anthropic). Matrix transposition and the tape coding conjecture. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Matrix-Transposition-and-the-Tape-Coding-Conjecture.

[10] Hans Brende and Claude Opus 5.5 (Anthropic). Nested elimination and half-rate bounds for storage codes. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision ea4c7869dd1f87f3a9cbd08ab1b91cadb2ddc482. URL: https:

//github.com/HansBrende/papers/tree/ea4c7869dd1f87f3a9cbd08ab1b91cadb2ddc482/Nested-Elimination-and-Half-Rate-Bounds-for-Storage-Codes.

[11] Hans Brende and GPT-6 Astra (OpenAI). Bit and word transposition on tapes. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Bit-and-Word-Transposition-on-Tapes.

[12] Hans Brende and GPT-6 Astra (OpenAI). Conditional elimination and circular storage codes. Memo to the fourth paper of the series, with Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes/memo.

[13] Hans Brende and GPT-6 Astra (OpenAI). Covers and local generation in linear storage codes. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Covers-and-Local-Generation-in-Linear-Storage-Codes.

[14] Hans Brende and GPT-6 Astra (OpenAI). Directed recovery structures in tape computations. Manuscript and partial Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Directed-Recovery-Structures-in-Tape-Computations.

[15] Hans Brende and GPT-6 Astra (OpenAI). Local coding barriers and the limits of tape geometry for matrix transposition. Manuscript and Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Local-Coding-Barriers-and-the-Limits-of-Tape-Geometry-for-Matrix-Transposition.

[16] Hans Brende and GPT-6 Astra (OpenAI). Subcubic storage bounds beyond scalar recovery. Memo to the fourth paper of the series, with Lean formalization, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes/memo-subcubic.

[17] Hans Brende and GPT-6 Astra (OpenAI). TCP and information-cheap depth reduction. Memo to the fourth paper of the series, written proofs, repository HansBrende/papers, October 2026. Pinned revision 69e711bab7f4d34995b8ecee807ebdb2c349349c. URL: https://github.com/HansBrende/papers/tree/69e711bab7f4d34995b8ecee807ebdb2c349349c/Curved-Motion-Depth-Reduction-and-Storage-Codes-for-Matrix-Transposition-on-Tapes/memo-information-depth.

[18] Peter J. Cameron, Anh N. Dang, and Søren Riis. Guessing games on triangle-free graphs. *Electronic Journal of Combinatorics*, 23(1):Paper 1.48, 2016. doi:10.37236/4731.

[19] P. M. Cohn. *Free Ideal Rings and Localization in General Rings*, volume 3 of *New Mathematical Monographs*. Cambridge University Press, 2006. doi:10.1017/CBO9780511542794.

[20] Vida Dujmović, Anastasios Sidiropoulos, and David R. Wood. Layouts of expander graphs. *Chicago Journal of Theoretical Computer Science*, 2016:Article 1, 2016. Preprint arXiv:1501.05020.

[21] Zeev Dvir, Alexander Golovnev, and Omri Weinstein. Static data structure lower bounds imply rigidity. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing (STOC 2019)*, pages 967–978, 2019. doi:10.1145/3313276.3316348.

[22] Zeev Dvir and Avi Wigderson. Monotone expanders: Constructions and applications. *Theory of Computing*, 6:291–308, 2010. Article 12. doi:10.4086/toc.2010.v006a012.

[23] Paul Erdős, Ronald L. Graham, and Endre Szemerédi. On sparse graphs with dense long paths. *Computers and Mathematics with Applications*, 1(3–4):365–369, 1975. doi:10.1016/0898-1221(75)90037-1.

[24] Zvi Galil, Ravi Kannan, and Endre Szemerédi. On 3-pushdown graphs with large separators. *Combinatorica*, 9(1):9–19, 1989. doi:10.1007/BF02122679.

[25] Zvi Galil, Ravi Kannan, and Endre Szemerédi. On nontrivial separators for k-page graphs and simulations by nondeterministic one-tape Turing machines. *Journal of Computer and System Sciences*, 38(1):134–149, 1989. Preliminary version in Proc. 18th ACM STOC, 1986, pp. 39–49. doi:10.1016/0022-0000(89)90036-6.

[26] Willem Haemers. On some problems of Lovász concerning the Shannon capacity of a graph. *IEEE Transactions on Information Theory*, 25(2):231–232, 1979.

[27] David Harvey and Joris van der Hoeven. Integer multiplication is at least as hard as matrix transposition. arXiv:2503.22848; also in Proc. FOCS 2025, pp. 1366–1381 (numbering as in arXiv), 2025. URL: https://arxiv.org/abs/2503.22848.

[28] John Hopcroft, Wolfgang Paul, and Leslie Valiant. On time versus space. *Journal of the ACM*, 24(2):332–337, 1977. doi:10.1145/322003.322015.

[29] Andrei Jaikin-Zapirain. Free $\mathbb{Q}$-groups are residually torsion-free nilpotent. *Annales Scientifiques de l'École Normale Supérieure (4)*, 57:1101–1133, 2024. Preprint arXiv:2212.00402; Proposition 3.5 cited. doi:10.24033/asens.2587.

[30] Andrei Jaikin-Zapirain and Henrique Souza. Sylvester domains and pro-p groups. *Documenta Mathematica*, 31(2):453–502, 2026. Preprint arXiv:2402.14130; Theorem 8.1 cited. doi:10.4171/DM/1034.

[31] Stasys Jukna and Georg Schnitger. Min-rank conjecture for log-depth circuits. *Journal of Computer and System Sciences*, 77(6):1023–1038, 2011. doi:10.1016/j.jcss.2009.09.003.

[32] Steffen Kionke. The growth of Betti numbers and approximation theorems. Lecture notes, Borel seminar 2017, arXiv:1709.00769, 2017. Lemma 2.6 cited. URL: https://arxiv.org/abs/1709.00769.

[33] Walter W. Kirchherr. Transposition of an $l \times l$ matrix requires $\Omega(\log l)$ reversals on conservative Turing machines. *Information Processing Letters*, 28(2):55–59, 1988. doi:10.1016/0020-0190(88)90163-9.

[34] Ralf Koetter and Muriel Médard. An algebraic approach to network coding. *IEEE/ACM Transactions on Networking*, 11(5):782–795, 2003.

[35] Richard J. Lipton and Robert Endre Tarjan. A separator theorem for planar graphs. *SIAM Journal on Applied Mathematics*, 36(2):177–189, 1979.

[36] Richard J. Lipton and Robert Endre Tarjan. Applications of a planar separator theorem. *SIAM Journal on Computing*, 9(3):615–627, 1980.

[37] Wolfgang Lück. Approximating L^2-invariants by their finite-dimensional analogues. *Geometric and Functional Analysis*, 4(4):455–481, 1994.

[38] Sivaramakrishnan Natarajan Ramamoorthy and Cyrus Rashtchian. Equivalence of systematic linear data structures and matrix rigidity. In *11th Innovations in Theoretical Computer Science Conference (ITCS 2020)*, volume 151 of *LIPIcs*, pages 35:1–35:20, 2020. doi:10.4230/LIPIcs.ITCS.2020.35.

[39] Wolfgang Paul and Rüdiger Reischuk. On alternation II. A graph theoretic approach to determinism versus nondeterminism. *Acta Informatica*, 14(4):391–403, 1980. doi:10.1007/BF00286494.

[40] Wolfgang J. Paul, Nicholas Pippenger, Endre Szemerédi, and William T. Trotter. On determinism versus non-determinism and related problems. In *24th Annual Symposium on Foundations of Computer Science (FOCS 1983)*, pages 429–438, 1983.

[41] Søren Riis. Information flows, graphs and their guessing numbers. *Electronic Journal of Combinatorics*, 14(1):Research Paper R44, 2007. doi:10.37236/962.

[42] Rahul Santhanam. On separators, segregators and time versus space. In *Proceedings of the 16th Annual IEEE Conference on Computational Complexity*, pages 286–294, 2001. doi:10.1109/CCC.2001.933895.

[43] Leslie G. Valiant. Graph-theoretic arguments in low-level complexity. In *Mathematical Foundations of Computer Science (MFCS)*, volume 53 of *Lecture Notes in Computer Science*, pages 162–176. Springer, 1977.

[44] Xindan Zhang, Baochun Li, and Zongpeng Li. Network coding can beat routing in undirected multiple-unicast networks. arXiv:2610.09367, 2026. URL: https://arxiv.org/abs/2610.09367.